BT601 - FINALTERM MERGED PPT (LECTURE 119 TO 213)

Virology

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Virology

Enzymatic Changes In Virus-Infected Cells

DNA Polymerase

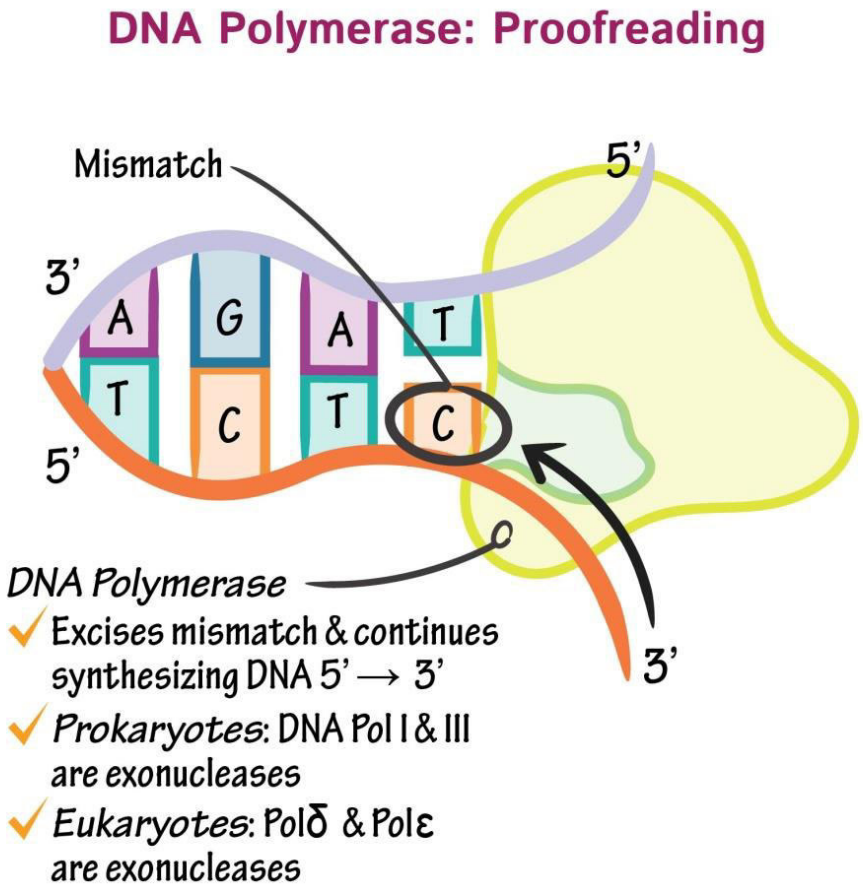
Enzymatic Changes

DNA Polymerase

- DNA polymerase is an enzyme that synthesizes DNA molecules from deoxyribonucleotides, the building blocks of DNA
- These enzymes are essential for DNA replication and usually work in pairs to create two identical DNA strands from a single original DNA molecule

Enzymatic Changes

STRUCTURE



Enzymatic Changes

POXVIRUS

- Poxvirus replicates in cytoplasm
- DNA polymerase does not exist in cytoplasm
- Survival advantage in viral genome is obvious

Virology

Enzymatic Changes in Virus-infected Cells

Thymidine Kinase

Enzymatic Changes

Thymidine Kinase

- Thymidine kinase is an enzyme,

  a phosphotransferase (a kinase): 2'-deoxythymidine kinase, ATP-thymidine 5'-phosphotransferase

- It can be found in most living cells

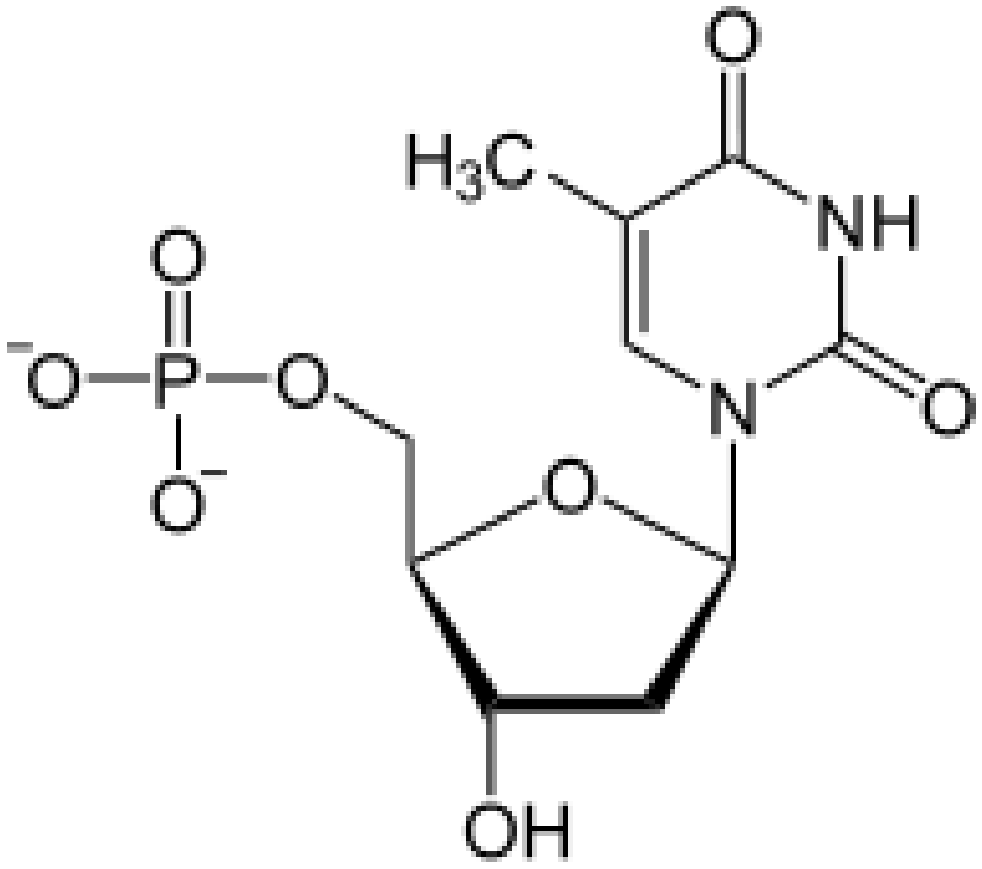
- Certain viruses also have genetic information for expression of viral thymidine kinases

- Thymidine kinase catalyzes the reaction:

Thd + ATP → TMP + ADP

Enzymatic Changes

STRUCTURE

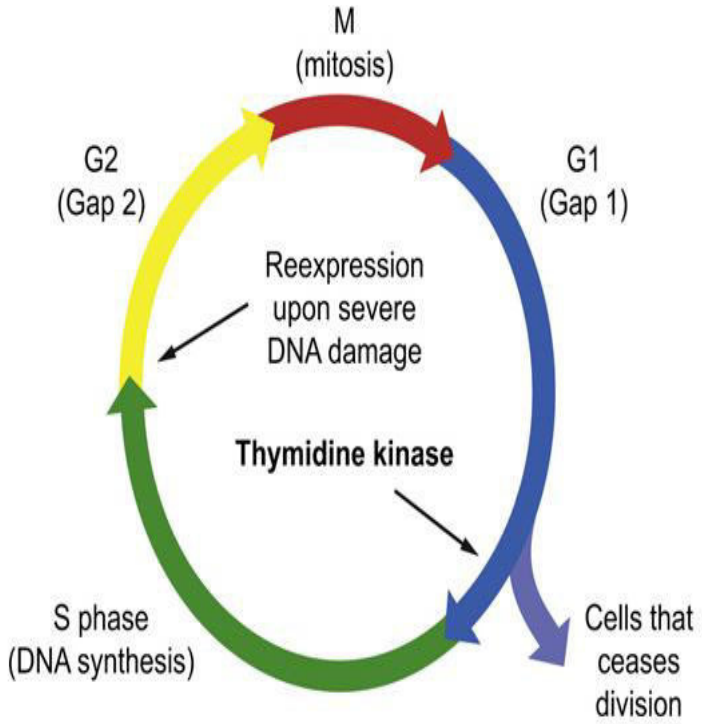


Enzymatic Changes

- Infection with herpesvirus and poxvirus
- Causes increase in thymidine kinase

Enzymatic Changes

Normal function in cell



- Not known until now
- It may phosphhorylate the free deoxyribonucleoside to a usable DNA precursor
- End result may be free thymidine which may be involved in DNA synthesis

Enzymatic Changes

In infected cell

- In vaccinia virus infection
- Enzyme production switched off after ten hours
- Can be inhibited by actinomycin or puromycin
- 40 fold increase in enzyme
- Mechanism yet unknown

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Enzymatic Changes in Virus-Infected Cells

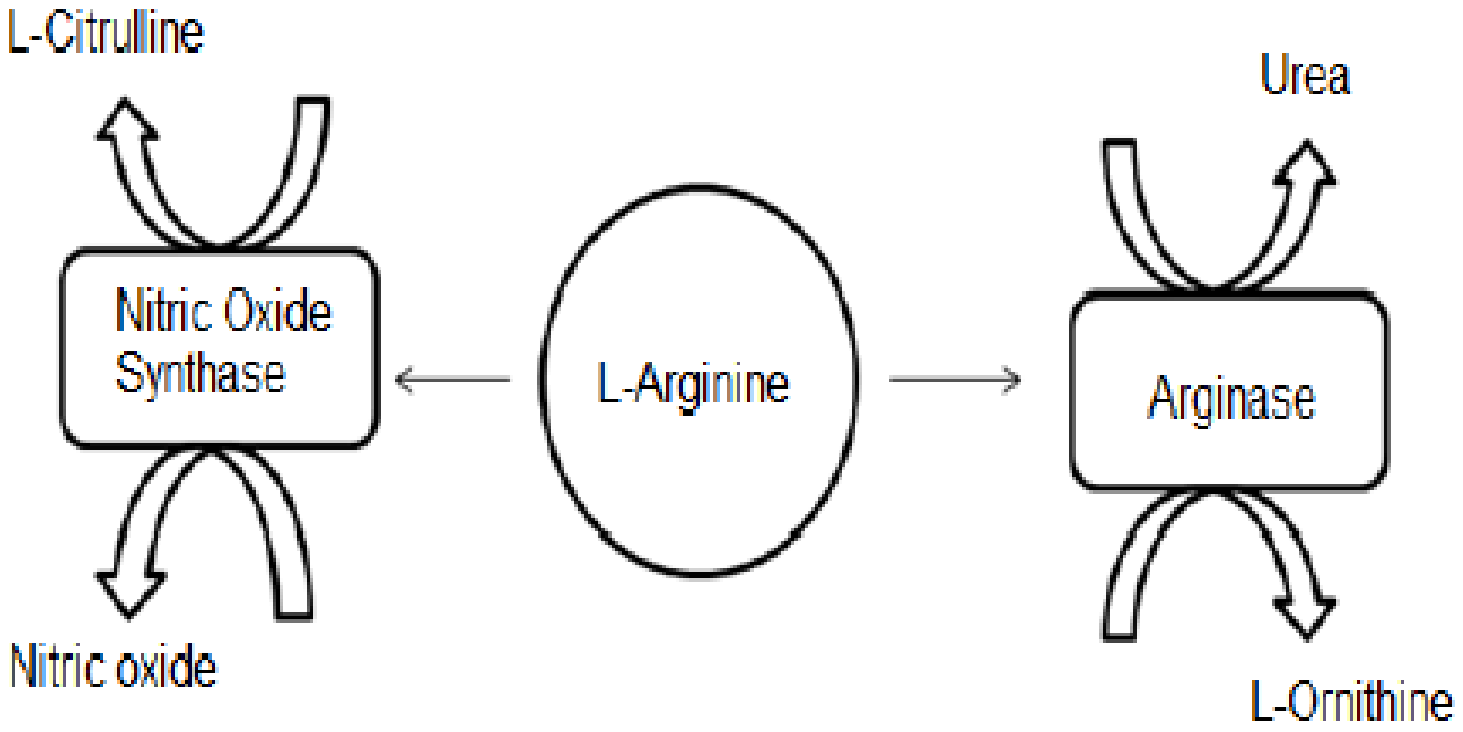
Arginase

Enzymatic Changes

Arginase

- Arginase (EC 3.5.3.1, arginine amidinase, canavanase, L-arginase, arginine transamidinase) is a manganese-containing enzyme
- The reaction catalyzed by this enzyme is: arginine + $H_2O \rightarrow$ ornithine + urea
- It is the final enzyme of the urea cycle

Enzymatic Changes

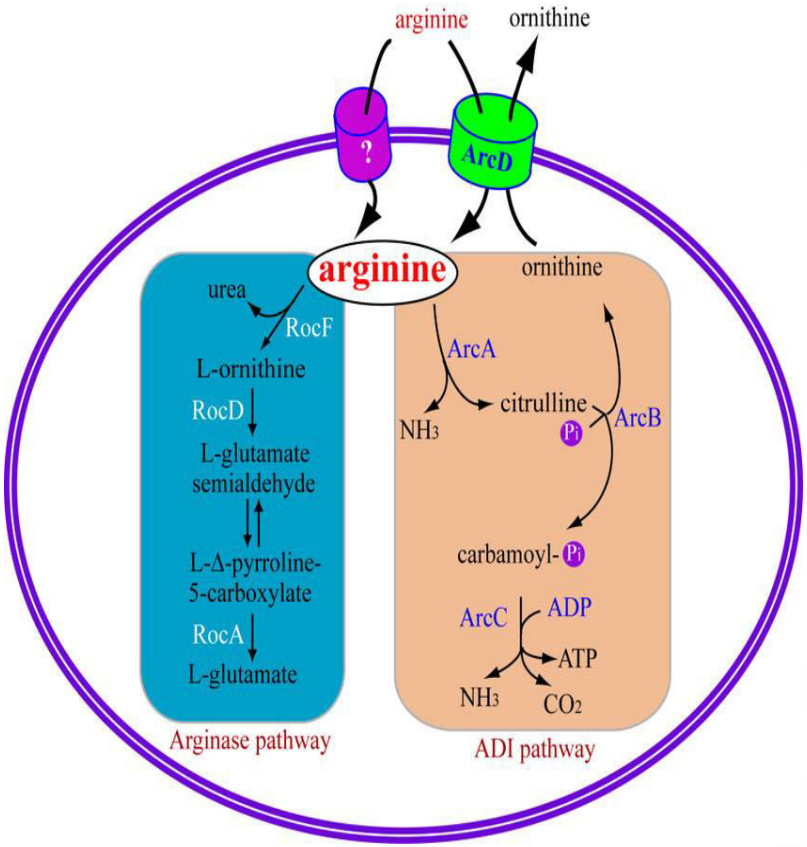


Enzymatic Changes



- Rabbit papilloma virus induces high levels of arginine in squamous epithelium
- Squamous epithelium normally contains arginine

Enzymatic Changes



- The induced enzyme differs immunologically and physiochemically from normal enzyme
- Released from normal cellular control mechanism

Enzymatic Changes

Speculation: net effect is the depletion of cellular arginine which results in reducing the amount of arginine rich histones synthesized

Changes in Structural Proteins

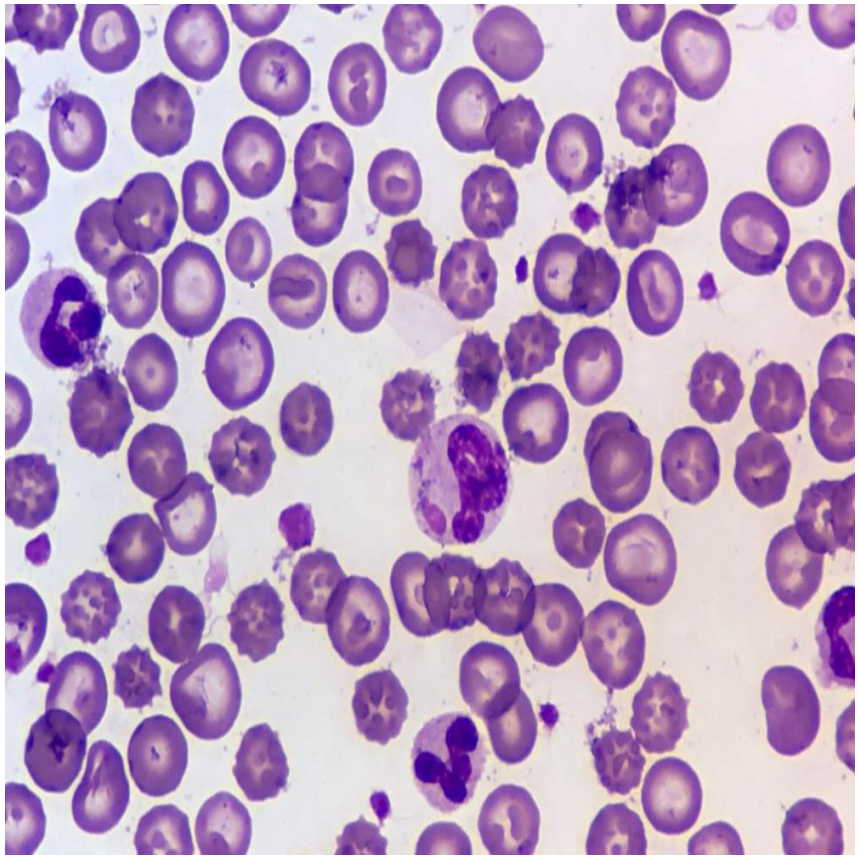
Structural Proteins

INCLUSIONS

- Examination of virus infected cells by light or electron microscope shows viral multiplication and maturation referred as ‘inclusions’ or ‘factories’

Structural Proteins

EXAMPLES

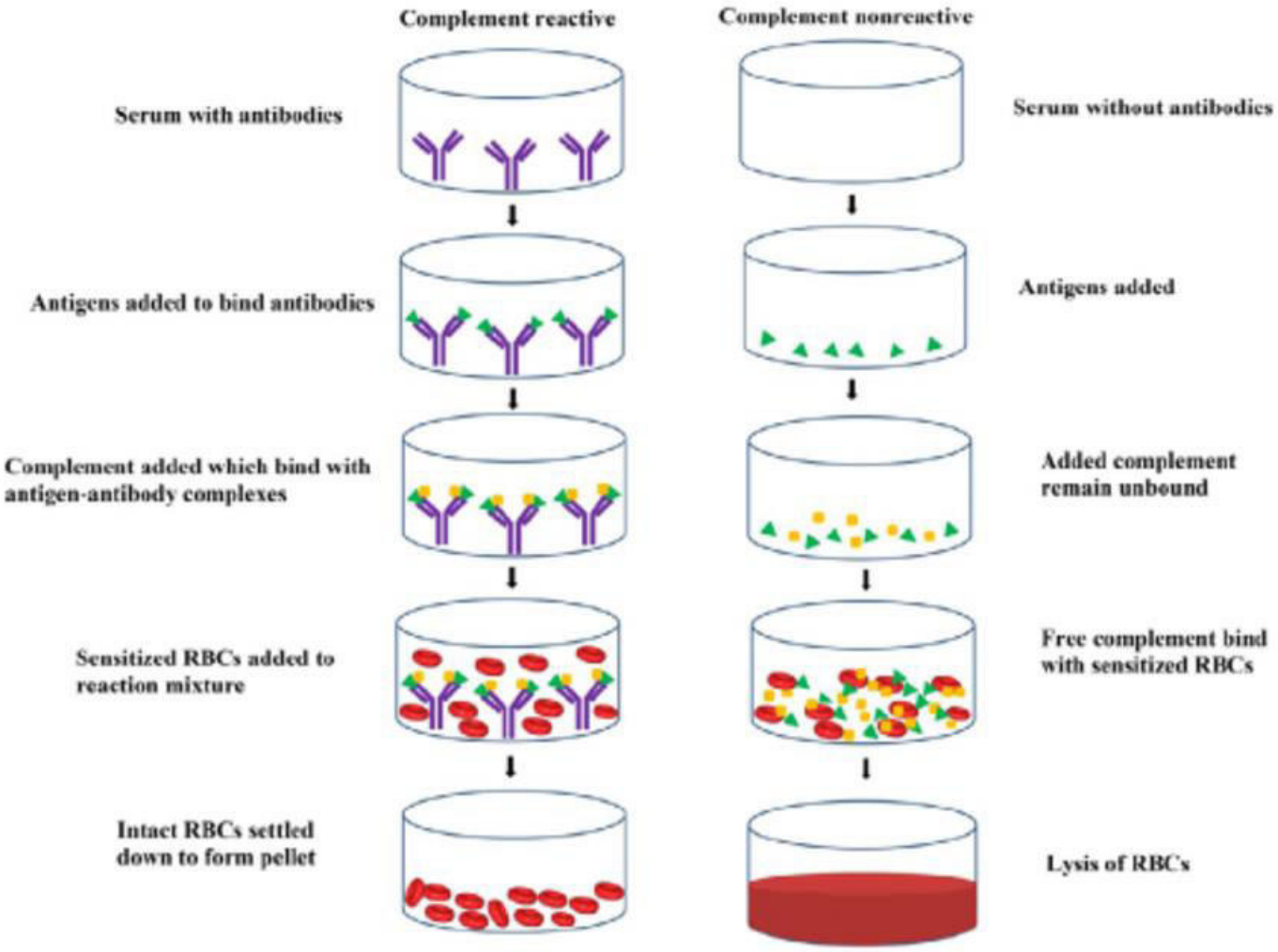


- Inclusion may be large, in case of poxvirus infection
- Or small incase of picornavirus infection
- Or maybe limited by a membrane

Structural Proteins

- New proteins are usually associated with tumors
- Property of oncogenic viruses

Structural Proteins



Structural Proteins

DETECTION

- Complement fixation
- Fluorescent antibody detection
- Graft detection

CONTROL

- Most probably under control of viral genome
- Possibility of cellular response products to virus

Virology

Factors Influencing Viral Multiplication

Factors Affecting Viral Replication

FACTORS

1. Viral concentration
2. Virus-cell systems
3. Elevated temperatures
4. Incomplete viral genomes

Factors Affecting Viral Replication

Viral Concentration

- One infectious unit equals to 10^3 to 10^5 noninfectious particles
- Low ratio associated with slow uncoating of viral particles
- Exception of vaccinia virus

Factors Affecting Viral Replication

Virus-cell Systems

- Blocks in viral development after viral genome starts to function
- E.g herpes simplex virus in dog kidney cells after development of cytopathic effects, inhibition of cell division, synthesis of viral

Factors Affecting Viral Replication

VIRUS-CELL SYSTEMS



- Failure of virus to replicate in dog kidney cells
- Possible that viral DNA acted upon cellular nuclease or that sufficient DNA polymerase not present

Factors Affecting Viral Replication

Elevated Temperatures

- At 40°C, poliovirus inhibits cellular RNA synthesis
- But viral RNA fails o replicate and no viral progeny produced
- Polymerization fails at this temp.

Factors Affecting Viral Replication

Incomplete Genomes

- Incomplete viral genomes
- Like influenza virus
- Produce nucleoproteins, hemagglutinin
- But no mature virus

Virology

Viral Interference

Viral Interference

Principle

- Demonstrated *invitro* and *invivo*
- By infecting cell with a virus followed by a second virus
- Demonstrated when the socnd virus fails to replicate or is inhibited

Viral Interference

Groups

1. Interference mediated by virus induced cellular protein called interferon
2. Interference that is non interforn mediated

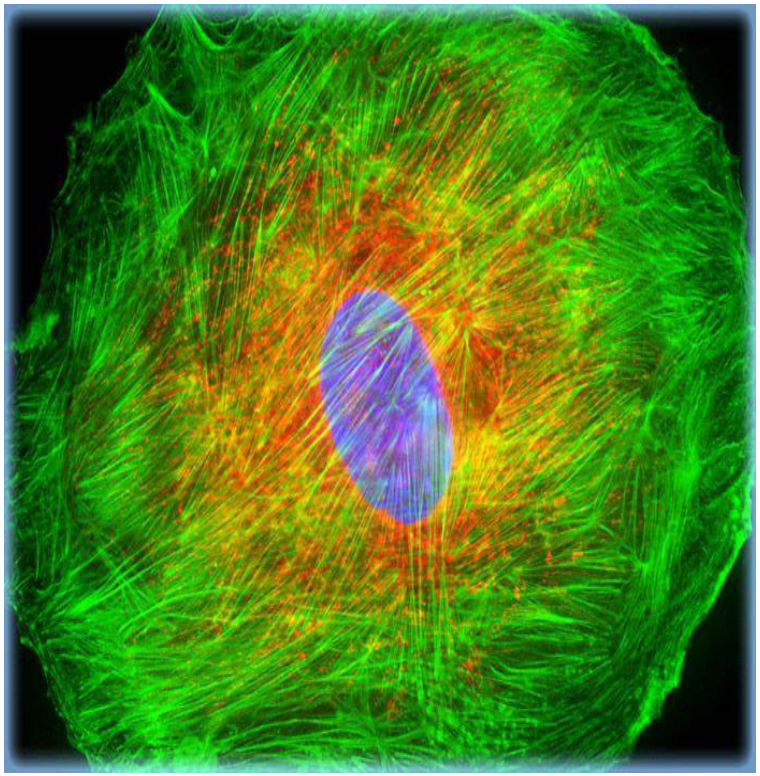
Viral Interference

Interferon

- Interferons are 20,000 molecular weight proteins
- Produced by cells in response to infection
- Interferon producing information resides in host cell genome

Viral Interference

INTERFERON



- Not virus-specific
- E.g chicken fibroblast cell interferon is inactive in mouse cells but inhibitory for viruses grown in chicken fibroblast cell cultures

Viral Interference

ROLE

- Interferons block translation of viral genome in early stages
- Basic mechanism however remains speculative

Viral Interference

- Interferon does not bind to naked viral RNA to reduce infectivity
- Suggests inhibitory action takes place at cellular site
- Role of interferons in recovering is being studied

Virology

Collection of Samples for Viruses

Viral Sample Collection

CONSIDERATIONS



1. What to collect?

✓ Type of specimen

- e.g septicemic infection → blood
- Urinary tract infection → urine
- Respiratory infection → nasal swabs

✓ Quantity of sample

2. When to collect?

✓ Before treatment with antibiotics

✓ Site may vary with time



Viral Sample Collection

TRANSPORT MEDIA

- Allows organisms (pathogens and contaminants) to survive

✓UTM (Universal Transport Medium)

✓PBS/Glycerol transport Medium

(NaCL, KCl, Na2HPO4 in distilled water)



Viral Sample Collection

Specimen	Transport media	Storage condition		Purpose/ Lab investigation
		Transport	Pending test	
Throat swab	VTM	2-8 ^{0}C	-20 ^{0}C	Isolation
NPA/ swab	VTM	2-8 ^{0}C	-20 ^{0}C	Isolation
CSF	No	2-8 ^{0}C	-20 ^{0}C	Isolation, serology
Stool	No	2-8 ^{0}C	-20 ^{0}C	Isolation
Urine	No	2-8 ^{0}C	-20 ^{0}C	Isolation
Serum/ Clotted blood	No	2-8 ^{0}C	-20 ^{0}C 2-8 ^{0}C	Isolation, serology
Whole blood	No	2-8 ^{0}C	2-8 ^{0}C	Isolation, serology

Virology

Preservation of Samples of Virus

Preservation of Viral Samples

Types of Samples

- Tissue samples
- Extracted DNA
- Extracted RNA
- cDNA
- PCR products

Preservation of Viral Samples

Tissue Samples

- can be kept at -80°C for years
- Temperature should be maintained
- Thawing repeatedly may contaminate the sample

Preservation of Viral Samples

Extracted DNA

- can be kept at -80°C
- Aliquots of sample
- Properly labeled and sealed

Preservation of Viral Samples

Extracted RNA

- Very sensitive due to presence of RNA everywhere
- Stores at -80°C for a long period of time
- Can also be kept at -20°C for short time periods
- Refreezing not recommended

Preservation of Viral Samples

cDNA

- Comparatively more stable than RNA
- Can be kept at -20°C for several weeks
- Can be kept at -80°C for many years

Preservation of Viral Samples

Specimen	Storage condition
	Pending test
Throat swab	-20 ^{0}C
NPA/ swab	-20 ^{0}C
CSF	-20 ^{0}C
Stool	-20 ^{0}C
Urine	-20 ^{0}C
Serum/	-20 ^{0}C
Clotted blood	2-8 ^{0}C
Whole blood	2-8 ^{0}C

Virology

Transport of Samples for Viruses

Viral Sample Transport

Purpose Of Sop

- Safety from bio-hazardous material
- Results depend on sample collection and handling



Viral Sample Transport

Biosafety consideration

- Zoonotic or transmissible
- Latex gloves, Lab coats, Masks
- New syringes and containers
- Sealed container
- Wash hands after removing gloves
- In event of injury, wash area with soap and water
- Always have a first aid safety kit



Viral Sample Transport

Handling and transport of blood sample

- ✓Collect in EDTA or Heparin containing containers
- ✓Store at 4 ^{o}C
- ✓Transport upright with cushion (to prevent hemolysis)
- ✓Never freeze blood



Viral Sample Transport

Nasopharyngeal swabs transport

- ✓All respiratory specimens should be transported in proper medium
- ✓Transport as quickly as possible to the laboratory to reduce overgrowth by oral flora
- ✓For transit periods up to 24 hours
  - ambient temperature for bacteria
  - 4-8°C for viruses

Viral Sample Transport

POSTMORTEM SAMPLES

- Collection
  - ✓Biopsy relevant tissues
    - in formalin for histopathology
    - in transport medium for microbiological testing
    - in sterile saline for isolation of viral pathogens
- Handling and Transport
    - in transport media within 24h at ambient temperature
    - in sterile saline at 4-8°C within 48h

Virology

FILTERATION METHOD

Berkefeld, Mandler and Pasteurchamberland filter

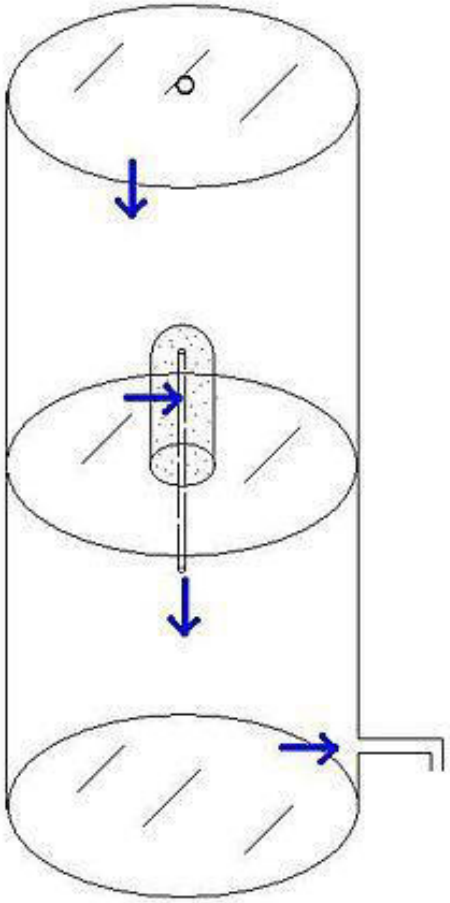
Filtration Method

The Berke Filter

- Manufactured of diatomaceous Earth
- Graded into three porosities
- The V (viel) grade is coarse
- Used for preliminary filtration to remove bacteria or large particles from a solution
- The N (normal) grade is most useful since it retains most bacteria, while all viruses will pass through
- The W (Weing) grade has very small pores that will retain all bacteria and large viruses

Filtration Method

The Berke Filter



Schematic diagram of
a Berkefeld filter

Filtration Method

The Mandler Filter

- Made of diatomaceous Earth
- Similar to berkefeld filters
- Three grades
  - Preliminary
  - Regular
  - Fine

Filtration Method

The Mandler Filter



Filtration Method

Pasteur chamberland Filter

- Made of porcelain
- Molded in candle form
- Graded in porosities
- L1 to L13
- L13 is the finest grade

Filtration Method

Pasteur Chamberland Filter



Virology

FILTERATION METHOD

Selas, Seitz and Boerner filter

Filtration Method

The Selas Filter

- Made up of porcelain
- Seven grades of porosity
- Graded as XF, XFF, No. 10, and No. 01 (coarse grade)
- Graded as No. 015, No. 02, No. 03 which retain bacteria

Filtration Method

The Selas Filter



Filtration Method

Seitz Filter

- Asbestos pads
- Manufactured in two grades
- K and EK (coarse)
- Pads in metal holder
- Discarded after one use

Filtration Method

The Seitz Filter



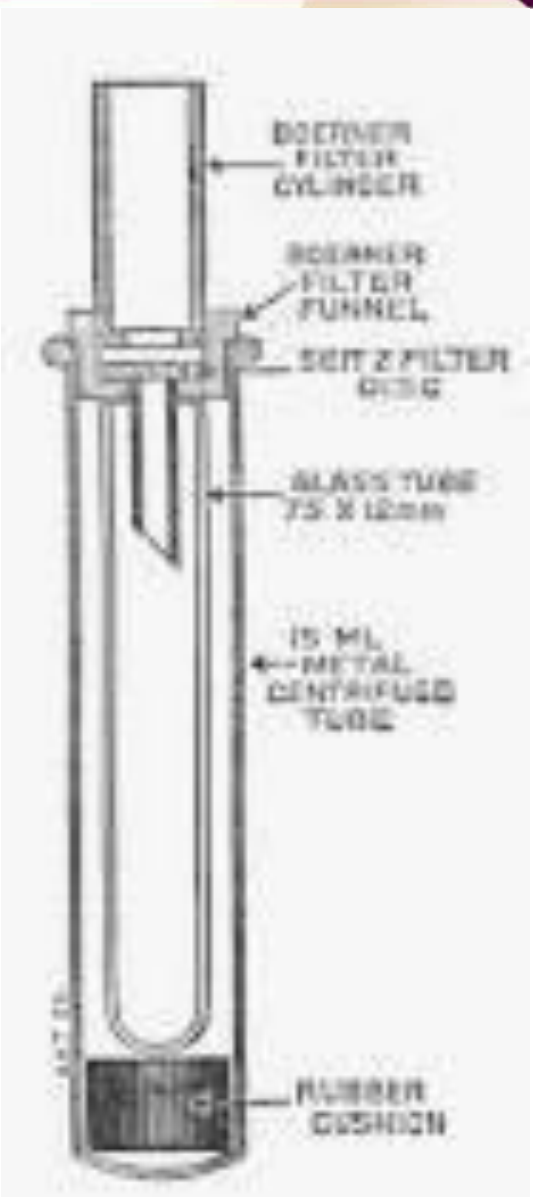
Filtration Method

Boerner Filter

- Same as Seitz
- Except they are small and fitted in small metal receptacle
- Adapts into small test tube
- Centrifuged to force liquid through them

Filtration Method

Boerner Filter



Virology

FILTERATION METHOD

Collodian filter and cellulose filter

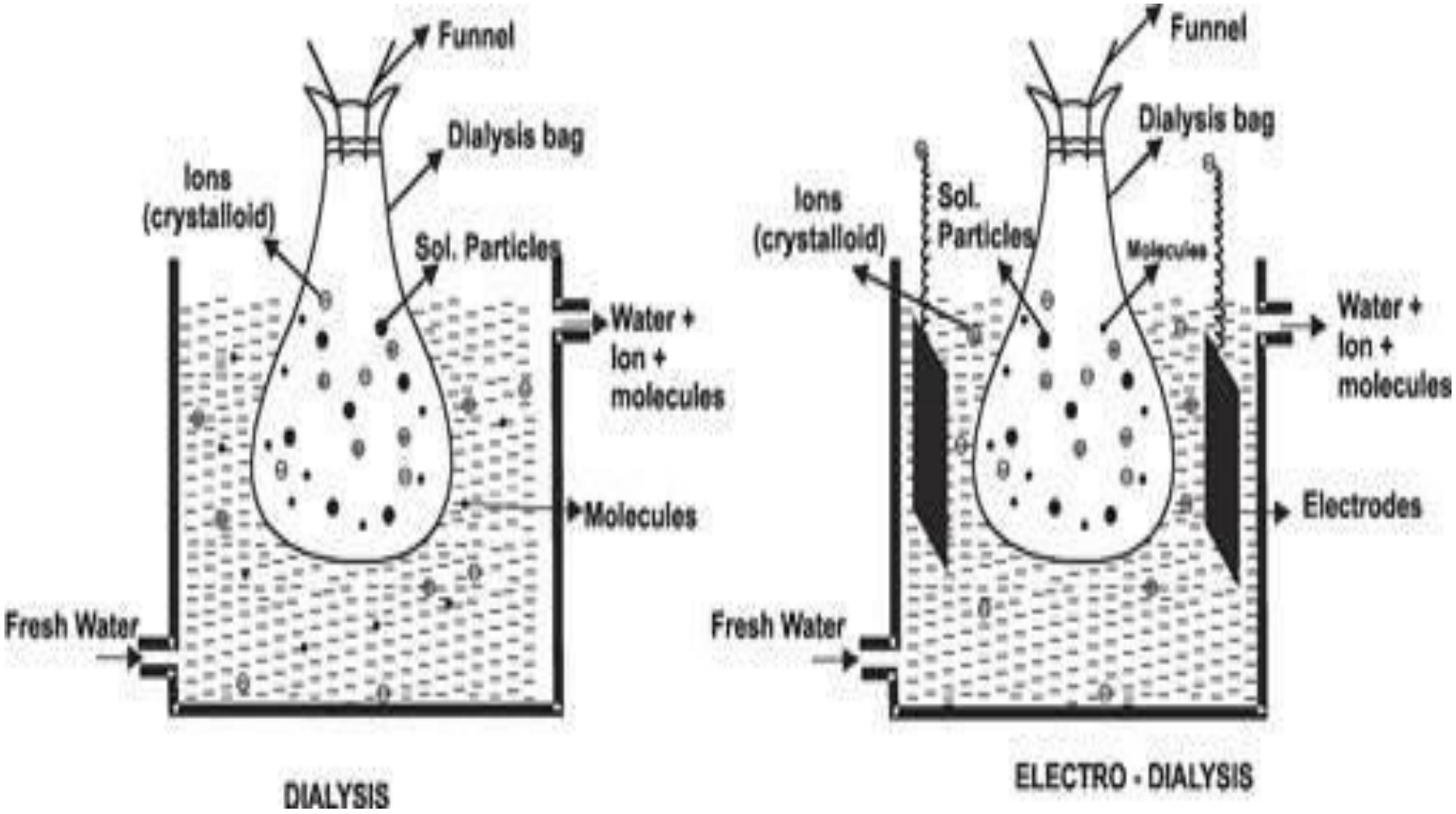
Filtration Method

Collodion Filter

- Made up of nitro-cellulose
- Pore diameter varies
- Variation by addition of various amounts of alcohol, ether, acetone and acetic acid
- Pore diameter: 3um- 10um
- Most useful in determining viral size

Filtration Method

Colloidan Filter



Filtration Method

Cellulose Filter

- Pure cellulose or cellulose derivatives
- Coarse with pore diameter 3um to 0.75 um
- Medium with pore diameter: 0.75um to 0.7um
- Dense with 0.2um
- Very dense below 0.2um

Filtration Method

The Cellulose Filter



Virology

Filteration Technique

Filteration Technique

Filtration

A technique used either to remove solid impurities from an organic solution or to isolate an organic solid

The two types of filtration commonly used

Gravity filtration

Vacuum or suction filtration

Filteration Technique

Gravity Filtration

- Gravity filtration is the method of choice to remove solid impurities from an organic liquid



Filteration Technique

Vacuum Filtration

Vacuum filtration is used primarily to collect a desired solid, for instance, the collection of crystals in a recrystallization procedure



Filteration Technique

- https://www.youtube.com/watch?v=r4GYLuqJV7Y

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Determination of Viral Size by Filteration

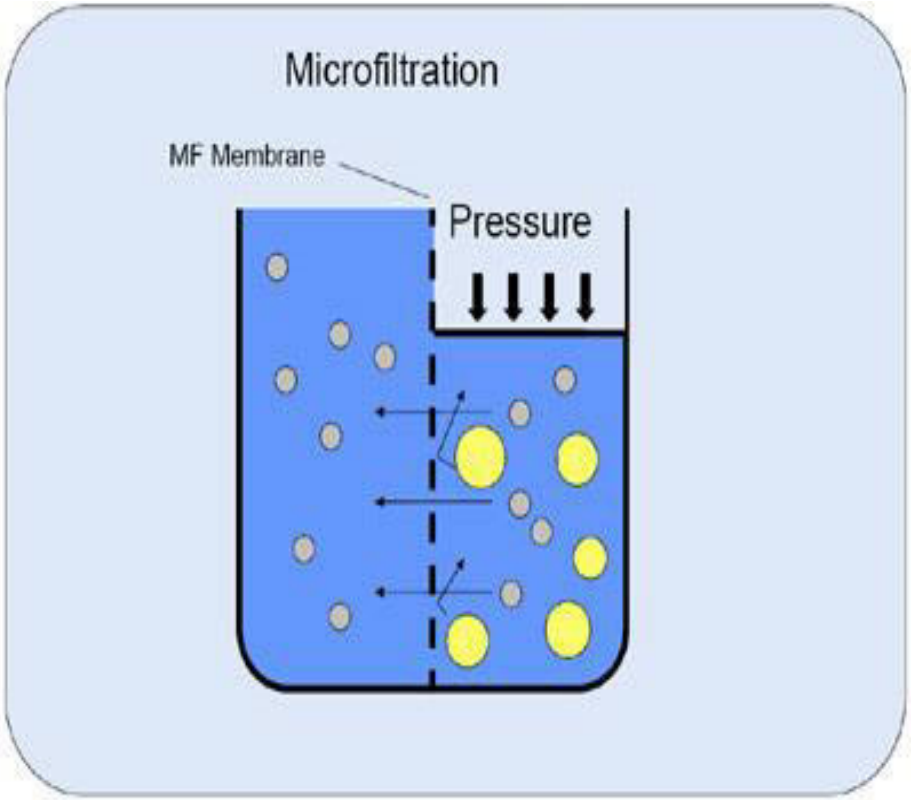
Viral Size

Membrane Technology

- Principle is physical separation
- The extent of removal depends on the size of the particles
- Microfilteration and ultracentrifugation

Viral Size

MICROFILTERATION



- Pore size 0.1 to 1um
- It removes all bacteria
- Only a part of viral contamination is removed

Viral Size

APPLICATIONS

Cold sterilization of beverages and pharmaceuticals

Clearing of fruit juice, wines and beer

Separation of bacteria from water

Separation of oil/ water emulsions

Reverse Osmosis

Solid-liquid separation for pharmacies or food industries

Viral Size

ULTRACENTRIFUGATION

Pore size: 0.001- 0.1 um from liquids

Examples

The dairy industry (milk, cheese)

The food industry (proteins)

The metal industry (oil/ water emulsions separation, paint treatment)

The textile industry

Virology

Centrifugation Techniques

Centrifugation Techniques

Centrifugation

A technique which involves the application of centrifugal force to separate particles from a solution according to their size, shape, density, viscosity of the medium and rotor speed

Centrifugation Techniques

Principle

More-dense components of the mixture migrate away from the axis of the centrifuge (move to the outside), while less-dense components of the mixture migrate towards the axis,

i. e., move to the center.



Centrifugation Techniques

Microcentrifuges

High-speed centrifuges

Ultrcentrifugations

Density

Gradient Centrifugation

Centrifugation Techniques

https://www.youtube.com/watch?v=KEXWd3_fM94

Virology

Determination of Essential Lipids

Determination of Essential Lipids

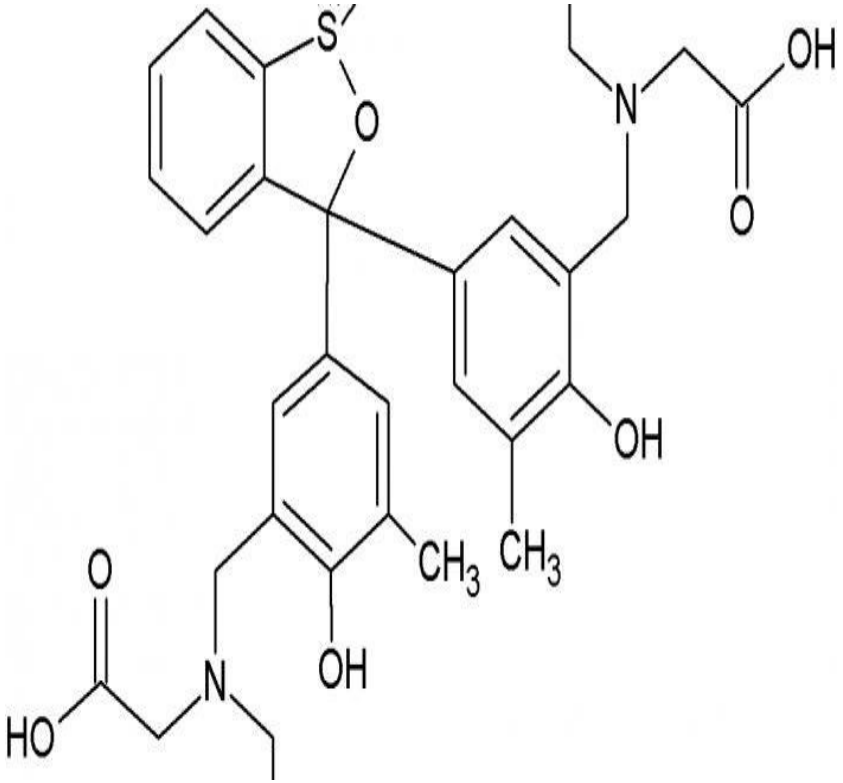
Why Essential Lipids

- Used for identification and classification of animal viruses
- On the basis of presence or absence of essential lipids

Determination of Essential Lipids

Principle

Presence of essential lipids determined by mixing virus preparations with either ether or chloroform



Determination of Essential Lipids

Procedure

- Sufficient ethyl ether is added to virus preparation to make a 20% solution.
- Mixture is incubated at 18-24hrs at 4 ^{o}C

Determination of Essential Lipids

- The resultant is poured into a petri dish.
- Ether is allowed to evaporate.



Determination of Essential Lipids



- To expose a virus to chloroform, 0.05ml chloroform is added to 1ml of virus preparation.
- Mix for ten minutes at 4°C.
- Chloroform is removed by centrifugation.

Determination of Essential Lipids

- Essential lipids can be determined by various methods used in labs.

Virology

Determination of Nucleic Acid Types

Determination of Nucleic Acid

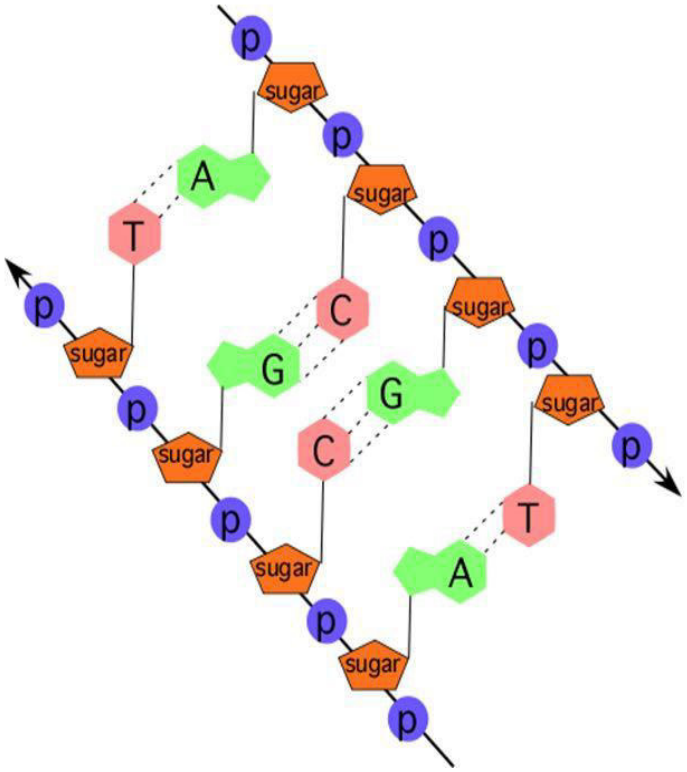
Nucleic acid types

- Double stranded DNA.
- Positive Sense single stranded RNA
- Negative sense single stranded RNA
- Double stranded RNA

Determination of Nucleic Acid

Principle

Use of DNA inhibitors determines type of Nucleic Acid a virus possesses



Determination of Nucleic Acid

DNA inhibitors

- e.g 5-floro-2’

  -deoxyuridine (FUDR)

- 5-bromo-2’

  -deoxyuridine (BUDR)

- 5-iodo-2’

  -deoxyuridine (IUDR)

Determination of Nucleic Acid

Action of DNA inhibitors

- Specific inhibition of viral DNA synthesis when added to cell culture medium

Determination of Nucleic Acid

Interpretation

- If growth of virus is inhibited, it is presumed to contain DNA
- If not, it possesses RNA

END

Virology

Concentration and Purification of Viruses

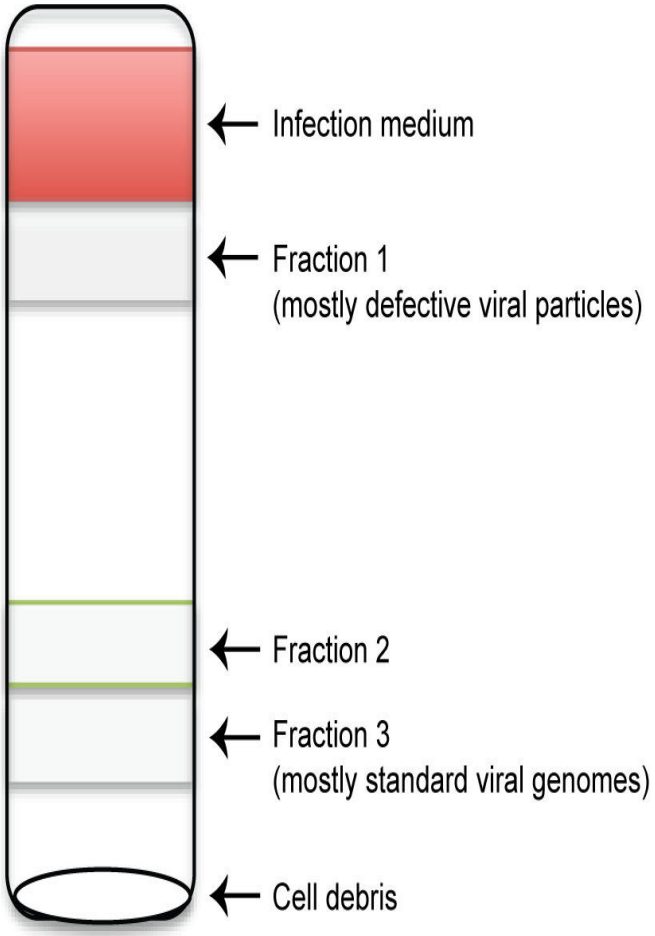
Concentration and Purification

Purpose

- To study many properties of viruses, a purified preparation must be used

Concentration and Purification

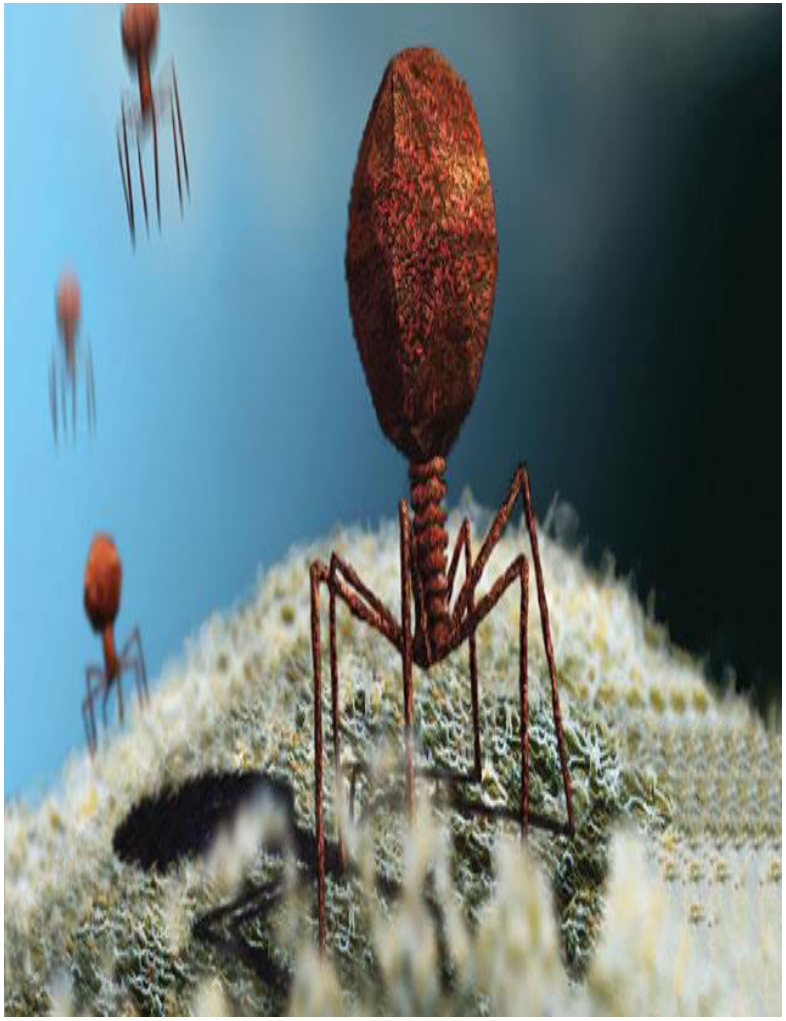
Challenges



- The separation of virus elementary bodies from the components of the cell in which the virus has been grown
- Centrifugation of crude virus containing material at various speed will help overcome this problem

Concentration and Purification

Removal of unwanted substances



- The speed at which majority of elementary bodies are sedimented can be determined
- Resuspension of virus in fresh diluents and resedimentation through several cycles

Concentration and Purification

Density gradient

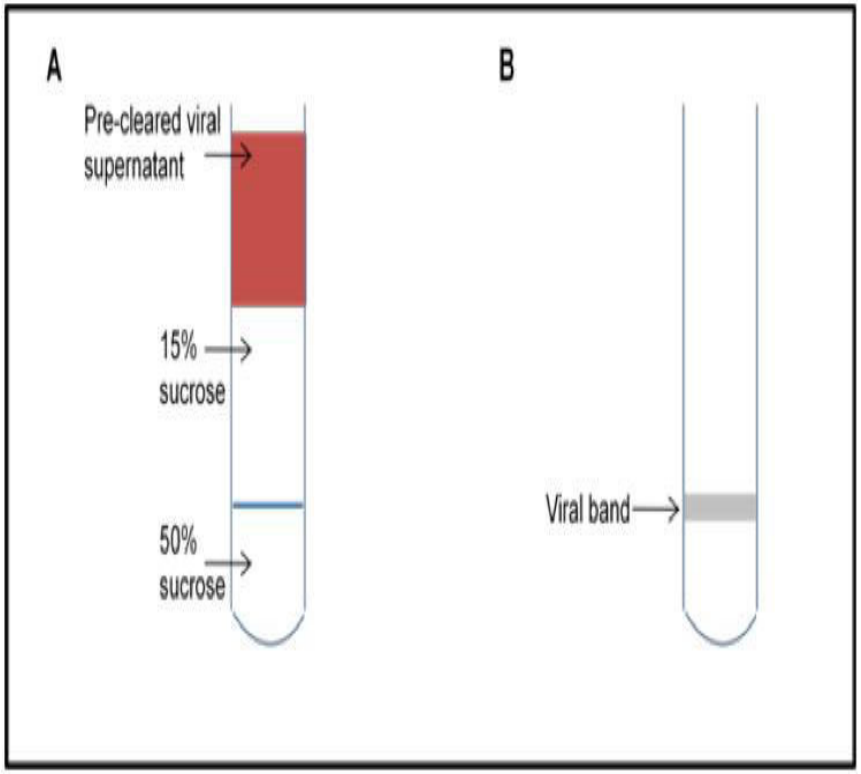
Composed of sucrose or glycerol enables separation of virus from tissue as well as different components

Made by layering fluids of different density in a centrifuge tube

Last layer has greatest density

Concentration and Purification

Principle



- Depending on the relative density, virus in tissue placed on top of the tube will be stratified
- Each settles in the zone of fluid of equal density

Concentration and Purification

Isolation4

Since plastic tubes are used for this technique, it is possible to use a syringe and a needle to draw out the desired layer

Concentration and Purification

Alternate method

Hemagglutination-elution can be applied to virus that attract erythrocytes

Concentration and Purification

Purification of viruses

1. Precipitation with cold ethanol
2. Extraction of tissues/fluids with florocarbons (Genetrons & Freons)

END

Virology

VIRAL DIAGNOSTIC TESTS

Enzyme Linked ImmunoSorbent Assay (ELISA)

Viral Diagnostic Procedures

1. Direct Examination of Specimen

- Electron Microscopy morphology / immune electron microscopy
- Light microscopy histological appearance - e.g. inclusion bodies
- Antigen detection immunofluorescence, ELISA etc.
- Molecular techniques for the direct detection of viral genomes

Viral Diagnostic Procedures

2. Indirect Examination

- Cell Culture - cytopathic effect, haemadsorption, confirmation by neutralization, interference, immunofluorescence etc.
- Eggs pocks on CAM - haemagglutination, inclusion bodies
- Animals disease or death confirmation by neutralization

3. Serology

- Detection of rising titres of antibody between acute and convalescent stages of infection, or the detection of IgM in primary infection.

Viral Diagnostic Procedures

Other Techniques

Classical Techniques	Newer Techniques
1. Complement fixation tests (CFT)	1. Radioimmunoassay (RIA)
2. Haemagglutination inhibition tests	2. Enzyme linked immunosorbent assay (EIA)
3. Immunofluorescence techniques (IF)	3. Particle agglutination
4. Neutralization tests	4. Western Blot (WB)
5. Single Radial Haemolysis	5. Recombinant immunoblot assay (RIBA), line immunoassay (Liatek) etc.

Viral Diagnostic Procedures

Enzyme Linked ImmunoSorbent Assay (ELISA)

ELISA (enzyme-linked immunosorbent assay) is a plate-based assay technique designed for detecting and quantifying substances such as peptides, proteins, antibodies and hormones.

- In an ELISA, an antigen must be immobilized on a solid surface and then complexed with an antibody that is linked to an enzyme.
- Detection is accomplished by assessing the conjugated enzyme activity via incubation with a substrate to produce a measureable product.
- The most crucial element of the detection strategy is a highly specific antibody-antigen interaction.

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Viral Diagnostic Procedures

The key step, immobilization of the antigen of interest, can be accomplished by direct adsorption to the assay plate or indirectly via a capture antibody that has been attached to the plate.

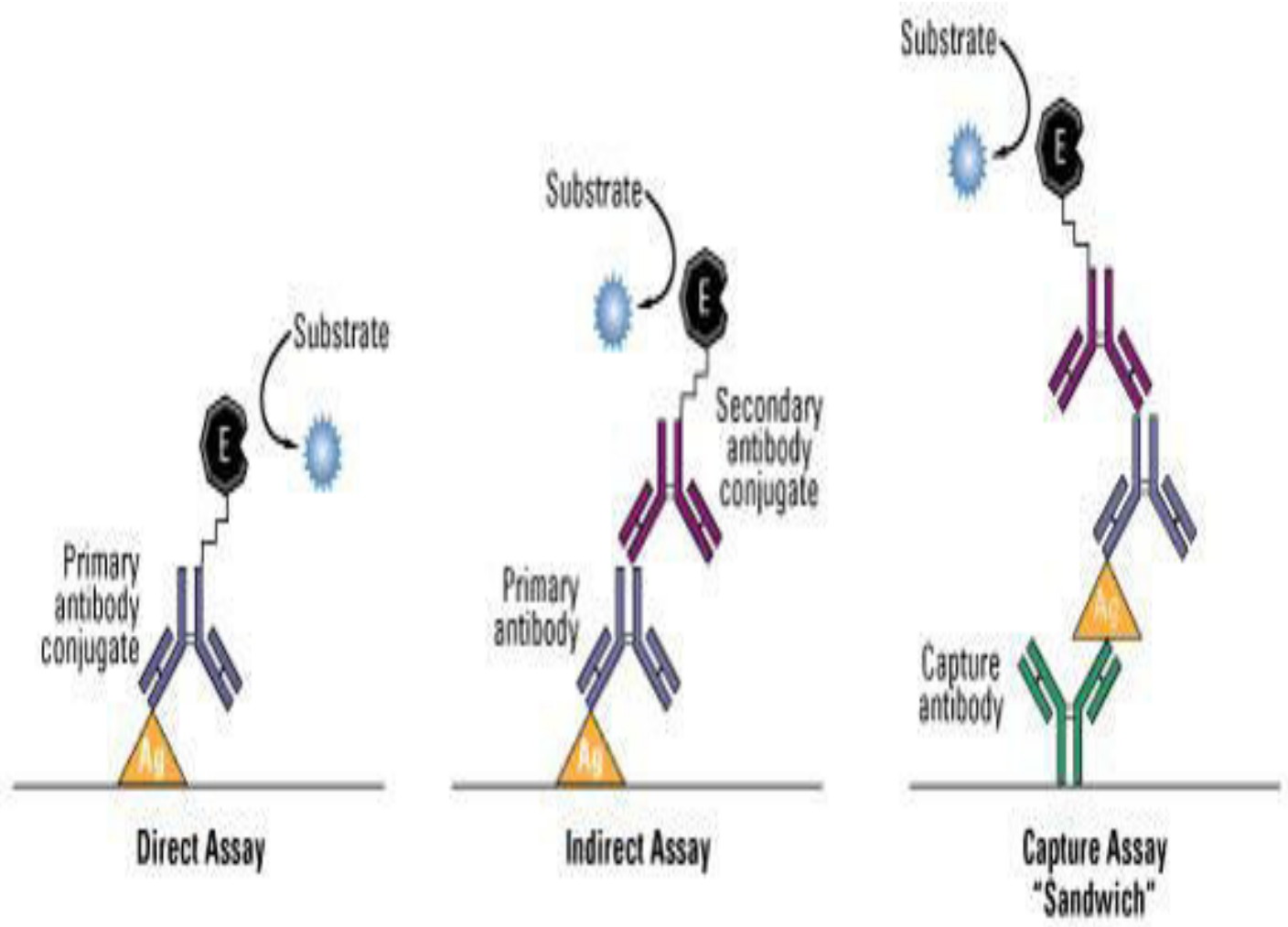
The antigen is then detected either directly (labeled primary antibody) or indirectly (labeled secondary antibody).

The most powerful ELISA assay format is the sandwich assay.

Viral Diagnostic Procedures

This type of capture assay is called a “sandwich” assay because the analyte to be measured is bound between two primary antibodies – the capture antibody and the detection antibody. The sandwich format is used because it is sensitive

Viral Diagnostic Procedures



Viral Diagnostic Procedures

ELISA kits contain pre-coated antibody-plates, detection antibodies, buffers, diluents, standards, and substrates

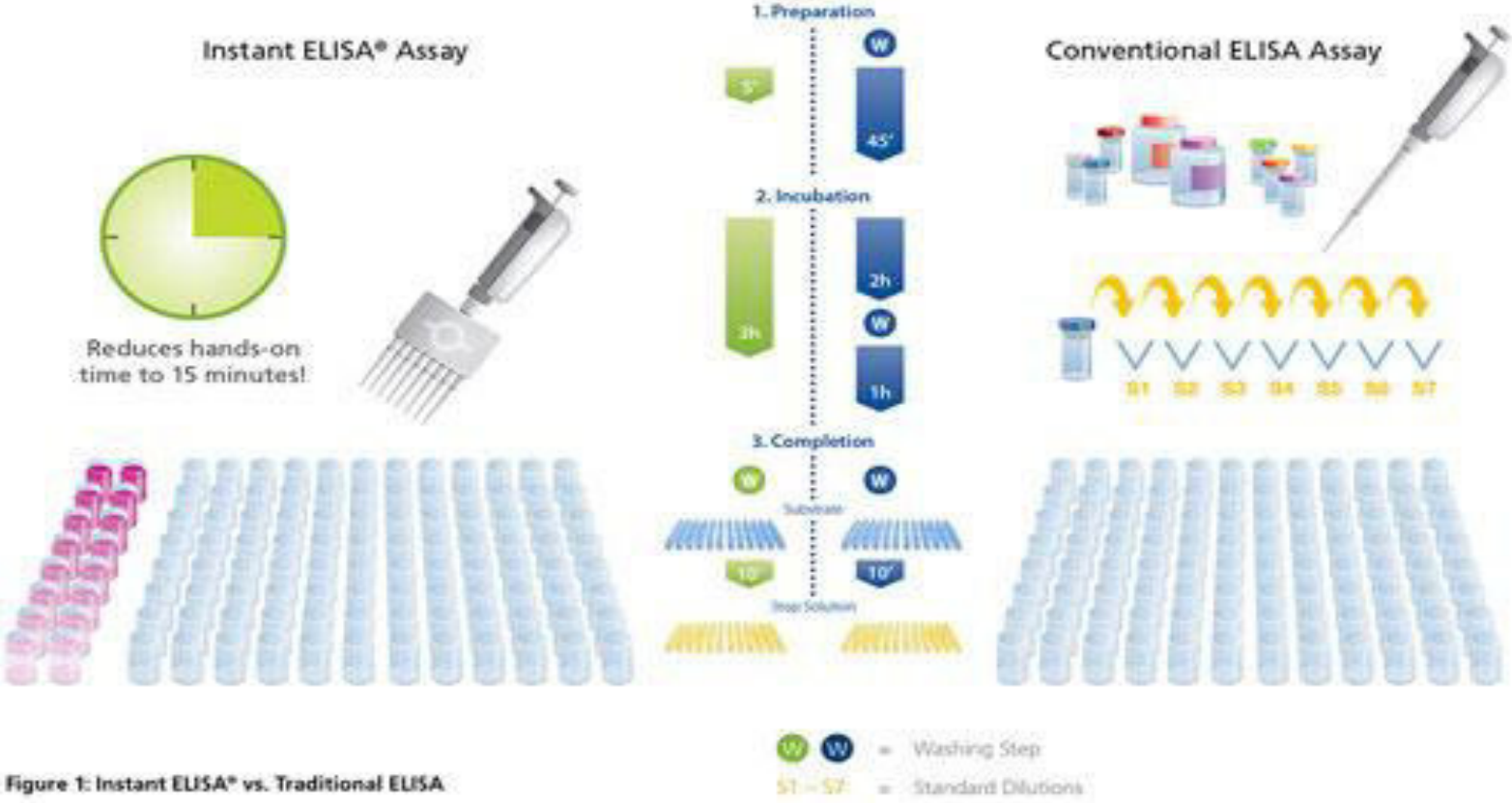


Figure 1: Instant ELISA® vs. Traditional ELISA

VIRAL DIAGNOSTIC PROCEDURES

Histological Examination of Virus infected cells

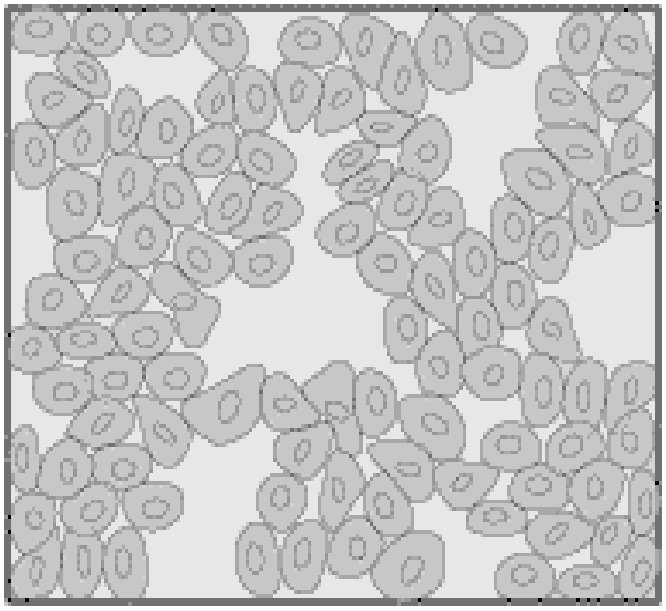
Viral Diagnostic Methods

CULTURE

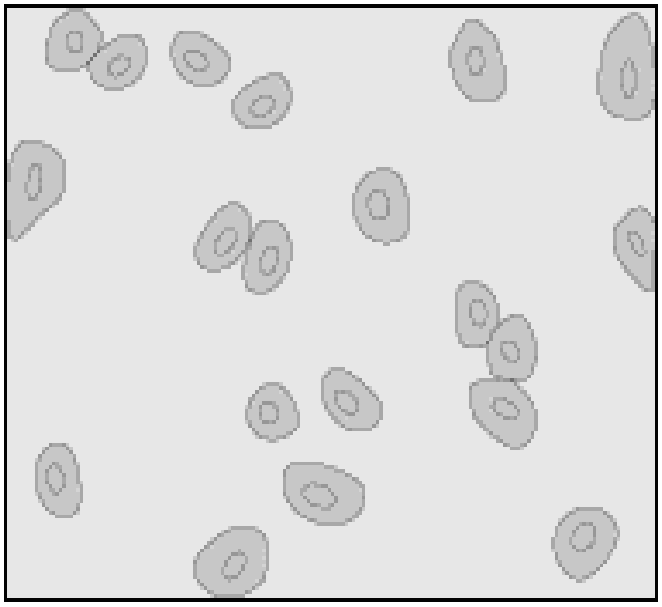
Grown in :

1. Primary cells
2. Semi-continuous cells
3. Continuous Cells

Viral Diagnostic Methods



Before Infection



After Infection

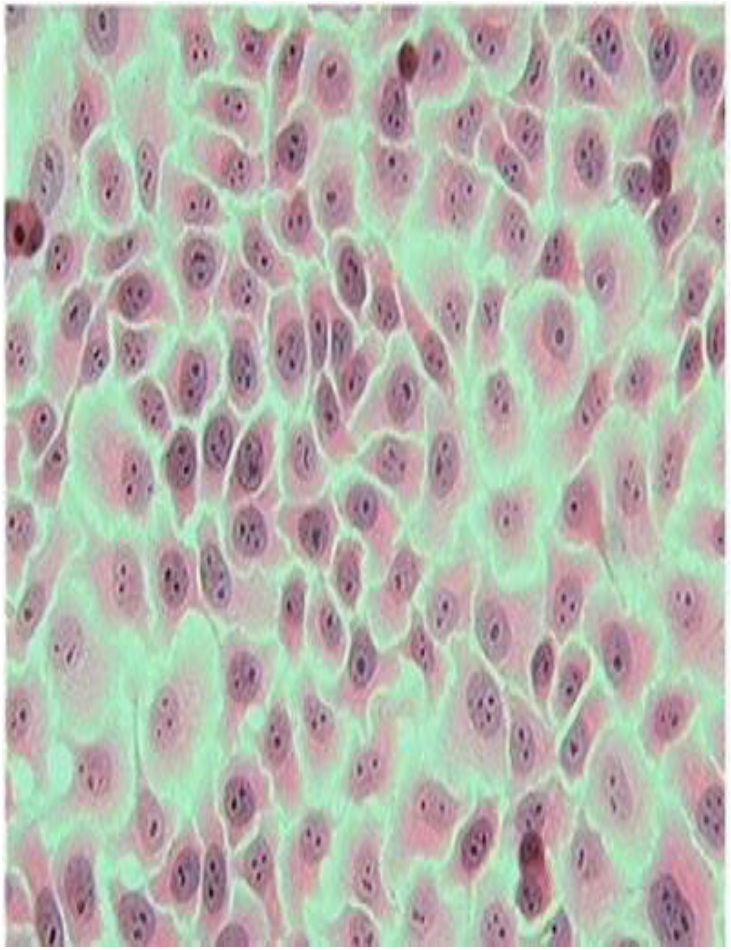
Viral Diagnostic Methods

Identification of growing viruses

- Cytopathic effects (CPE)
- Haemadsorption

Viral Diagnostic Methods

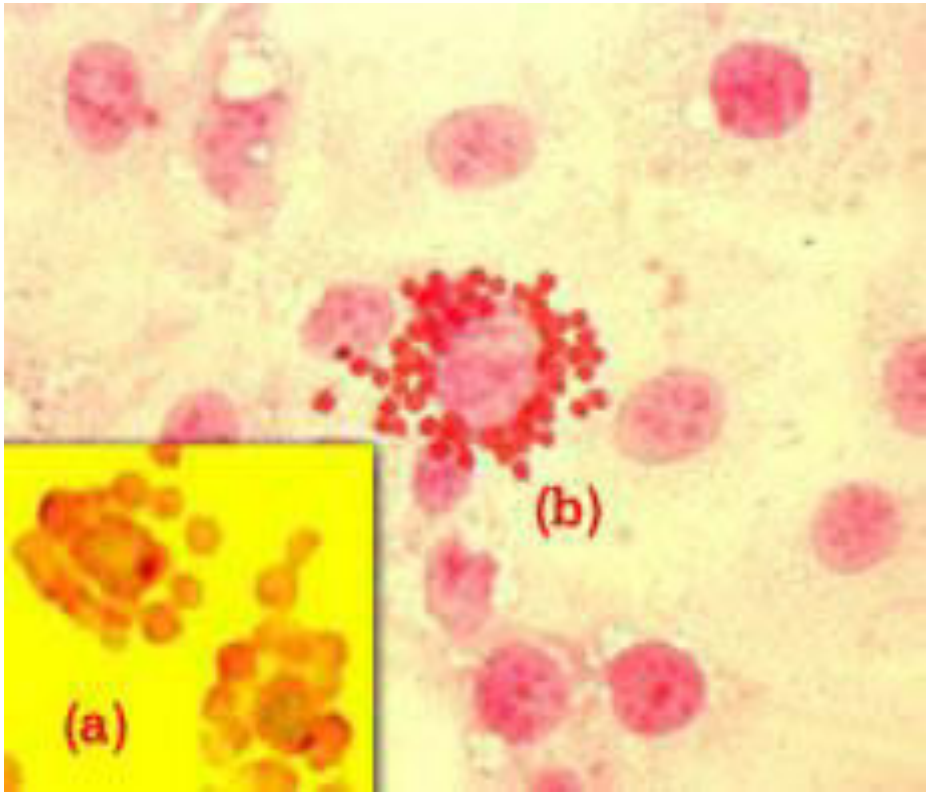
Cytopathic Effects



- Specific or non-specific signs
- Herpes Simplex Virus produces specific CPE
- Enterovirus don’t produce CPE

Viral Diagnostic Methods

Haemadsorption



- Cells acquire ability to stick to mammalian RBCs
- Mainly used for influenza and parainfluenza

Viral Diagnostic Methods

Confirmation of virus

- Neutralization
- Haemadsorption-inhibition
- Immunofluoresence
- Molecular tests
  - PCR
  - ELISA

END

Virology

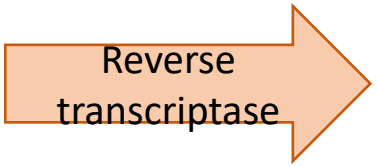
VIRAL DIAGNOSTIC PROCEDURES

Polymerase Chain Reaction

Viral Diagnostic Procedures

PCR

- Polymerase chain reaction is a laboratory technique used to amplify segments of DNA or RNA

- RNA 

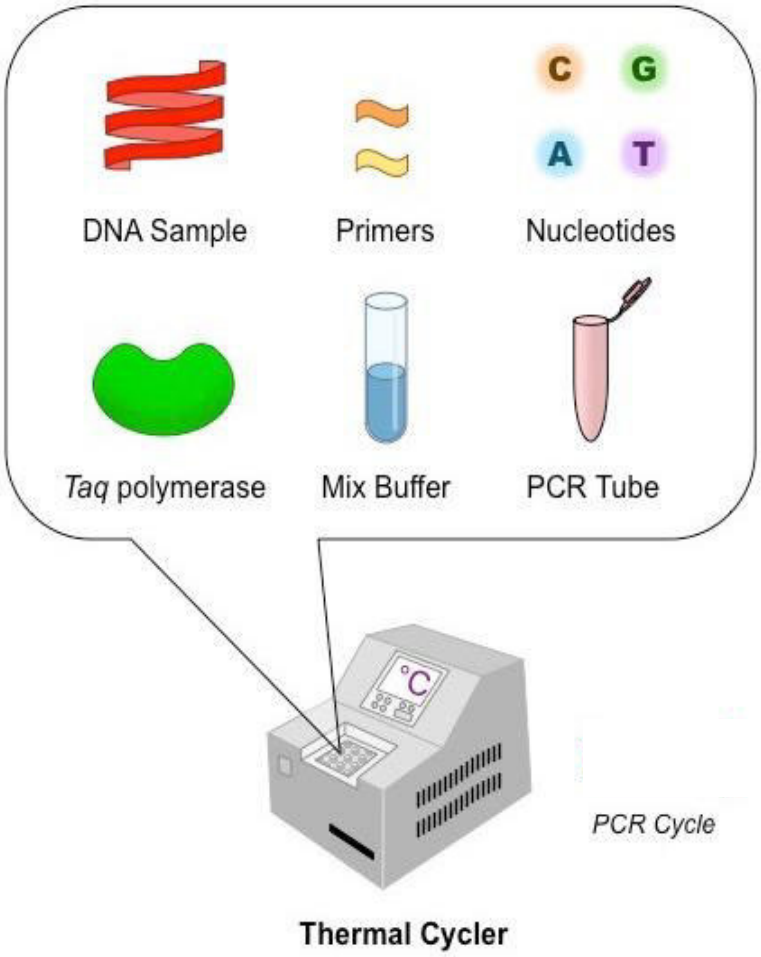
cDNA

- Cycles: 25-30

Viral Diagnostic Procedures

ESSENTIALS

1. DNA sample
2. Primers (Reverse and forward)
3. Nucleotides (dNTP's)
4. Buffer
5. MgCl2
6. Heat stable Taq Polymerase
7. Distilled water
8. PCR tubes and therma cycler



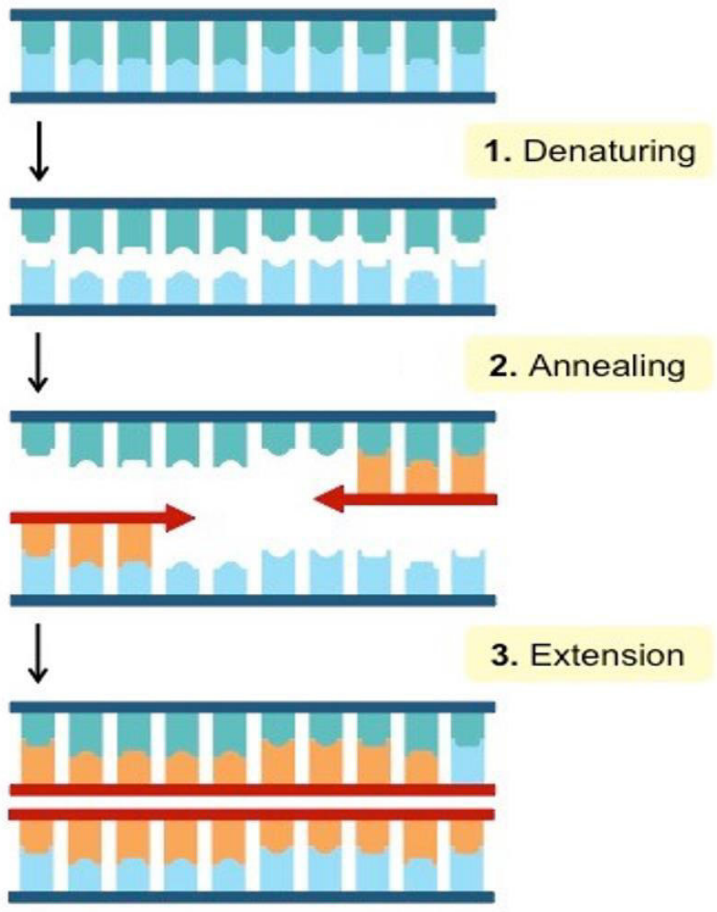
Viral Diagnostic Procedures

PROCEDURE

1. Step 1:
- Pre-denaturation

2. Step 2:
- Denaturation
- Annealing
- Extension

3. Step 3:
- Final polymerization



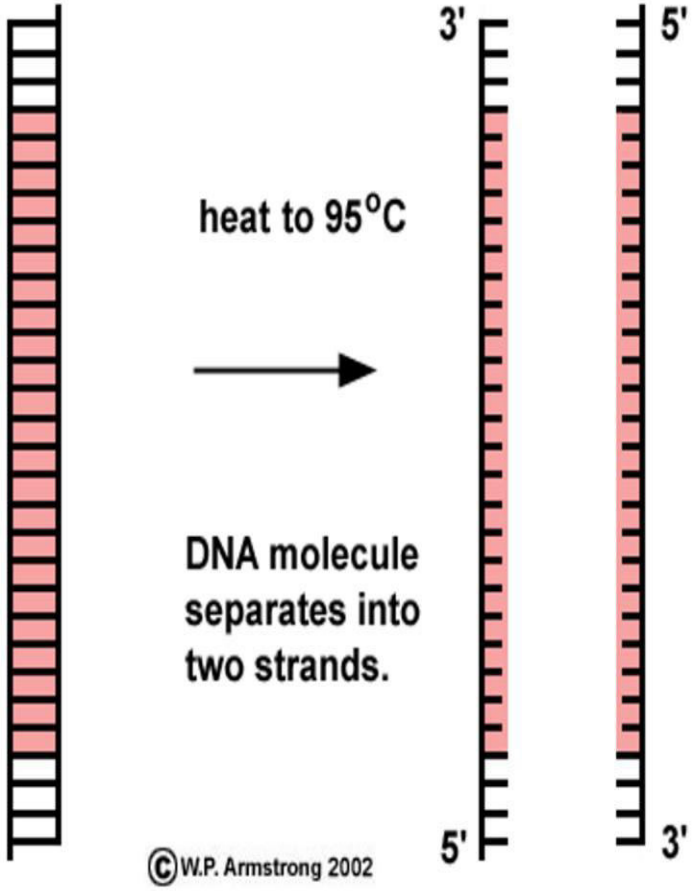
Viral Diagnostic Procedures

DENATURATION

Predenaturation: DNA boiled at 95°C for 5 minutes

Denaturation: 95°C for 15-30 seconds

Hydrogen bonds break & the complementary strands of the double-stranded DNA separate

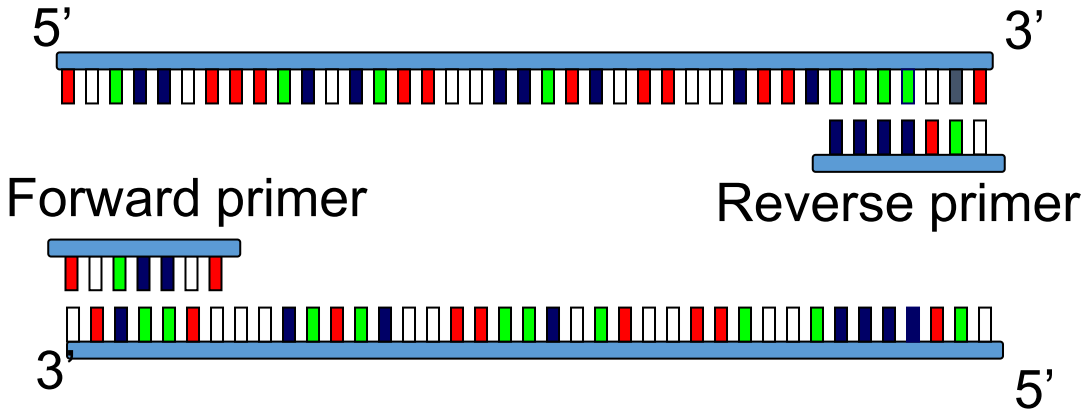


Viral Diagnostic Procedures

ANNEALING

Both primers (i.e. Reverse and forward) bind to their complementary sequences.

Temperature: 40-60°C



Viral Diagnostic Procedures

DENATURATION

- Temperature: 72°C
- MgCl2 facilitates Taq in adding up the nucleotides to the strand.
- Taq polymerase binds to the annealed primers and extends DNA at the 3’ end of the chain.
- Final polymerization: Fills gaps

Viral Diagnostic Procedures

Types of PCR

1. Real-Time PCR
2. Nested PCR
3. Multiplex PCR
4. Arbitrary Primed PCR
5. Hot start PCR
6. Inverse PCR

END

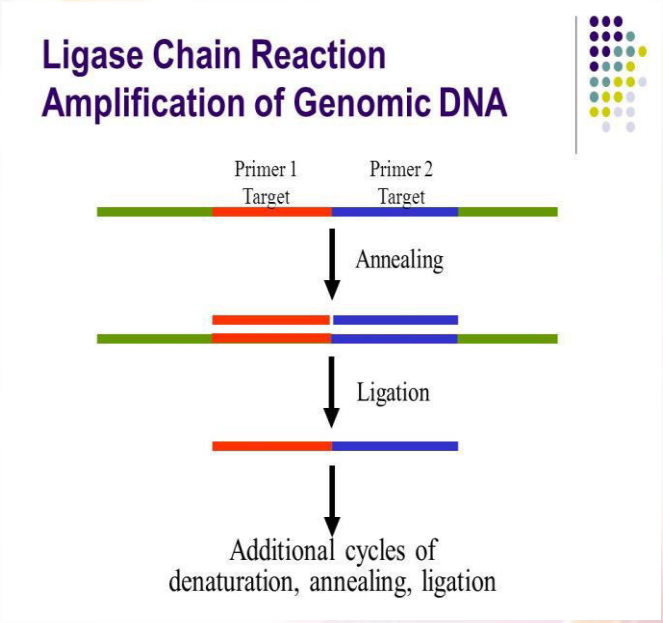
VIRAL DIAGNOSTIC TESTS

Ligase Chain Reaction

Ligase Chain Reaction

Ligase Chain Reaction

A method of DNA amplification.



Ligase Chain Reaction

- Differs from PCR
- Greater specificity
- Requires two different enzymes

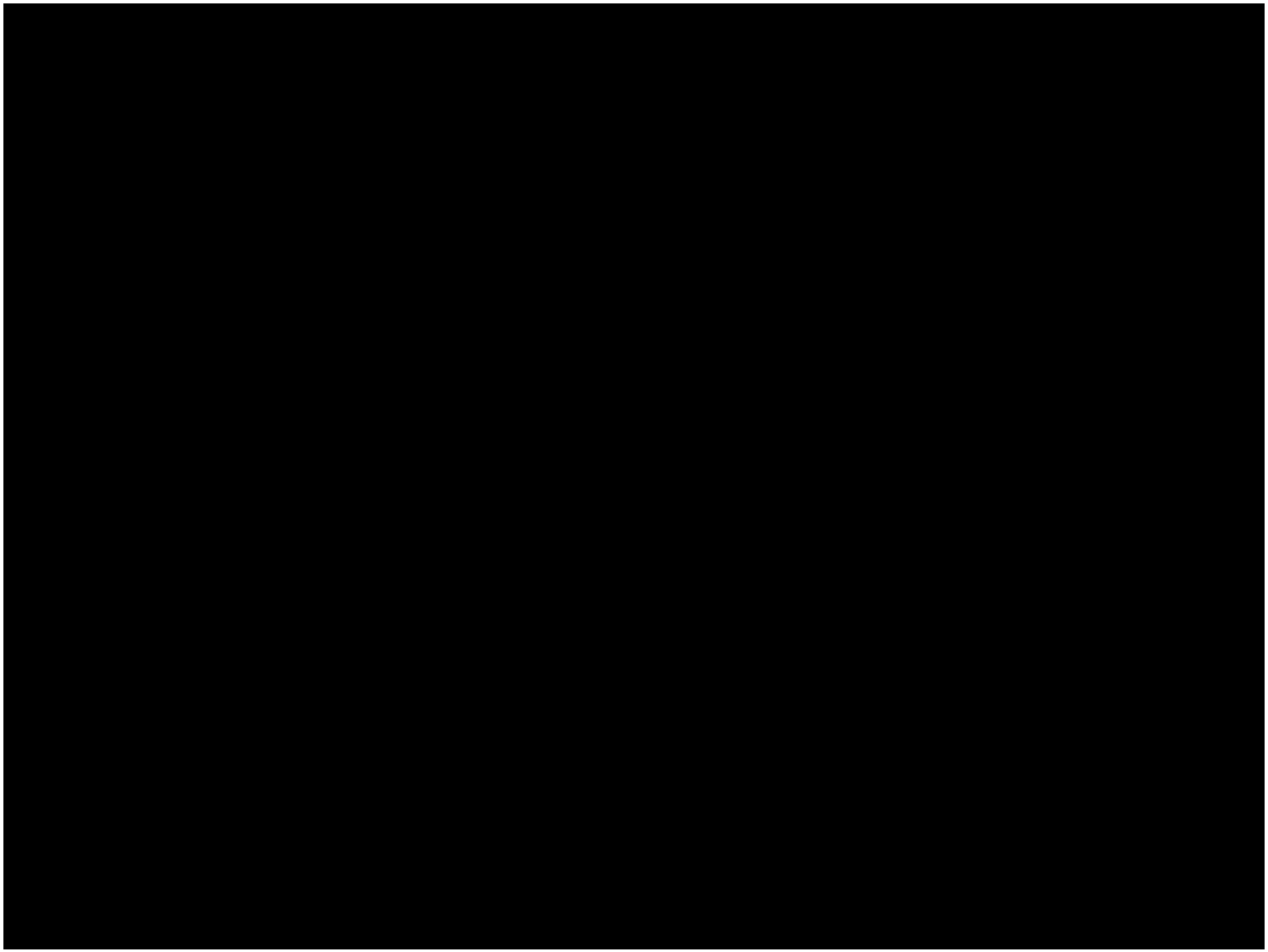
❖Ligase

❖Taq polymerase enzyme

Ligase Chain Reaction

- https://www.youtube.com/watch?v=dvW-kznCzy0

Viral Diagnostic Tests



Virology

VIRAL DIAGNOSTIC PROCEDURES

Nucleic acid sequence based amplification

Viral Diagnostic Procedures

1. Direct Examination of Specimen

- Electron Microscopy morphology / immune electron microscopy
- Light microscopy histological appearance - e.g. inclusion bodies
- Antigen detection immunofluorescence, ELISA etc.
- Molecular techniques for the direct detection of viral genomes

Viral Diagnostic Procedures

2. Indirect Examination

- Cell Culture - cytopathic effect, haemadsorption, confirmation by neutralization, interference, immunofluorescence etc.
- Eggs pocks on CAM - haemagglutination, inclusion bodies
- Animals disease or death confirmation by neutralization

3. Serology

Detection of rising titres of antibody between acute and convalescent stages of infection, or the detection of IgM in primary infection.

Viral Diagnostic Procedures

Classical Techniques	Newer Techniques
1. Complement fixation tests (CFT)	1. Radioimmunoassay (RIA)
2. Haemagglutination inhibition tests	2. Enzyme linked immunosorbent assay (EIA)
3. Immunofluorescence techniques (IF)	3. Particle agglutination
4. Neutralization tests	4. Western Blot (WB)
5. Single Radial Haemolysis	5. Recombinant immunoblot assay (RIBA), line immunoassay (Liatek) etc.

Viral Diagnostic Procedures

- Strand displacement Amplification
- Polymerase Chain Reaction (PCR)
- Ligase Chain Reaction (LCR)
- Transcription based amplification

VIRAL DIAGNOSTIC PROCEDURES

Strand Displacement amplification

Viral Diagnostic Procedures

Strand/Multiple displacement amplification (MDA)

- A non-PCR based DNA amplification technique
- This method can rapidly amplify minute amounts of DNA samples to a reasonable quantity for genomic analysis.

Viral Diagnostic Procedures

https://www.youtube.com/watch?v=j8YxNBw2CiM&t=4s

VIRAL DIAGNOSTIC TESTS

Branched Chain DNA

VIRAL DIAGNOSTIC PROCEDURES

Branched DNA

Signal amplification assay (as opposed to a target amplification assay) that is used to detect nucleic acid molecules

VIRAL DIAGNOSTIC PROCEDURES

METHOD

Base

Capture Probe

Extender

Target

label extender

pre-amplifier

amplifier

VIRAL DIAGNOSTIC PROCEDURES

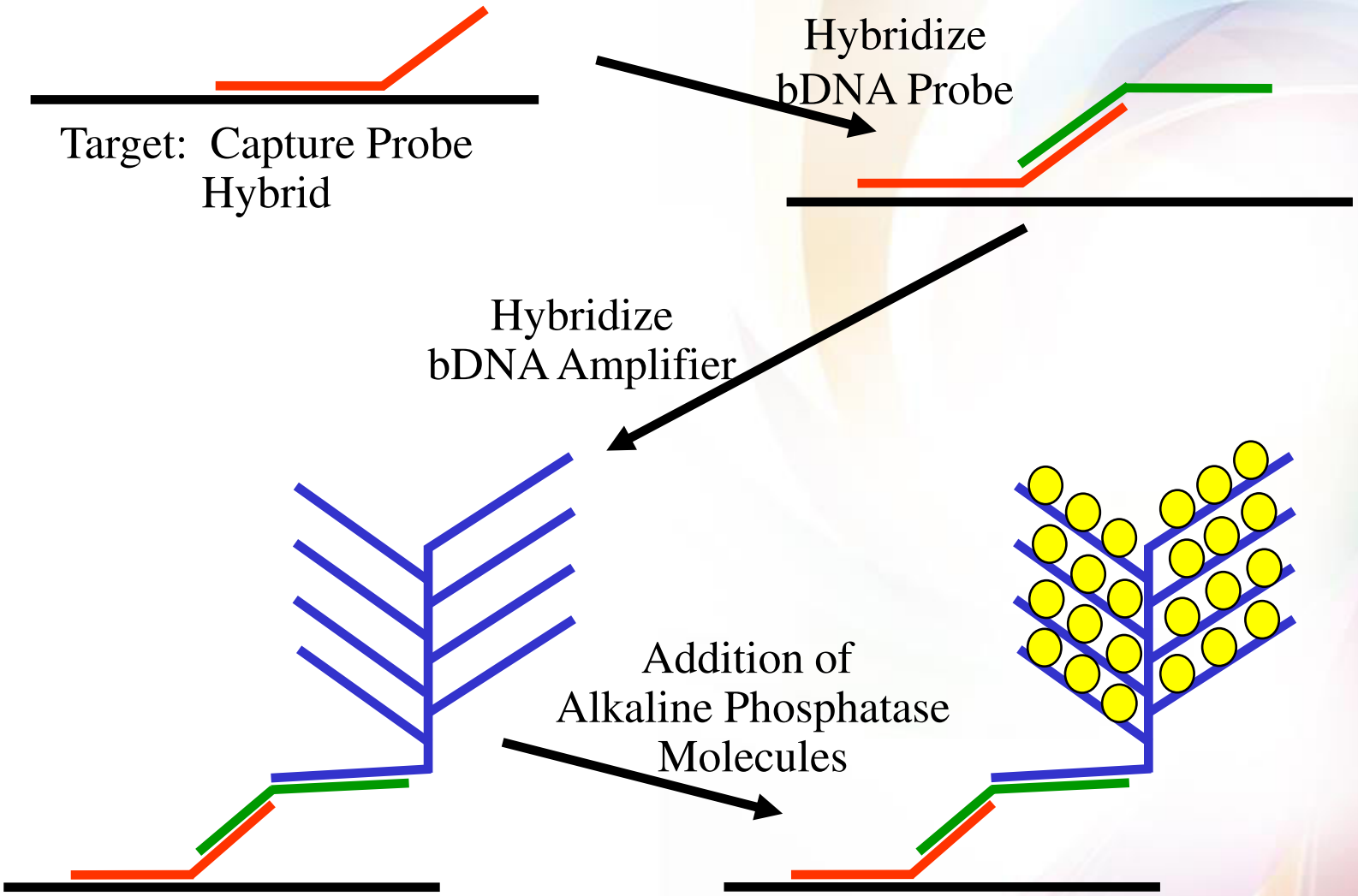
BRANCHED DNA DETECTION

- Target nucleic acid sequences are not replicated through C.
- Uses “capture probes,” “bDNA probes” and “bDNA amplifier probes.”
- Assay is based upon microtiter plate technology.

VIRAL DIAGNOSTIC PROCEDURES

- Solid phase signal amplification system
- Multiple sets of synthetic oligonucleotide probes
- Capture probes bound to well
- Target specific probes
- Amplifier molecule with 15 identical branches, each of which can bind to 3 labeled probes

VIRAL DIAGNOSTIC PROCEDURES



Virology

VIRAL DIAGNOSTIC PROCEDURES

Solid phase immunoassay

Viral Diagnostic Procedures

Solid phase immuno assay

Most common used tests

Cheap

Less complex

For both composition and antigen detection

Viral Diagnostic Procedures

INCLUDES

1. ELISA (Enzyme linked immunosorbent Assay)
2. ELIspot (Enzyme Linked Immunospot)
3. GCSPR (grating-coupled surface plasmon resonance)
4. Solid phase radio immunoassay

Viral Diagnostic Procedures

APPLICATIONS

highly sensitive means to measure the presence of antigen in defined and homogeneous samples such as purified proteins in buffer, as well as in undefined heterogenous biological samples such as cell lysates, tissue culture supernatants, blood, and other clinical samples

Viral Diagnostic Procedures

RADIO IMMUNO ASSAY

https://www.youtube.com/watch?v=70TPrfL_8-M

VIRAL DIAGNOSTIC TESTS

Immunodiffusion

Viral Diagnostic Tests

IMMUNODIFFUSIO

A diagnostic test which involves diffusion through a substance such as agar which is generally soft gel agar (1%) or agarose(1%), used for the detection of antibodies or antigen.

Viral Diagnostic Tests

TYPES

- Single diffusion in one dimension (Oudin procedure)
- 2. Double diffusion in one dimension (Oakley Fulthorpe procedure)
- 3. Single diffusion in two dimension (radial immunodiffusion or Mancini method)
- 4. Double diffusion in two dimensions (Ouchterlony double immunodiffustin)

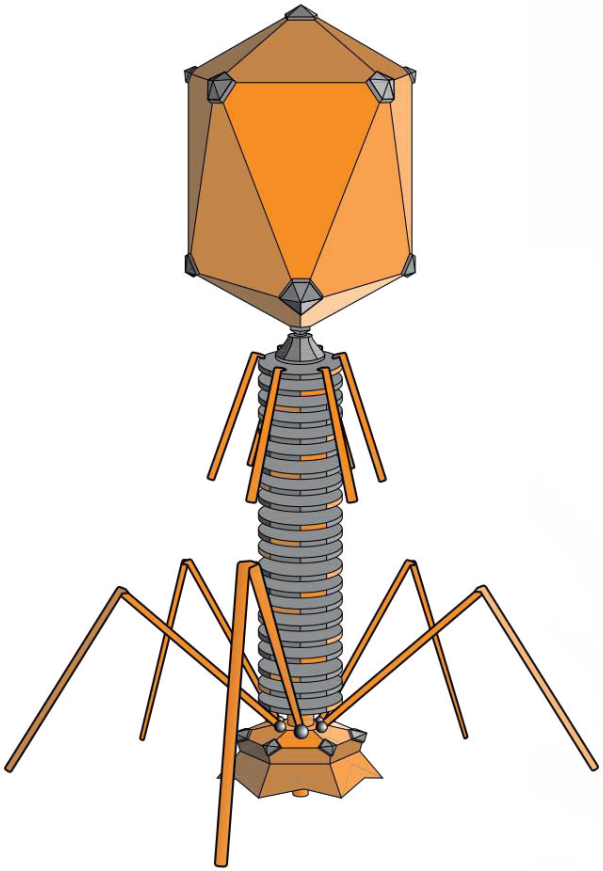
Viral Diagnostic Tests

https://www.youtube.com/watch?v=ADtT2dA1lsc

Virology

BACTERIOPHAGES
Properties

Bacteriophages



A **bacteriophage**

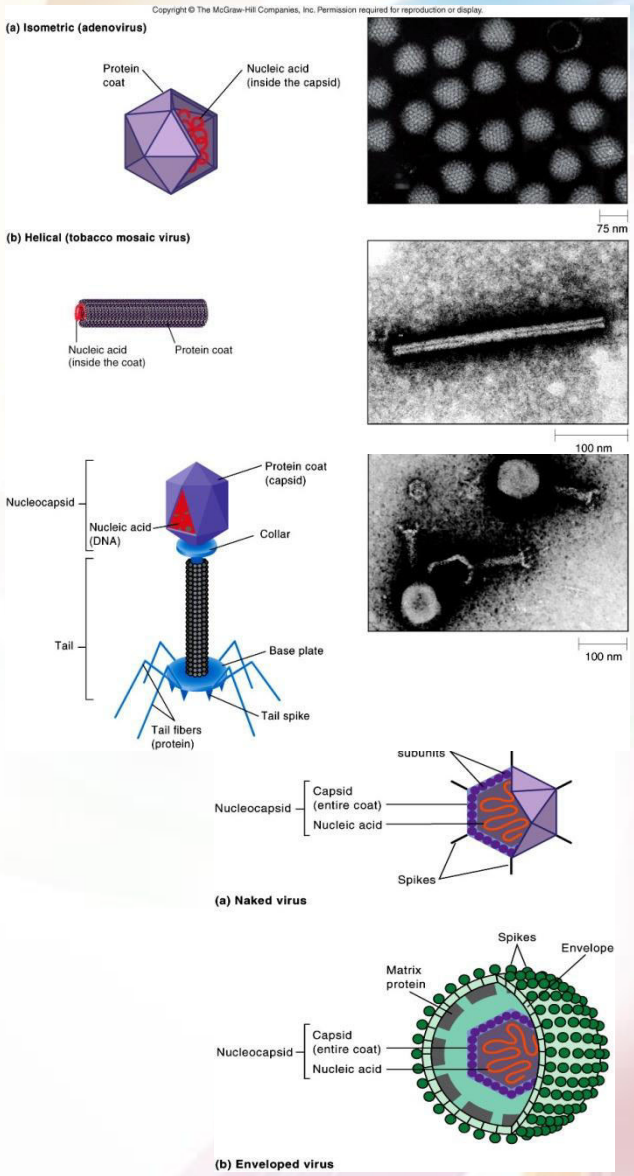
also known informally as a ***phage*** is a virus that infects and replicates within bacteria and archaea

Bacteriophages

General Characteristics

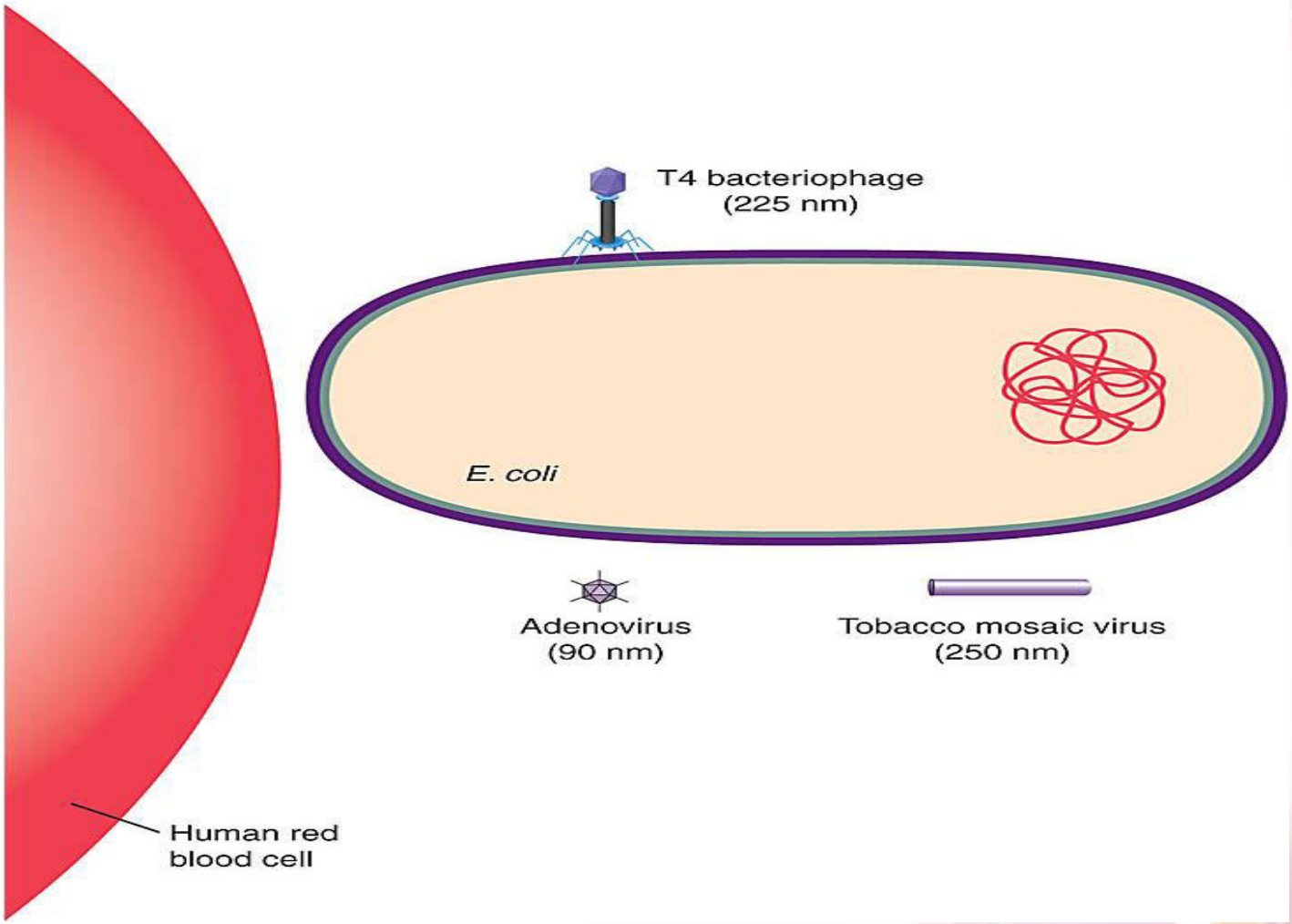
Virus architecture

- Virus particle called virion
  - Consists of nucleic acid surrounded by protein coat
    - Capsid
- Shapes
  - Isometric
  - Helical
  - Complex
- Two types of virion
  - Naked – without envelope
  - Enveloped – surrounded by lipid membrane



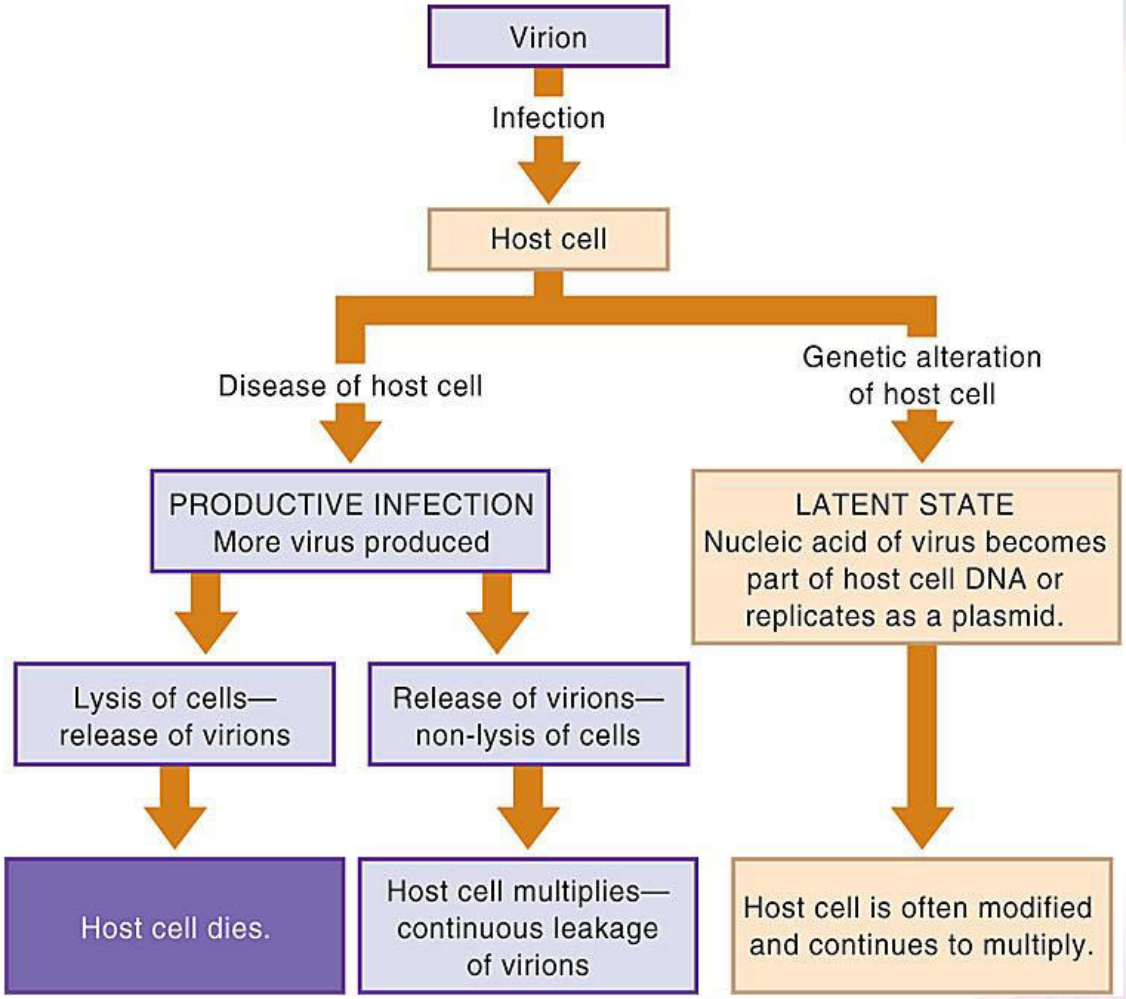
Bacteriophages

SIZE



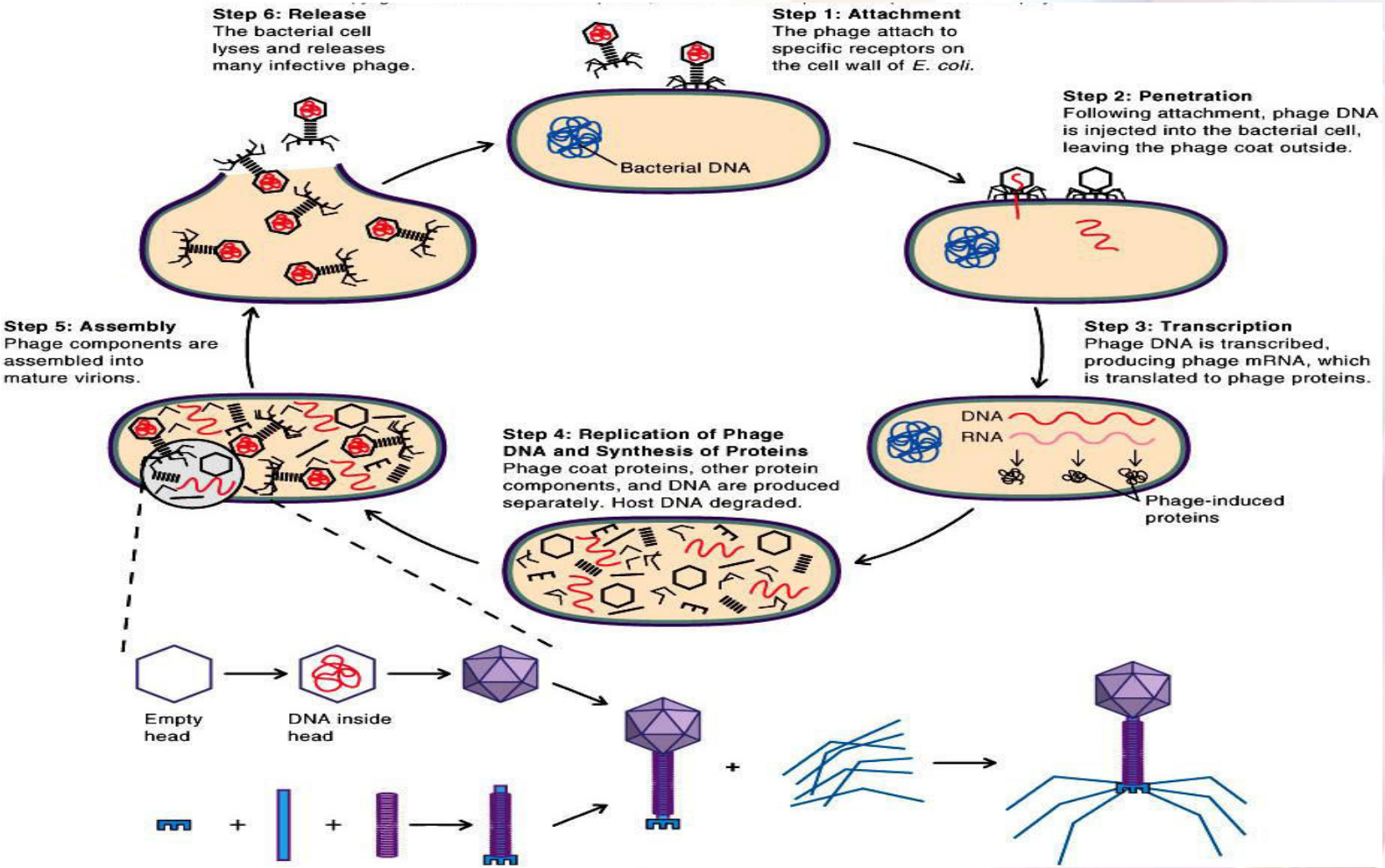
Bacteriophages

VIRUS INTERACTIONS WITH HOST CELLS



Bacteriophages

LYTIC CYCLE



Bacteriophages

https://www.youtube.com/watch?v=i9L-RoQ1frc

Virology

Relication of
T-Phages

T-phages

Replication

- Attachment
- Penetration
- Uncoating
- Biosynthesis
- Release

T-phages

Attachment

Specific receptors

Techoic acid

Lipids

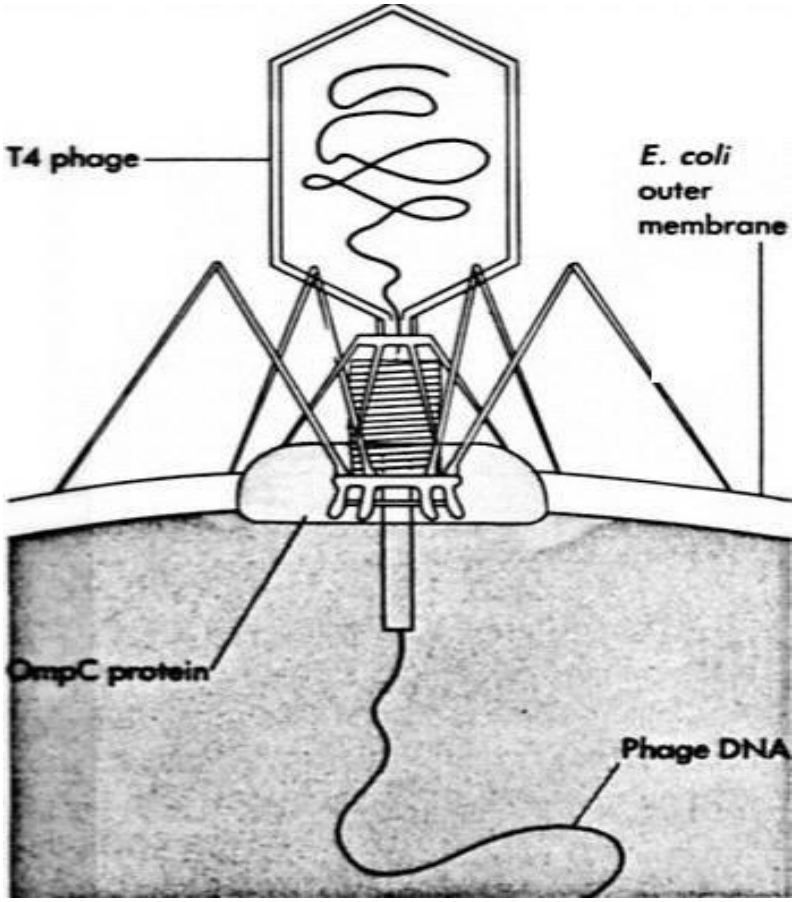
Proteins

flagella

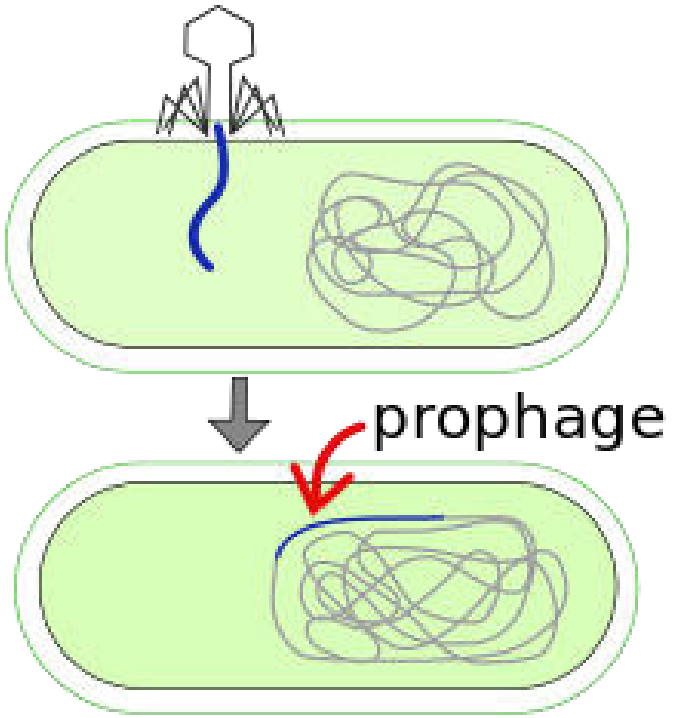
On specific bacterial hosts

T-phages

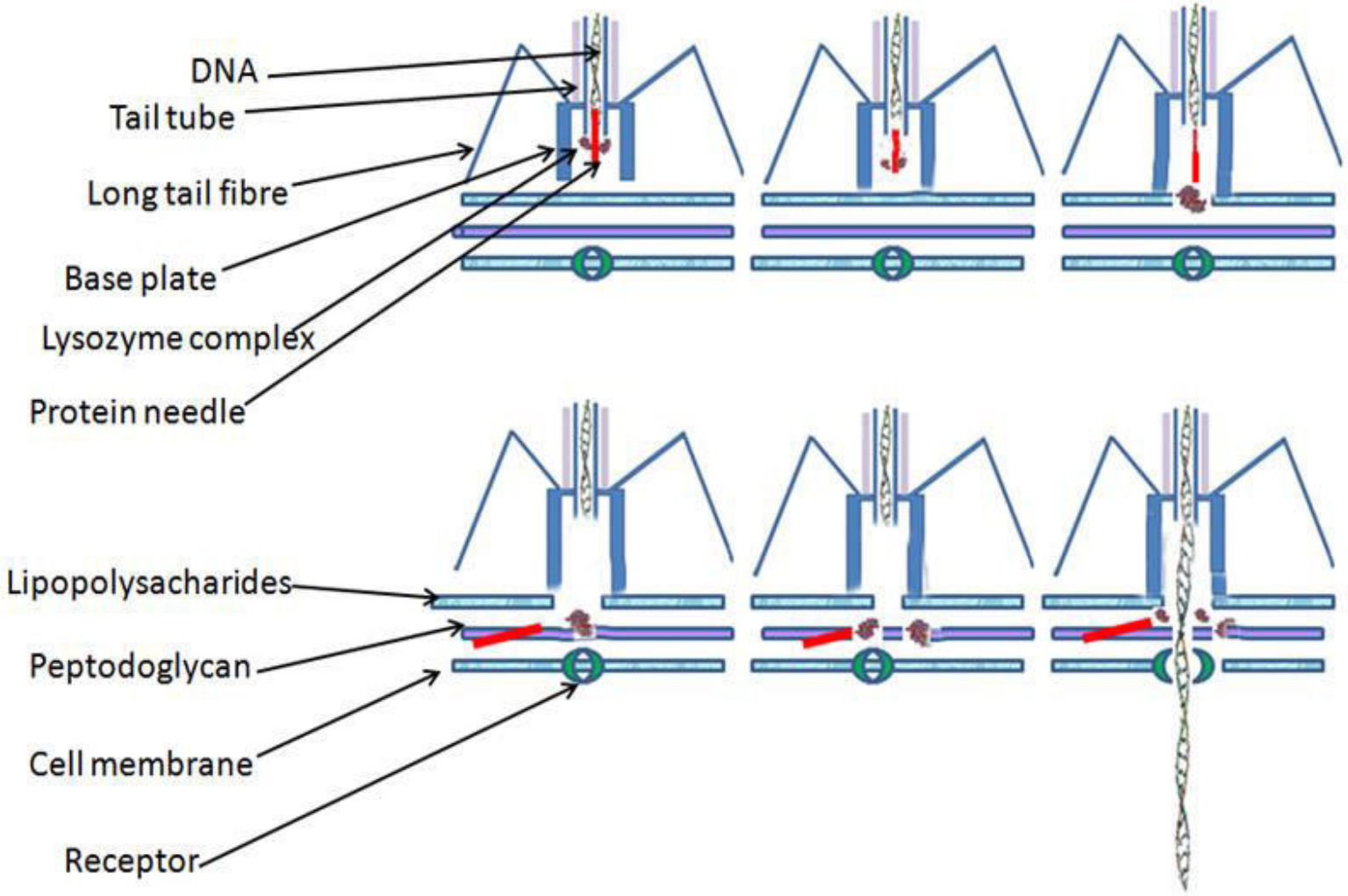
Penetration



- Leave capsid outside

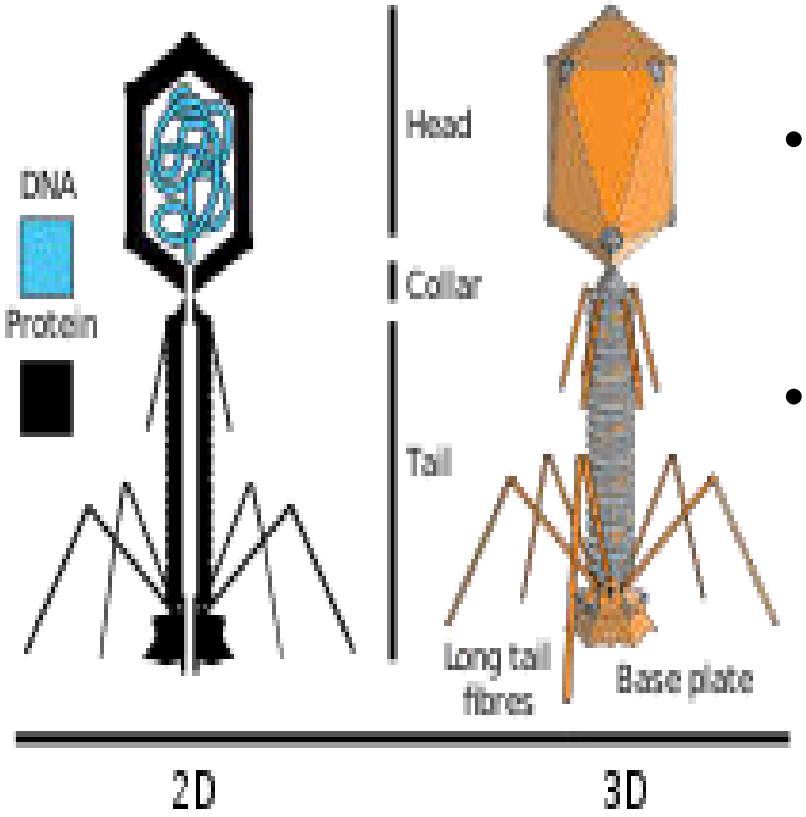


T–phages


DNA
Tail tube
Long tail fibre
Base plate
Lysozyme complex
Protein needle
Lipopolysacharides
Peptodoglycan
Cell membrane
Receptor

T-phages

Biosynthesis



- Bacterial ribosomes translate viral proteins
- Proteins modify bacterial RNA polymerase
- Proteins for new virions

T-phages

Assembly

- T4: helper proteins required
- Base plates assembled first
- Tails built afterwards
- Heads constructed separately
- Whole process takes 15 minutes

T-phages

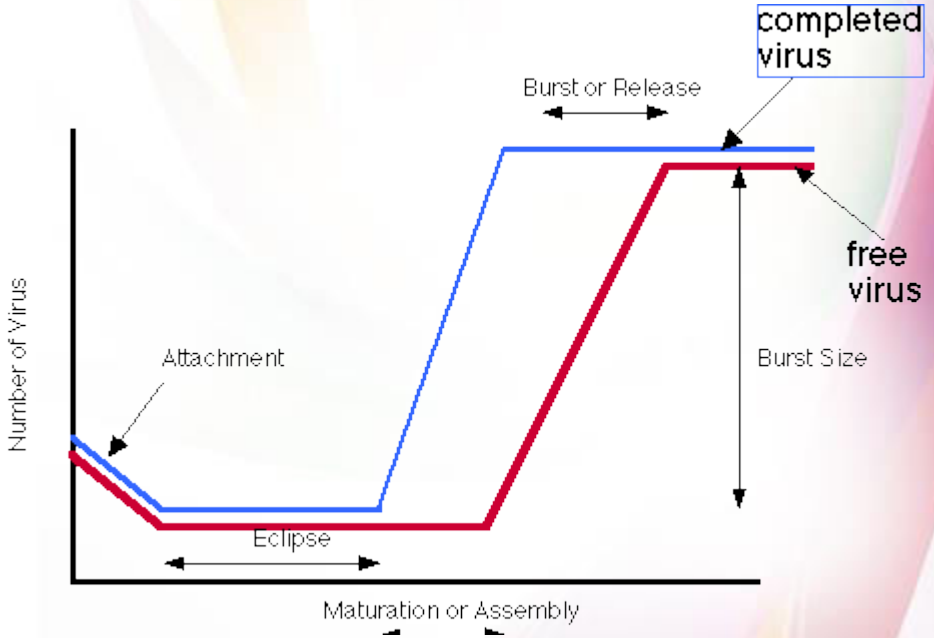
Release

- Lysis
  - ❖Tailed phages
  - ❖endolysin
- Extrusion
  - ❖enveloped
- Budding
- Secretion

END

Virology

Phage Growth Curve



Phage Growth Curve

GRAPH

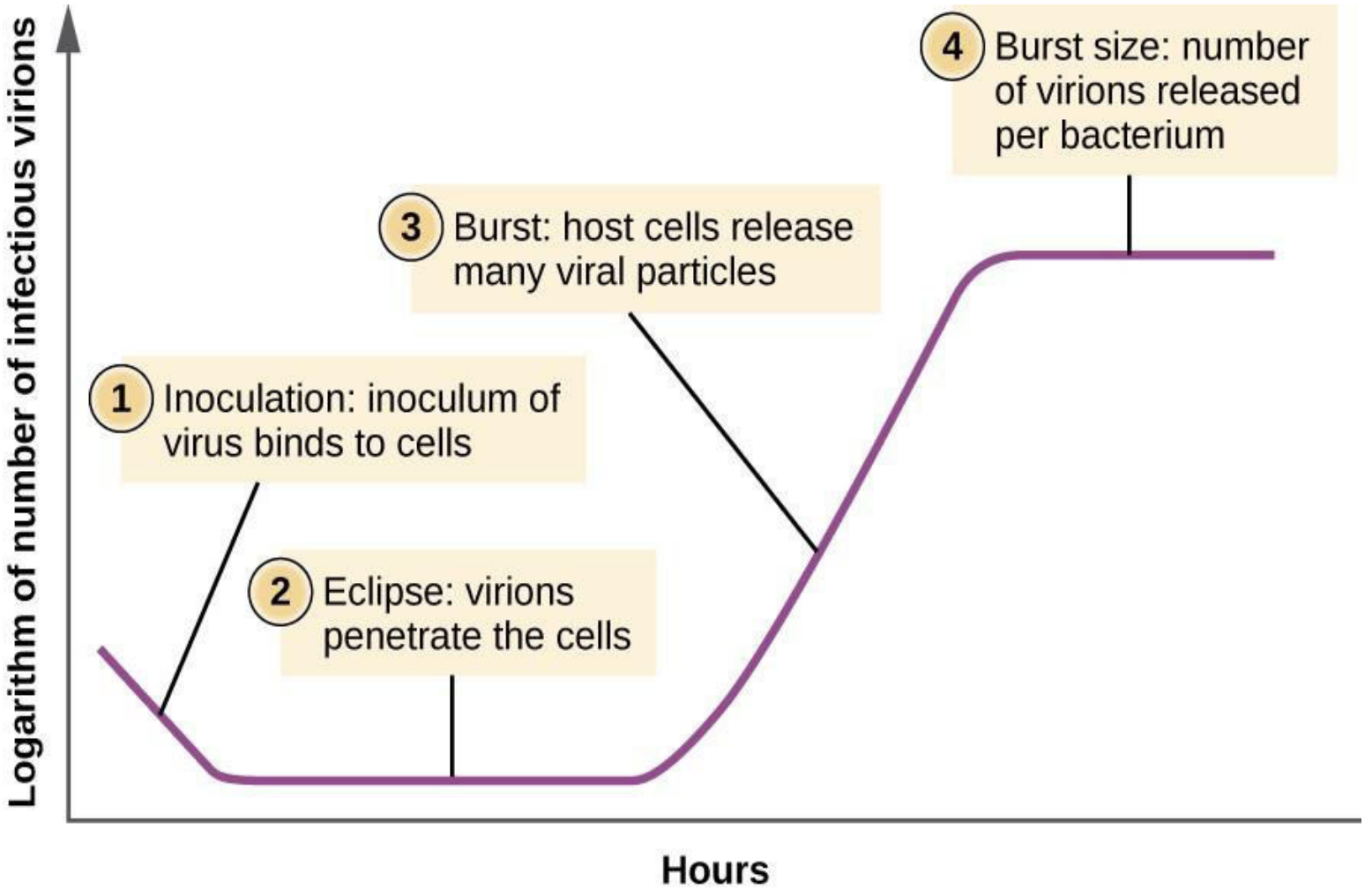
- Different from bacterial growth
- Not sigmoidal curve
- Consists of four steps

Phage Growth Curve

STEPS

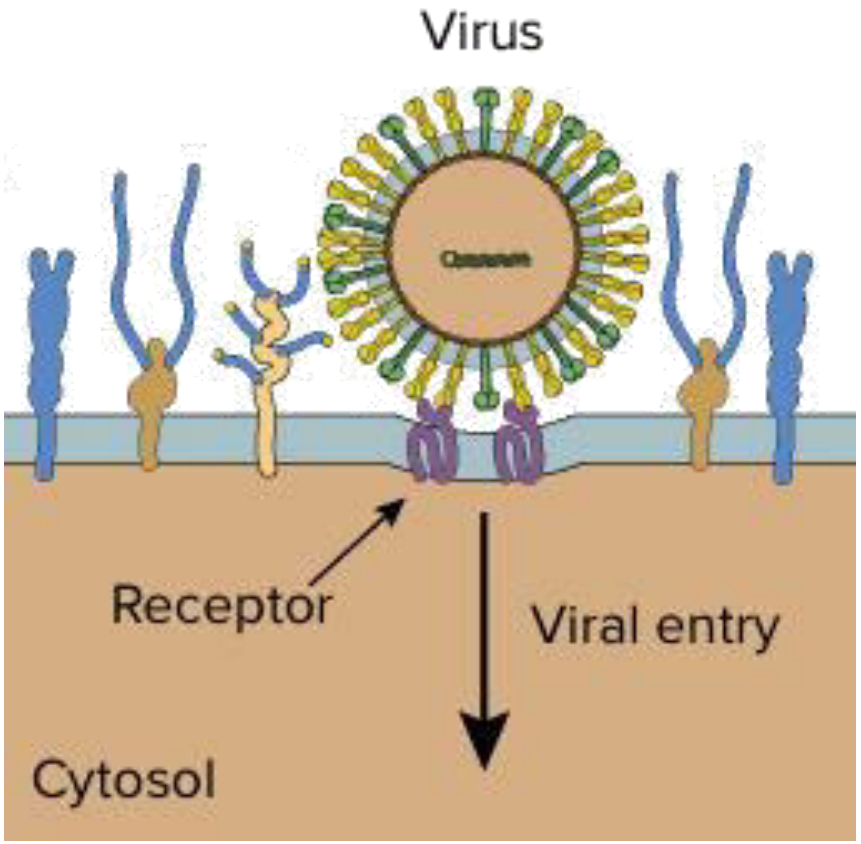
1. Inoculation
2. Eclipse
3. Burst
4. Burst size

Phage Growth Curve



Phage Growth Curve

INOCULATION/ ATTACHMENT



- Virus in blood stream attaches to host cell
- Beginning to decrease in number (negative gradient)

Phage Growth Curve

Eclipse/ Latent Period

More and more viruses replicate

Thus virus titre decreases

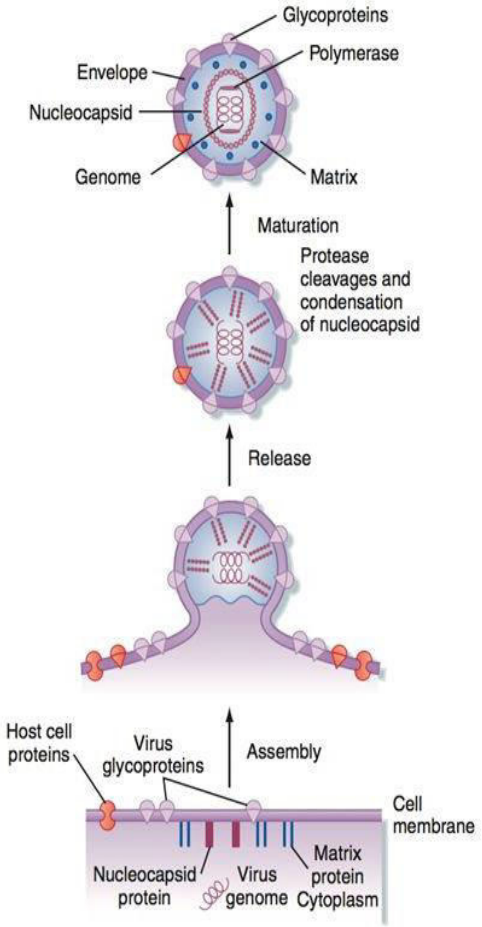
Reaches at 0 gradient

No virus in external medium

Phage Growth Curve

BURST

Release



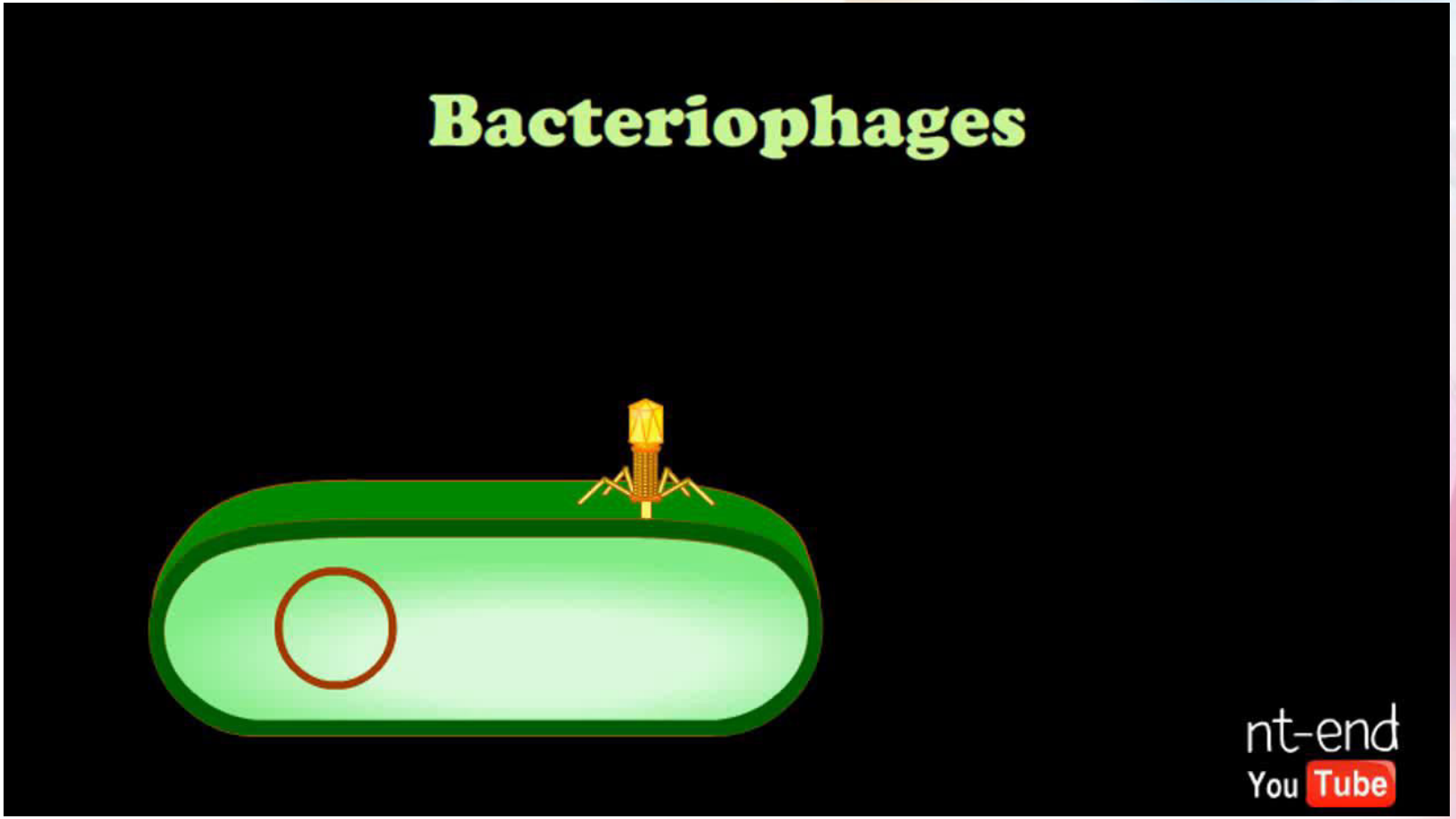
- Virions released
- Host cell releases many viral particles
- Host cells lyse
- Very steep rise in viral titre

Phage Growth Curve

Burst Size

- Number of virions released per bacterium
- Becomes constant

Phage Growth Curve



Virology

Phage Growth

Phage Growth

Bacteriophage Growth

- Phage collision with an adsorption-susceptible bacterium,
- Virion attachment,
- Virion nucleic acid uptake,

Phage Growth

Bacteriophage Growth

- Synthesis of phage proteins and nucleic acid,
- A period during which virions mature,
- A virion release step

Phage Growth

Phage Relationship with bacteria

The Red Queen hypothesis

- The interaction of phage with bacteria is the primary concern of phage community
- Bacteria have developed mechanisms that prevent phages from having an effect on them, which has led to this evolutionary arms race between the phages and their host bacteria
- Bacterial resistance to phages puts pressure on the phages to develop stronger effects on the bacteria.

Phage Growth

The Red Queen hypothesis

- The Red Queen hypothesis describes this relationship, as the organisms must constantly adapt and evolve in order to survive.
- This relationship is important to understand as phages are now being used for more practical and medicinal purposes.

Phage Growth

Restriction-Modification System

- Bacteria have developed multiple defense mechanisms
- Foreign DNA trying to enter the bacterial host is restricted by endonucleases that recognize specific base pairs within the DNA

Phage Growth

Restriction-Modification System

- DNA of the cell is protected from restriction due to methylase
- RM systems have evolved to keep up with the ever-changing bacteria and phage.

Phage Growth

- In general, these RM types differ in the nucleotide sequences that they recognize.
- There is an occasional slip where the endonuclease misses the DNA sequence of the phage and the phage DNA is able to enter the cell anyway, becoming methylated and protected against the endonuclease.

Phage Growth

Clustered Regularly Interspersed Palindromic Repeats

- CRISPR means that the immunity to phages by bacteria has been acquired via adding spacers of DNA that are identical to that of the DNA from the phage.
- Some phages have been found to be immune to this mechanism as well.
- In some way or another, the phages have managed to get rid of the sequence that would be replicated.
- Phages are also capable of interacting with species other than bacteria, e.g., such as phage-encoded exotoxin interaction with animals
- Phage therapy is an example of applied phage community ecology.

Virology

Phage Therapy

Phage Therapy

Phage Therapy

- ❖Therapeutic use of bacteriophages for bacterial infections
- ❖Phage therapy has many potential applications in
  - Human medicine
  - Dentistry
  - Veterinary science
  - Agriculture
- ❖Biocontrol

Phage Therapy

Benefits

- Bacteriophages are much more specific than antibiotics
- They are typically harmless not only to the host organism but gut flora
- They have a high therapeutic index
- Few side effects
- A smaller dose will be effective

Phage Therapy

Obstacles

- The high bacterial strain specificity
- Different cocktails for treatment of the same infection
- Bacteria can evolve different receptors either before or during treatment

Phage Therapy

Obstacles

- In the case all bacteria is eliminated, the specific phage might evolve and attack good bacteria
- The need for banks of phages makes regulatory testing for safety harder and make it more expensive

Phage Therapy

Collection.

- The simplest method of phage treatment involves collecting local samples of water likely to contain high quantities of bacteria and bacteriophages
- for example effluent outlets, sewage and other sources.[7]

Phage Therapy

Collection.

- The samples are taken and applied to the bacteria that are to be destroyed which have been cultured on growth medium.
- If the bacteria die, as usually happens, the mixture is centrifuged; the phages collect on the top of the mixture and can be drawn off

Phage Therapy

Obstacles

- The high bacterial strain specificity
- Different cocktails for treatment of the same infection
- Bacteria can evolve different receptors either before or during treatment

Phage Therapy

Obstacles

- In the case all bacteria is eliminated, the specific phage might evolve and attack good bacteria
- The need for banks of phages makes regulatory testing for safety harder and make it more expensive

Phage Therapy

Administration

- Application is pill form

❖Temperature stability up to 55 °C

❖shelf lives of 14 months

- Application in liquid form

❖ possible, stored preferably in refrigerated vials

- Applicatioin in Oral form

❖ when an antacid is included, as this increases the number of phages surviving passage through the stomach.

Virology

Estimation of Growth Numbers/Phage Ecology

Estimation Of Growth Numbers/Phage Ecology

- The most numerous "organisms" on Earth
- The viruses of bacteria (more generally, of prokaryotes
- Phage ecology is the study of the interaction of bacteriophages with their environments.

Estimation Of Growth Numbers/Phage Ecology

- Phages are Obligate intracellular parasites meaning that they are able to reproduce only while infecting bacteria
- Phages therefore are found only within environments that contain bacteria.
- Most environments contain bacteria, including our own bodies (called normal flora).
- Often these bacteria are found in large numbers. As a consequence, phages are found almost everywhere.

Estimation Of Growth Numbers/Phage Ecology

- As a rule of thumb, phage population will exceed bacterial densities by a ratio of 10-to-1 or more
- As there exist estimates of bacterial numbers on Earth of approximately 1030, there consequently is an expectation that 1031 or more individual virus (mostly phage) particles exist
- Making phages the most numerous category of “Orgamnisms" on our planet.

Estimation Of Growth Numbers/Phage Ecology

- Bacteria (along with archaea) appear to be highly diverse and there possibly are millions of species.
- Phage-ecological interactions therefore are quantitatively vast: huge numbers of interactions.
- Phage-ecological interactions are also qualitatively diverse
- There are huge numbers of environment types, bacterial-host types, and also individual phage types

Estimation Of Growth Numbers/Phage Ecology

Phage Ecology

- Phage population ecology
- Phage community ecology
- Phage ecosystem ecology

Estimation Of Growth Numbers/Phage Ecology

Phage population ecology

- A population is a group of individuals which either do or can interbreed interbreeding
- Population ecology considers characteristics that are apparent in populations of individuals
- These characteristics include so-called intraspecific interactions, that is between individuals making up the same population
- For example competition as well as cooperation

Estimation Of Growth Numbers/Phage Ecology

Phage community ecology

- A community consists of all of the biological individuals found within a given environment (more formally, within an ecosystem),
- Particularly when more than one species is present
- Community ecology studies those characteristics of communities that either are not apparent or less apparent
- Community ecology thus deals with interspecific interactions.
- Interspecific interactions, like intraspecific interactions, can range from cooperative to competitive but also to quite antagonistic (as are seen, for example, with predator-prey interactions).
- An important consequence of these interactions is coevolution.

Estimation Of Growth Numbers/Phage Ecology

END

Phage ecosystem ecology

- An ecosystem consists of both the biotic and abiotic components of an environment.
- Ecosystem ecology naturally differs from community ecology in terms of the impact of the community on these abiotic entities, and vice versa.
- In practice, the portion of the abiotic environment of most concern to ecosystem ecologists is inorganic nutrients and energy.

Virology

LYSOGENY

Properties

Lysogeny

Definition

Lysogeny is characterized by integration of the bacteriophage nucleic acid into the host bacterium's genome or formations of a circular replicon in the bacterial cytoplasm

Lysogeny

Properties

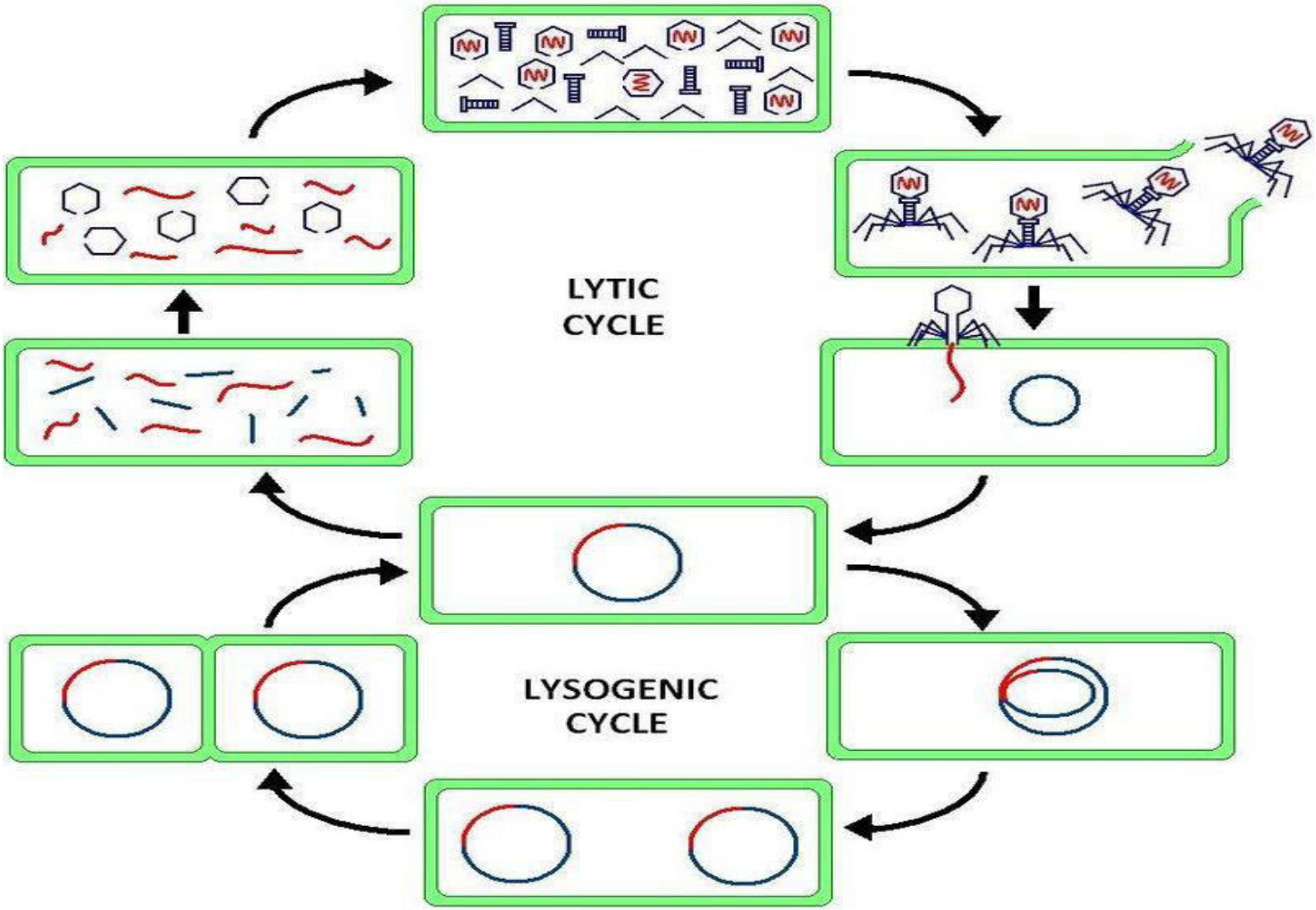
- Slow replication of viruses
- Mostly subclinical infection
- No harm to host
- Number of virions increases slowly

Lysogeny

Advantages

- Genome incorporated in prophage
- Repressor prevents viral replication
- Lyse bacteria for release

Lysogeny



Lysogeny

Lysogenic conversion

A phage induces a change in the phenotype of the infected bacteria that is not part of a usual phage cycle.

Bacterial survival

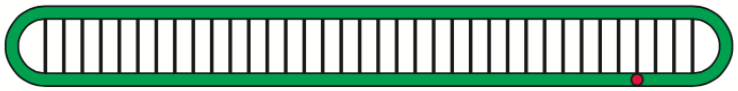
Lysogenic conversion has shown to enable biofilm formation in Bacillus anthracis

Bacterial virulence

Non-virulent bacteria have also been shown to transform into highly virulent pathogens through lysogenic conversion with the virulence factors carried on the lysogenic prophage

Virology

SATELLITES



Satellites

Definition

Satellite

viruses are particles

that contain nucleic acid

genomes encoding a

structural protein that

encaptivates the satellite

genome.

Satellites

Properties

- Subviral agents composed of nucleic acid
- Satellites are dependent on the presence of a helper virus for their propagation

Satellites

Properties

Commonly associated with plants

- Could be found in mammals, arthropods, and bacteria
- Have basic components to make their own protein shell to enclose their genetic material

Satellites

Satellite Virus Vs. Virus

Virus	Satellite Virus	
Replication	Able to direct host cell to replicate genome	Depends on presence of helper virus for replication of genome
Nucleic acid	Contain DNA or RNA or both at different points in life cycle	Contain DNA or RNA
Genome Size	<10kbp to >2000kbp	0.22-1.5kbp
Structure	Contain protein shell or capsidPackaged genome with a capsid Envelope-not specific to all viruses	Satellite viruses contain the protein to encode own capsid with aid of helper virusSatellite RNA's and DNA's do not have capsids, rely on helper virus to enclose their genome
Host Range	Can infect all types of organism; animals, plants, fungi, bacteria, archaea	Plants (most common), mammals, arthropods, bacteria

Satellites

Satellite Rans

- **Satellite RNAs** do not encode capsid protein, but are packaged by a protein encoded in the helper virus genome. Satellite genomes may be single-stranded RNA or DNA or circular RNA, and are replicated by enzymes provided by the helper virus

Satellites

Examples

- Satellite viruses may infect plants, animals, or bacteria
- An example of a satellite virus is satellite tobacco necrosis virus, which encodes a capsid protein that forms an icosahedral capsid that packages only the 1,260 nucleotide satellite RNA
- The helper virus, tobacco necrosis virus, encodes an RNA polymerase that replicates its genome and that of the satellite

Satellites

In Plants

- may attenuate or exacerbate disease caused by the helper virus
- Examples: necrosis and systemic chlorosis, or reduced chlorophyll production leading to leaves that are pale, yellow, or yellow-white
- Symptoms by satellite RNAs are thought to be a consequence of silencing of host genes

Virology

Viroids

Viroids

Viroids

- The smallest infectious pathogens known so far
- They are composed solely of a short strand of circular, single-stranded RNA

Viroids

Viroids

- RNA that has no protein coating
- All known viroids are inhabitants of plants in which they are causing diseases

Viroids

Transmission

- By aphids
- By cross contamination
- Due to mechanical damage to plants as a result of horticultural or agricultural practices
- From plant to plant by leaf contact

Viroids

Replication

❖Viroids replicate

- in the nucleus (Pospiviroidae)
- or chloroplasts (Avsunviroidae)

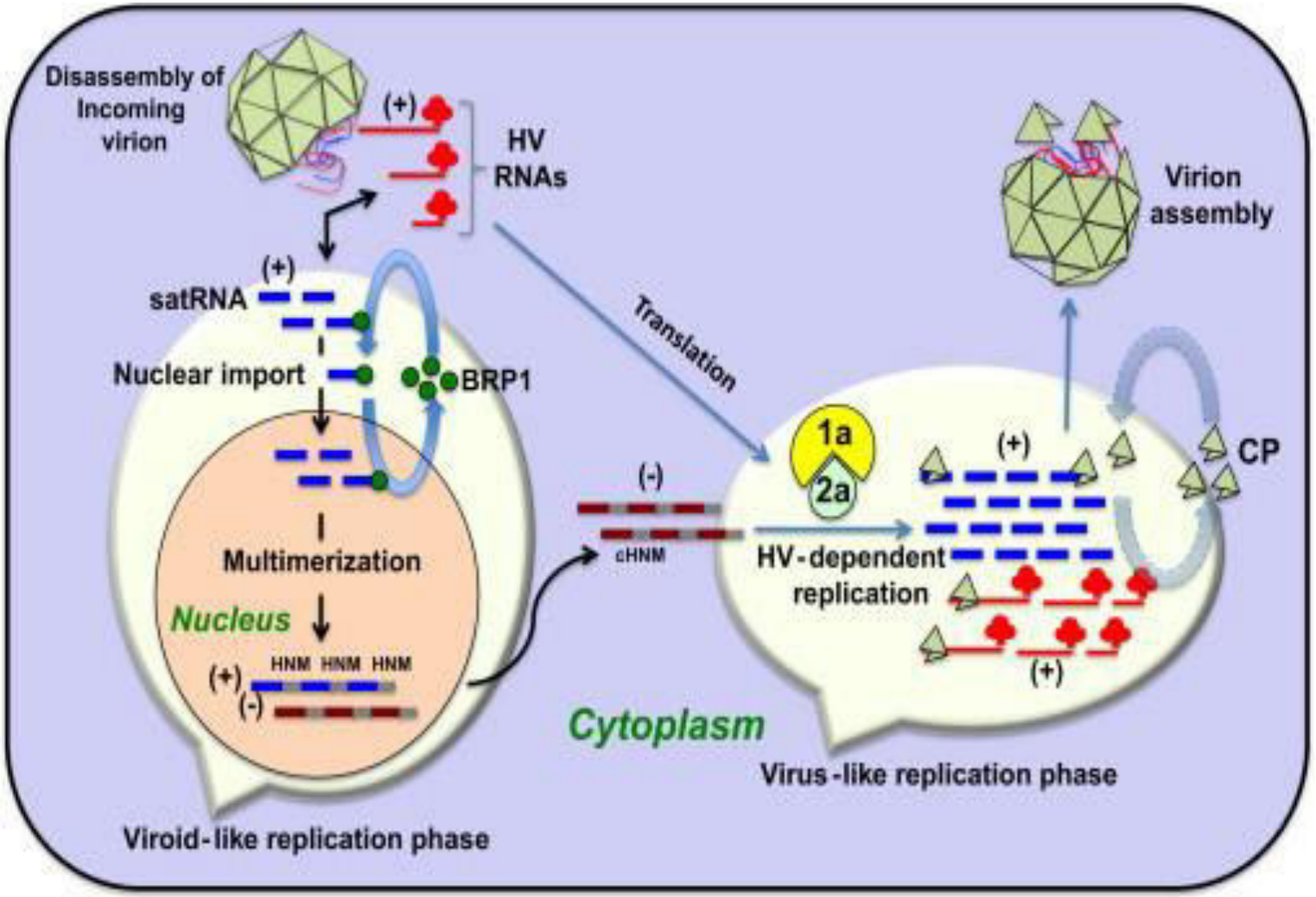
Viroids

Replication

❖Three steps

- Through an RNA-based mechanism. They require RNA polymerase II, a host cell enzyme normally associated with synthesis of messenger RNA from DNA, which instead catalyzes "rolling circle" synthesis of new RNA using the viroid as template

Viroids


Disassembly of Incoming virion
(+)
HV RNAs
Virion assembly
satRNA
(+)
Translation
Nuclear import
BRP1
1a
2a
(+)
CP
(-)
cHNM
Multimerization
HV-dependent replication
Nucleus
HNM HNM HNM
(+)
(-)
(+)
Cytoplasm
Virus-like replication phase
Viroid-like replication phase

Virology

Prions

Prions

Prions

- Misfolded proteins
- Prions for proteinaceous infectious particle
- Characterize several fatal neurodegenerative diseases in animals and humans

Prions

Importance

Cause of

- Transmissible spongiform encephalopathies (TSEs)
- Scrapie in sheep
- Chronic wasting disease (CWD) in deer
- Bovine spongiform encephalopathy (BSE) in cattle (commonly known as "mad cow disease")
- Creutzfeldt-Jakob disease in humans

Prions

Structure

The protein that prions are made of (PrP) is found throughout the body, even in healthy people and animals.

PrPC

PrPC is a normal protein found on the membranes of cells.

It has 209 amino acids (in humans)

One disulfide bond,

A molecular mass of 35–36 kDa

A mainly alpha-helical structure.

The normal protein is not sedimentable; meaning that it cannot be separated by centrifuging techniques.

Prions

Structure

PrPres

Protease-resistant PrPSc-like protein (PrPres) is an isoform of PrPc which is structurally altered and converted into a misfolded proteinase K-resistant form in vitro.

PrPSc

The infectious isoform of PrP, known as PrPSc, is able to convert normal PrPC proteins into the infectious isoform by changing their conformation, or shape; this, in turn, alters the way the proteins interconnect. PrPSc always causes prion disease.

Virology

Difference Between Satellites, Prions and Viroids

Satellites, Prions and Viroids

Satellite

- Smallest
- Most simple infectious
- RNA molecules
- Parasites of viruses

Satellites, Prions and Viroids

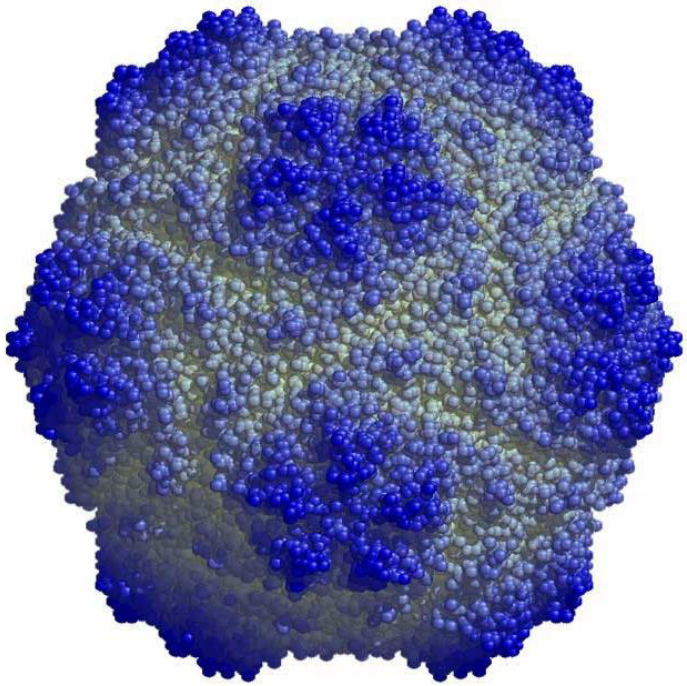
Satellite

- Satellites require helper viruses for replication
- Satellites do encode protein
- Circular RNA genome absent
- Satellites range between 475-1700 bases

Satellites, Prions and Viroids

Satellite

Tobacco necrosis satellite virus PDB_ID: 2STV



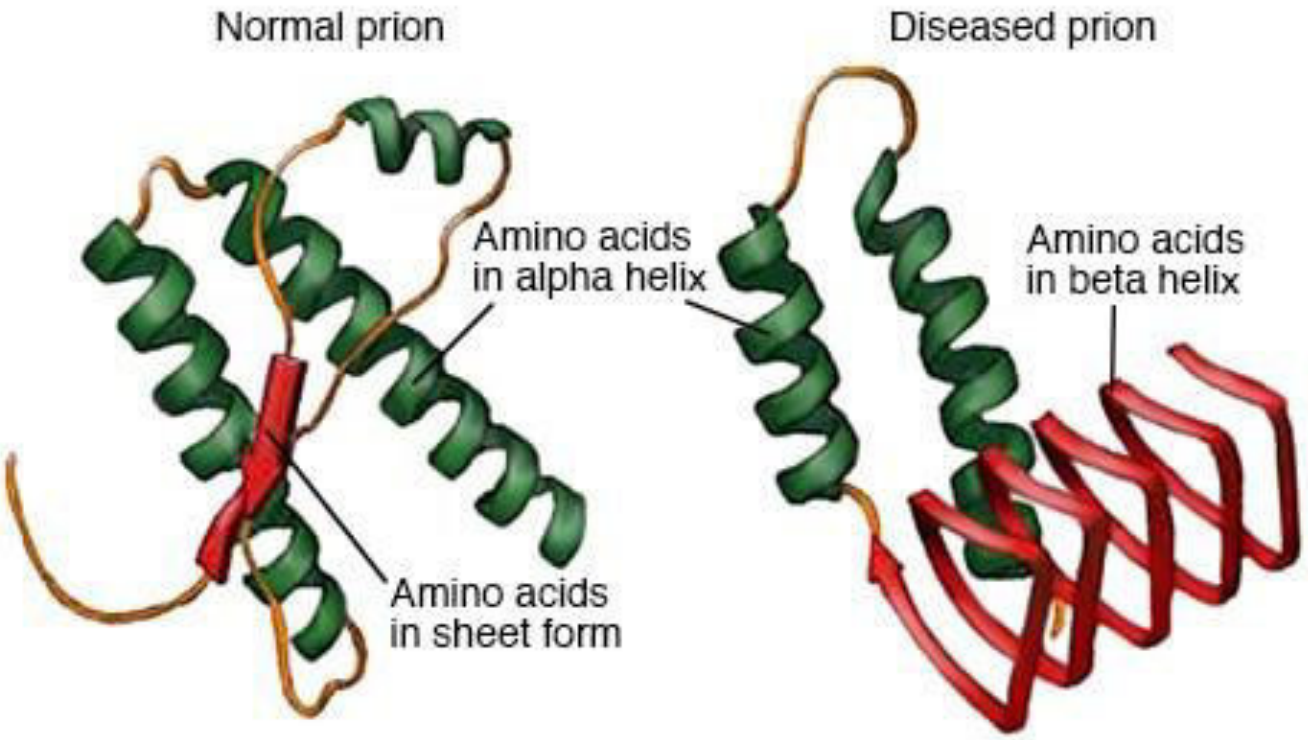
Satellites, Prions and Viroids

Prions

- Neither have DNA or RNA
- Abnormal or mutated form of protein
- Causes several lethal diseases

Satellites, Prions and Viroids

Prions



Satellites, Prions and Viroids

Viroids

- Autonomous
- Circular RNA genome present
- 250 to 400 bases
- Positive sense RNA
- Smaller than virus

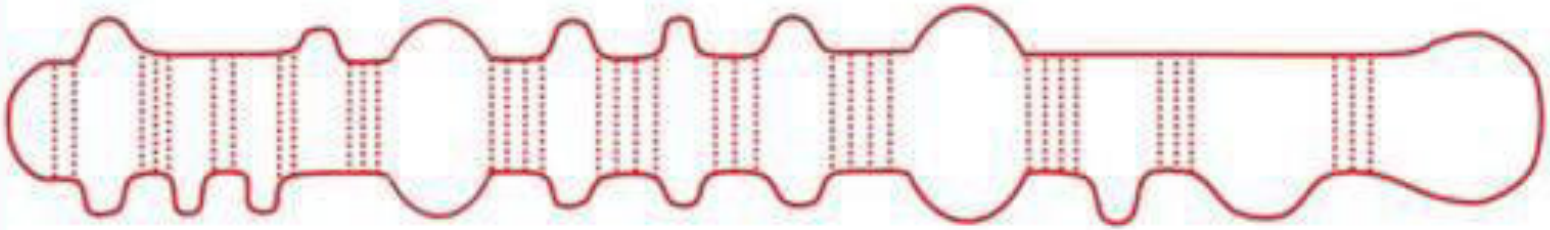
Satellites, Prions and Viroids

VIROIDS

- Do not need helper virus for replication
- Two families: *Pospoviridae* and *Avsunviridae*
- Usually affects plants

Satellites, Prions and Viroids

Viroids



Structure of a viroid – circular single-stranded RNA with some pairing between complementary bases and loops where no such pairing occurs

Virology

SATELLITES

Properties

Satellites

Satellite

- Smallest subviral agent
- Most simple infectious
- RNA molecules
- Parasites of viruses

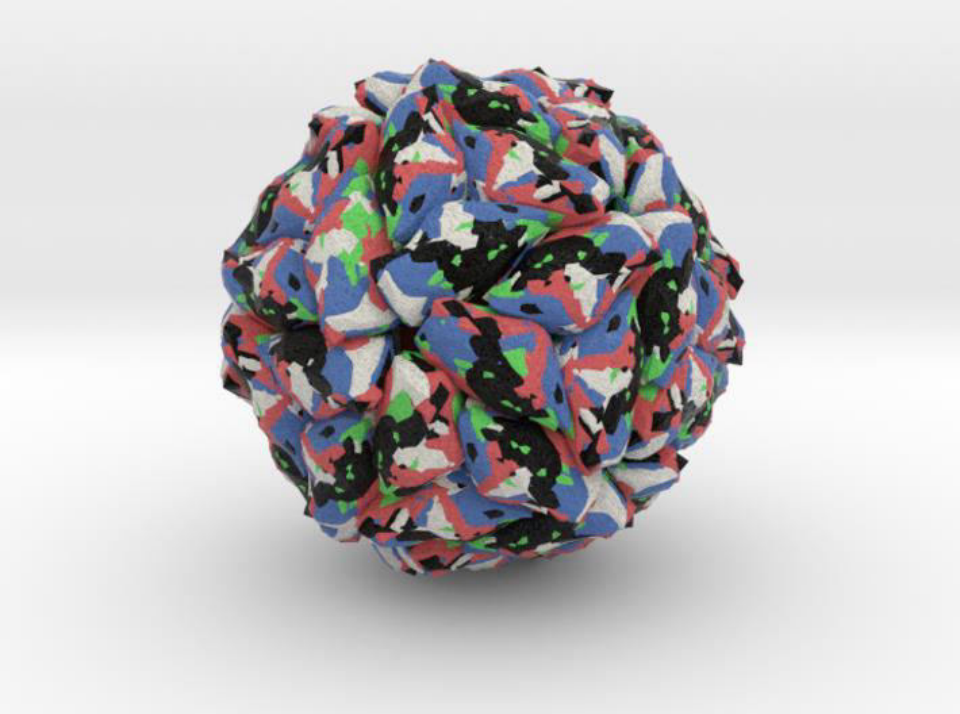
Satellites

Satellite

- Require helper viruses for replication
- Satellites do encode protein
- Satellites range between 475-1700 bases

Satellites

Satellite



- Present in plants, mammals, arthropods
- Symbiotic relation with helper virus
- 0.22- 1.5kbp genomic size

Satellites

Classification

- Single stranded RNA satellite virus
- Double stranded DNA satellite virus
- Single stranded satellite DNAs
- Double stranded satellite RNAs

Satellites

Applications

- Application of mild strains to prevent plant viral diseases
- Transgenic plant production
- To study cell to cell movements in plants

Satellites

Advantages To Study Satellites

- Ease in manipulation
- High in-vivo stability
- High expression level

END

Virology

Satellites

Examples

Satellites

Necrosis and Systemic Chlorosis

- Reduced chlorophyll production
- Pale, yellow or yellowish white leaves

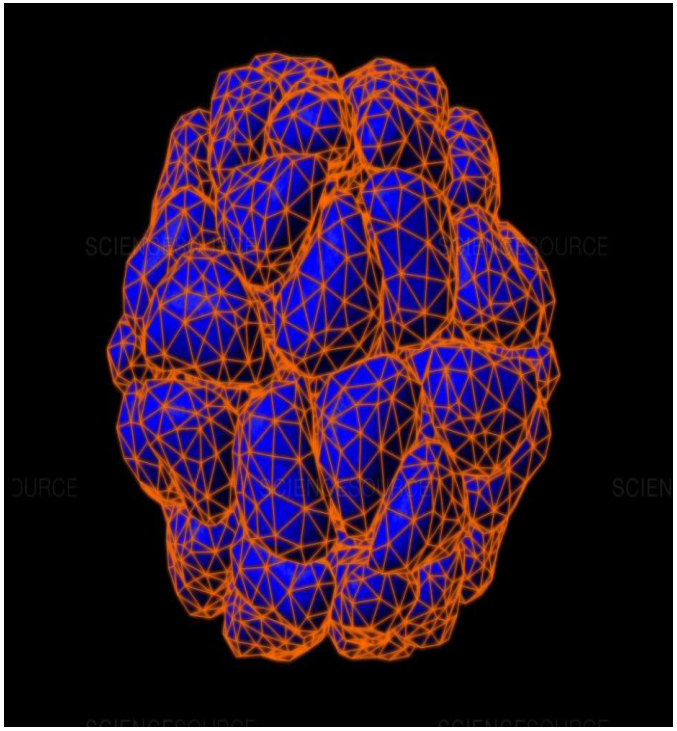
Satellites

Hepatitis Delta Virus

- Associated with human helper virus, hepatitis B virus
- 1.7 kb genome
- Circular single stranded RNA
- Replicated by cellular RNA polymerase

Satellites

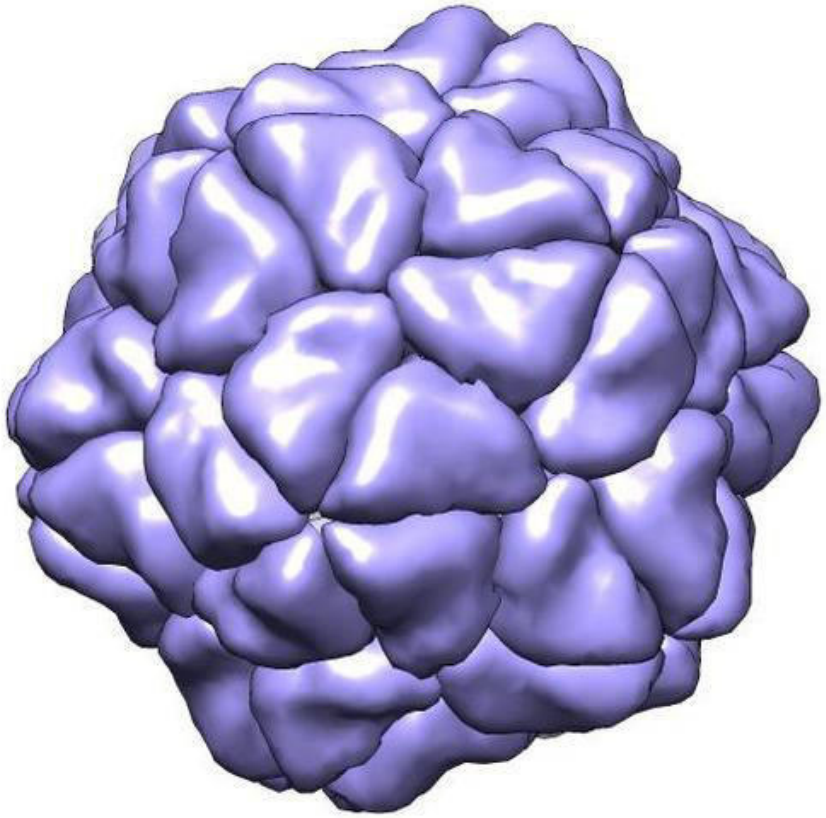
Tobacco Necrosis Virus



- First identified satellite
- 18nm isomeric particles
- Linear positive single stranded RNA genome
- Icosahederal
- 60 identical copies of identical proteins

Satellites

Satellite Tobacco Mosaic VIRUS



- 60 identical copies of identical proteins
- 1063 nucleotides
- Single stranded RNA genome

Satellites

Satellite Panicum Mosaic Virus

- Only affects grasses
- Icosahederal virus
- Eight stranded coat protein
- Induces mild mottle in millet plant

Virology

VIROIDS

Properties

Viroids

Properties

- The smallest infectious pathogens known so far
- They are composed solely of a short strand of circular, single-stranded RNA

Viroids

Properties

- RNA that has no protein coating
- All known viroids are inhabitants of plants in which they are causing diseases

Viroids

Nomenclature

- Family Pospiviroidae
- Genus *Pospiviroid* species: *Potato spindle tuber viroid* 361 nucleotides
- Genus *Pospiviroid* another species *Citrus exocortis viroid* 368–467 nt
- Genus *Hostuviroid*

  species: *Hop stunt viroid*, 294– 303 nt

Viroids

END

Nomenclature

- Family Pospiviroidae
- Genus *Cocadviroid* species: *Coconut cadang-cadang viroid* 246–247 nt
- Genus *Apscaviroid* type species *Apple scar skin viroid* 329–334 nt
- Genus *Coleviroid* type species: *Coleus blumei viroid*

Virology

VIROIDS

Structure

Viroids

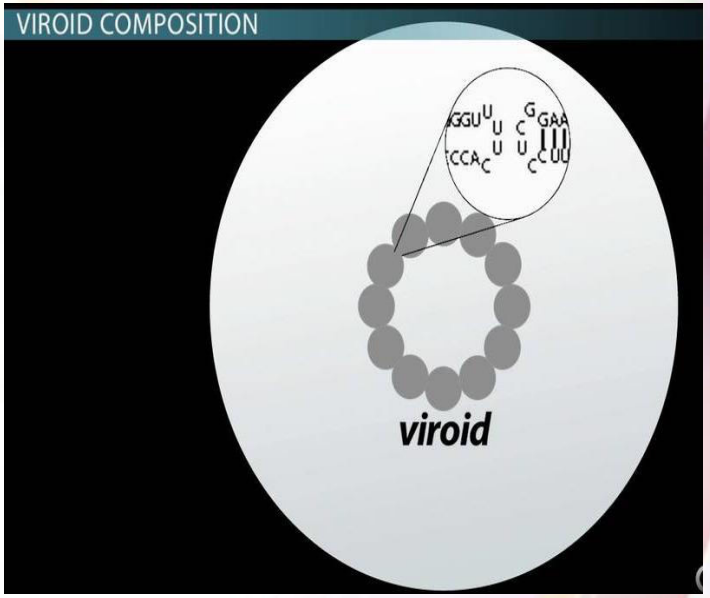
Structure

- Low molecular weight
- Unique structure
- Cause of disease

Satellites

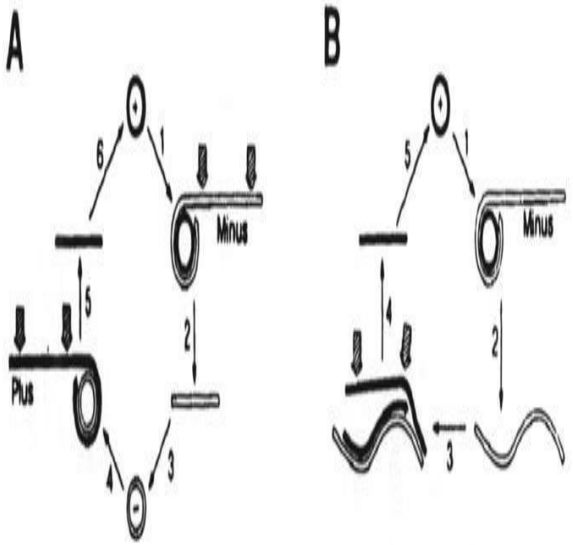
Structure

- Not encapsidated
- Replicate autonomously



Satellites

Genome



- Single stranded
- Covalently closed, circular
- As well as, linear RNA molecules
- Intramolecular complimentarity

Satellites

- Do not manufacture any proteins
- Infectious
- Highly structured
- Produce single specific RNA molecule

Satellites

Replication

- Rolling circle mechanism in-vivo
- Of two types
- Non-hydrolytic reaction

END

Virology

VIROIDS

Transmission

Viroids

Transmission

- Mechanical means
- Contaminated machinery, tools, pruning and mechanic agriculture techniques

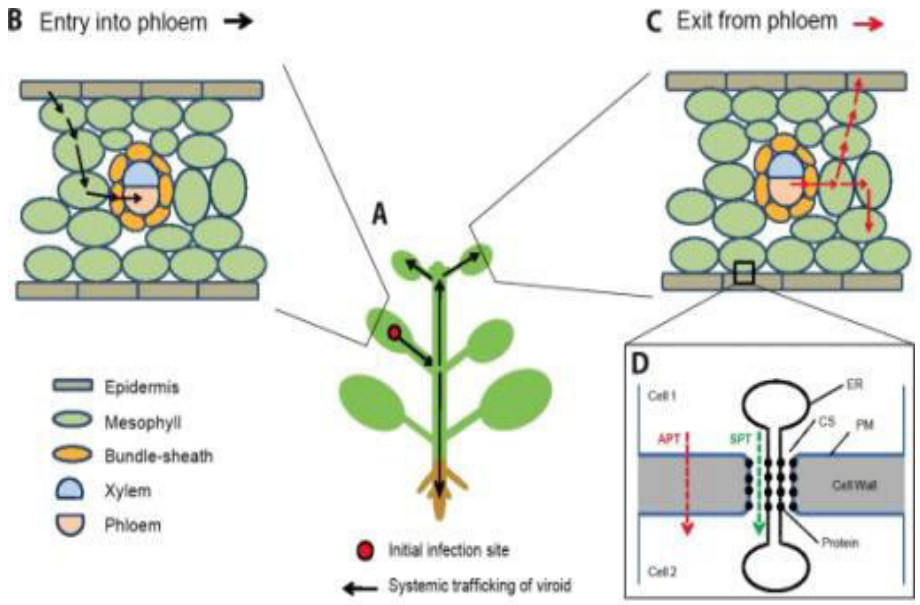
Viroids

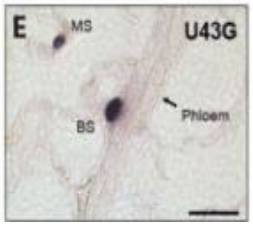
Transmission

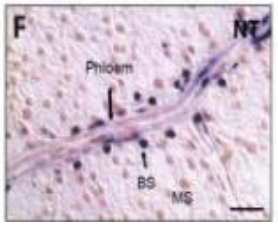
- Transmitted sometimes through seeds and pollens
- No important natural vectors

Viroids

Through plants







- Viroids enter pollen and ovule
- Transmitted through seed
- When seed germinates, the new plant gets infected

Viroids

- Transmitted through aphids
- Mechanical damage
- Leaf to leaf contact

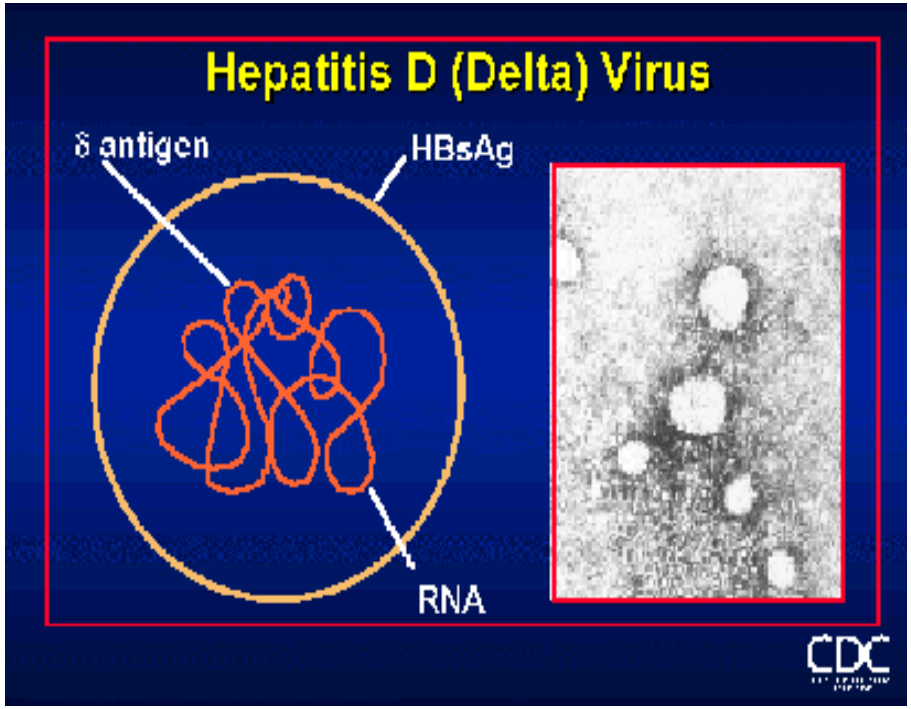
Viroids

- Often spread by vegetative propagation
- Agriculture and horticulture practices
- Both natural and human driven transmission

Viroids

- Both horizontal and vertical spread
- Horizontal: parasitic fungi, nematodes, insects
- Vertical: seed transmission,

Viroids



- Hepatitis D viroid enter human liver cells, if binding protein is present
- Obtains it from hepatitis B virus
- Delta virus enters blood stream and spreads by blood transfusion

Virology

VIROIDS

Examples

Viroids

Cucumber Green mottle virus

- Cucumber, melons and watermelons
- Affects young leaves with green and light green spots
- Veins remain green
- With heavy infection, leaves are deformed
- Small fruits1

Viroids

Carnation mottle virus

- Carnation flower plants
- Mild mottling of young leaves
- Faint chlorosis
- Yellow ring spots in heavy infection
- Vein chlorosis 8 to 10 days post infection

Viroids

Tomato bushy stunt virus



- Economically significant plant pathogen
- Stunted growth, leaf mottling, deformed or absent fruits
- Broad host range
- Necrotic or chlorotic lesions

Viroids

Potato Spindle Tuber Viroid

- Small circular RNA molecule
- Natural host: potato and tomato
- Transmission by seeds and aphids
- Necrosis of leaves and veins

Virology

PRIONS

Structure

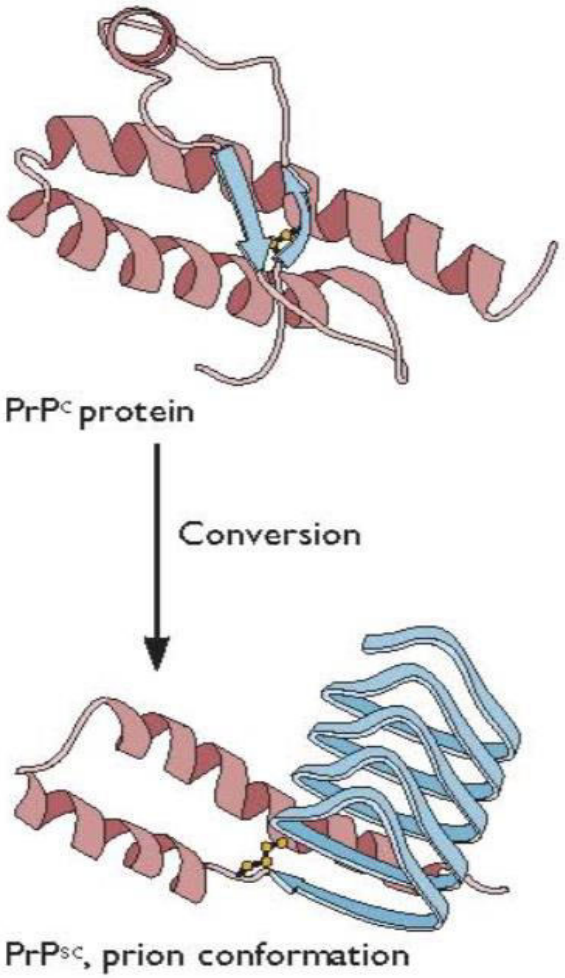
Prions

Normal proteins

- 200 to 250 amino acids
- Twisted into three telephone cord-like coils
- Named as HELICES
- With tails of more amino acids

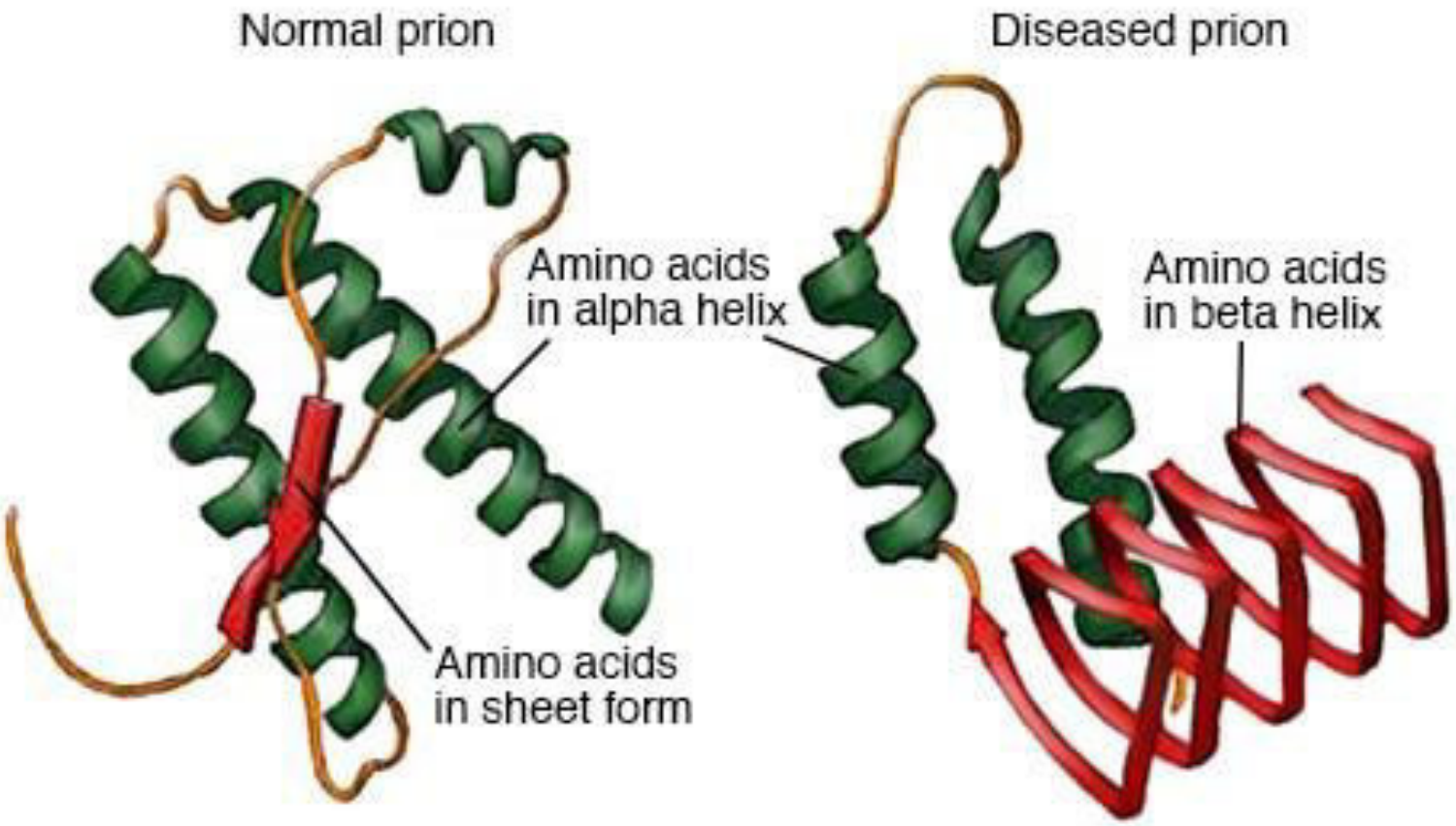
Prions

Mutated proteins



- Built from same amino acids
- But take different shape
- 100 times smaller than smallest virus

Prions


Normal prion
Amino acids
in alpha helix
Amino acids
in sheet form
Diseased prion
Amino acids
in beta helix

Prions

Difference from Bacteria and Viruses

- Prions don't contain nucleic acids
- Resistant to heat and chemicals
- Difficult to decompose biologically
- Survive in soil for many years

Prions

Structure

- Misfolded/ abnormally folded protein
- Pathogenic
- Causes similar proteins to misfold also

Prions

Replication

- Unique
- Replicate on their own and become infectious
- Occur in two forms: PrP-sen and PrP-res

END

Viroids

Chrysanthemum stunt viroids



- Affects chrysanthemum and pinks
- Light colored flowers
- Leaves reduce in number and size
- Wavy or crumpled leaf surface

Viroids

Hepatitis D viroids

- Only human disease
- Simultaneous infection with hepatitis B virus
- Causes liver cell death
- Sequestering 7S RNA and/or cleaving it

END

Virology

PRIONS

Genome

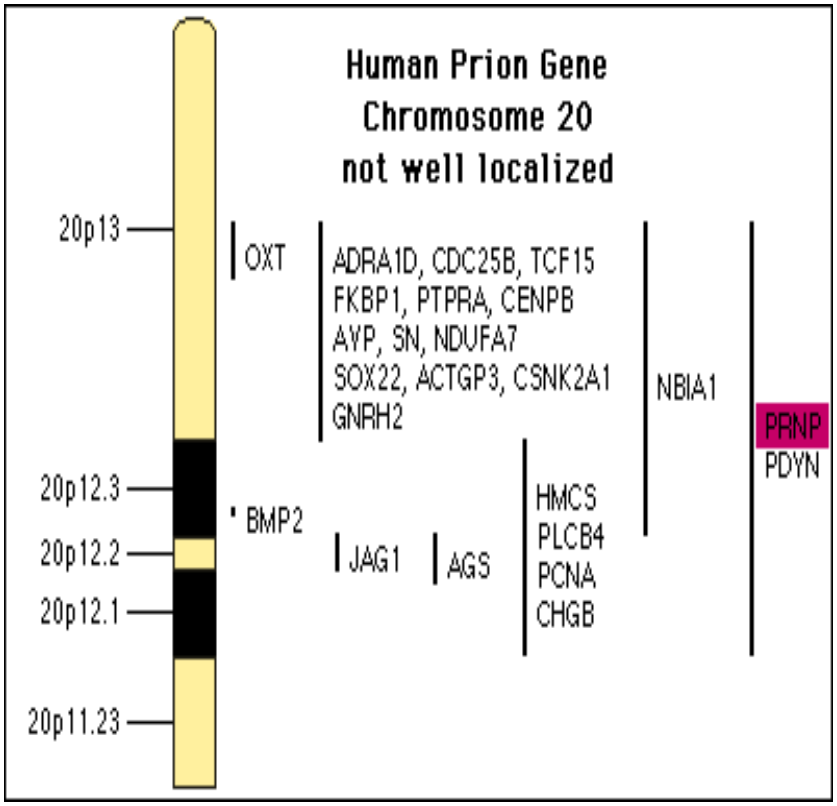
Prions

Genome

- Genome is absent
- No DNA
- No RNA

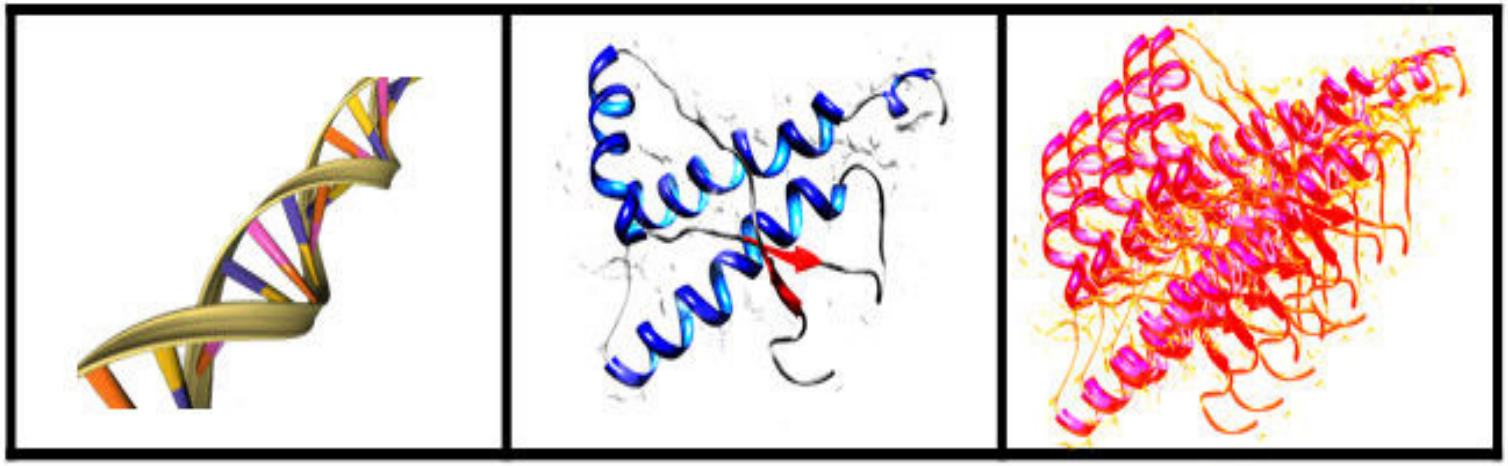
Prions

Gene



- single exon of a single-copy chromosomal gene
- PrP gene in mammals contain one or two 5'-non-coding exon

Prions



PRNP is a gene in your DNA which encodes for prion protein

Prion protein or **PrP** is a protein on the surface of your cells

A **prion** is an infectious particle made up of misfolded prion proteins

Prions

Protein

- Itself a protein
- Misfolded
- That makes other alike proteins misfolded
- causes disease condition in host

Prions

Structure

- Misfolded/ abnormally folded protein
- Pathogenic
- Causes similar proteins to misfold too

END

Virology

PRIONS

Pathology of Disease

PRIONS: Pathology of Disease

- TSE are caused by abnormal folding of prions
- The normal prion protein, which is designated as PrPc, is a 35kD membrane glycoprotein, which is water-soluble and proteinase-sensitive.
- Abnormal prions, designated as PrPSc or PrPTSE

PRIONS: Pathology of Disease

Pathogenesis Of Prion Diseases

- The unique feature of prion diseases is that they are self-propagating and transmissible.
- Once PrPSc is generated endogenously or introduced into the body from the environment, it converts normal prions into abnormal ones.
- This conversion begins with the initial production of a small polymer of misfolded prions, (a seed), perhaps no more than 28 molecules.

PRIONS: Pathology of Disease

Pathogenesis Of Prion Diseases

- This seed converts normal adjacent prions into abnormal ones by an unknown mechanism.
- As more PrPSc polymers are produced, they, in turn, act as seeds, propagating the conversion of normal to abnormal prions.
- This novel mechanism of disease is also implicated in the propagation of protein misfolding that is involved in Alzheimer's disease, Parkinson's disease, and amyotrophic lateral sclerosis.

PRIONS: Pathology of Disease

Pathology Of Prion Diseases

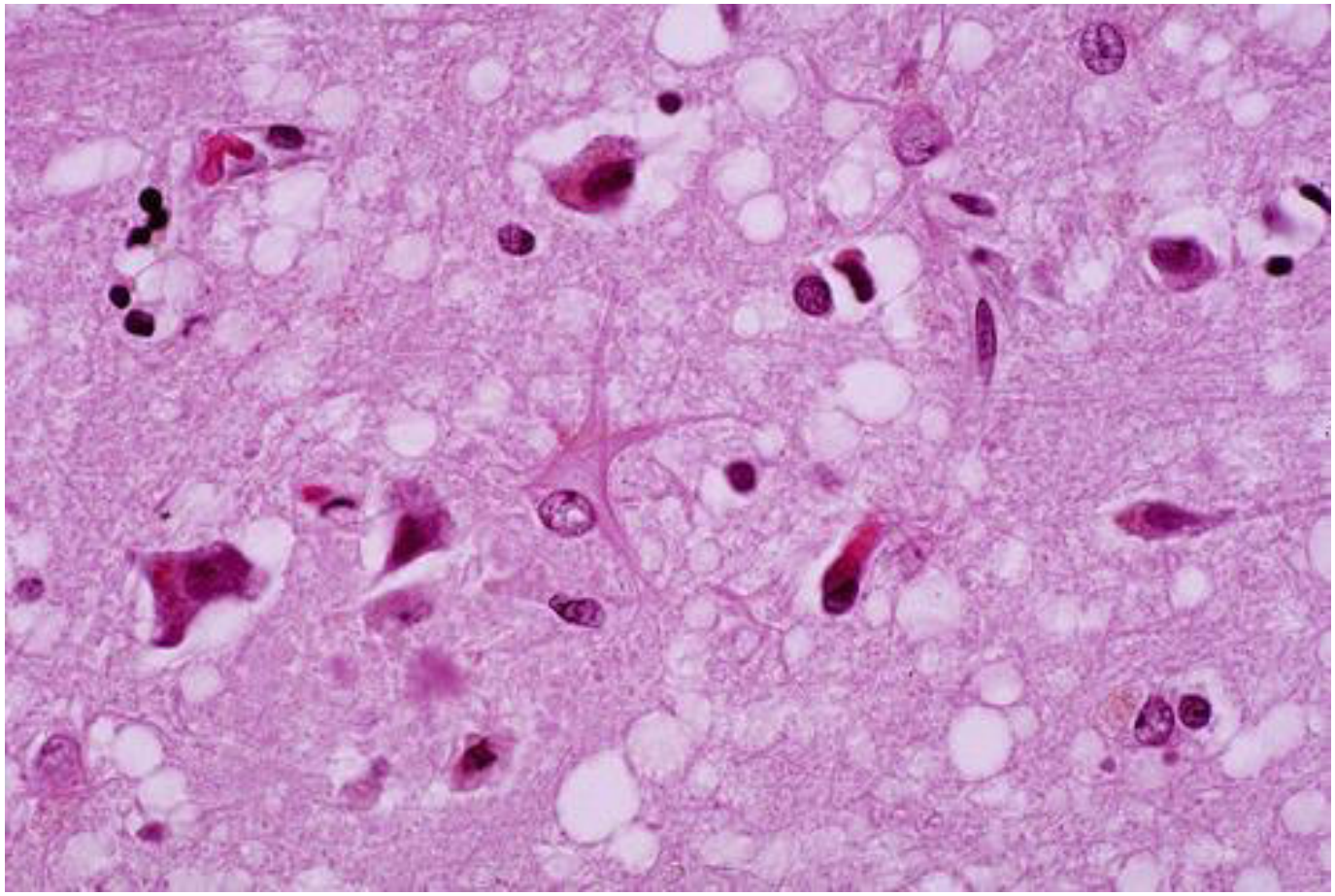
- During the incubation period and active clinical phase of prion diseases, PrPTSE is probably present (and can transmit the disease) in all tissues and body fluids, especially brain, nerves, and skeletal muscle.
- Pathology develops only in the brain.
- Initially, intracytoplasmic vacuoles appear in neurons.
- As the disease progresses, vacuolization becomes more pronounced and the cortical neuropil develops a spongy appearance, hence the term spongiform encephalopathy.

PRIONS: Pathology of Disease

Pathology Of Prion Diseases

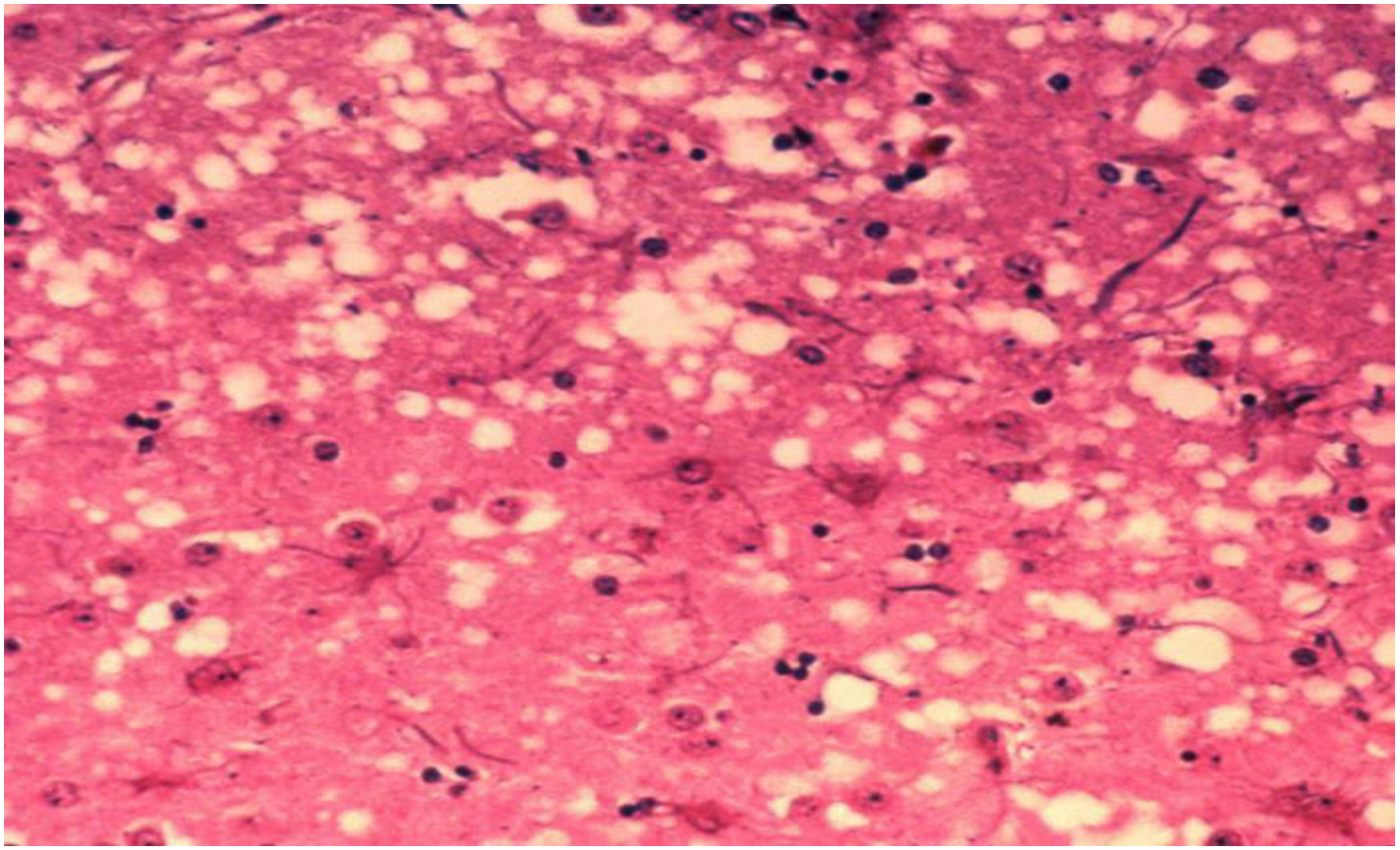
- Advanced cases show nerronal loss, gliosis, and brain atrophy.
- Cerebellar atrophy is usually severe.
- Unlike most cerebellar degenerations, there is more pronounced loss of granular neurons
- There is no inflammation. The CSF is normal.
- The changes are confined to the gray matter (the primary pathology involves the neuronal body).

PRIONS: Pathology of Disease



Spongyform encephalophathy

PRIONS: Pathology of Disease



Virology

TSE BY PRIONS

Scrapies

Prions

Etiology

- Neurodegenerative disease caused by prions
- Replicate by converting normal proteins into copies of prions
- non-conventional disease agents

Prions

Host species



- Sheep
- Goat
- Mice
- Rat
- Monkey
- And other Lab animals

Prions

Geographical distribution



- World wide
- More in developing countries

Prions

Transmission

- Well understood in sheep
- To goats if kept with sheep
- Within herd
- During pregnancy
- Through urine

Prions



Prions

Symptoms

- Affected sheep will apart from flock
- Trembling
- Convulsions
- Scraping body with wall
- Drink small quantities of water often
- Blindness

END

Virology

TSE BY PRIONS

Transmissible
Mink
Encephalopathy

Prions

Etiology

- Caused by prions
- Progressive and fatal neurodegenerative disease

Prions

Host species



- Minks
- Once signs are evident, death is inevitable
- Raccoons
- Skunks
- Ferrets
- Hamsters etc

Prions

Geographical distribution



- In USA from 1947 to 1985
- Canada
- Finland
- Germany

Prions

Transmission

- Orally
- Wounds on tongue
- Mucous membrane
- Skin

Prions



Prions

Symptoms

- Difficulty in eating and swallowing
- Hyperexcitable
- Incoordination
- Clenching of jaw
- Covulsions
- Death

END

Virology

TSE BY PRIONS

Feline Spongiform Encephalopathy

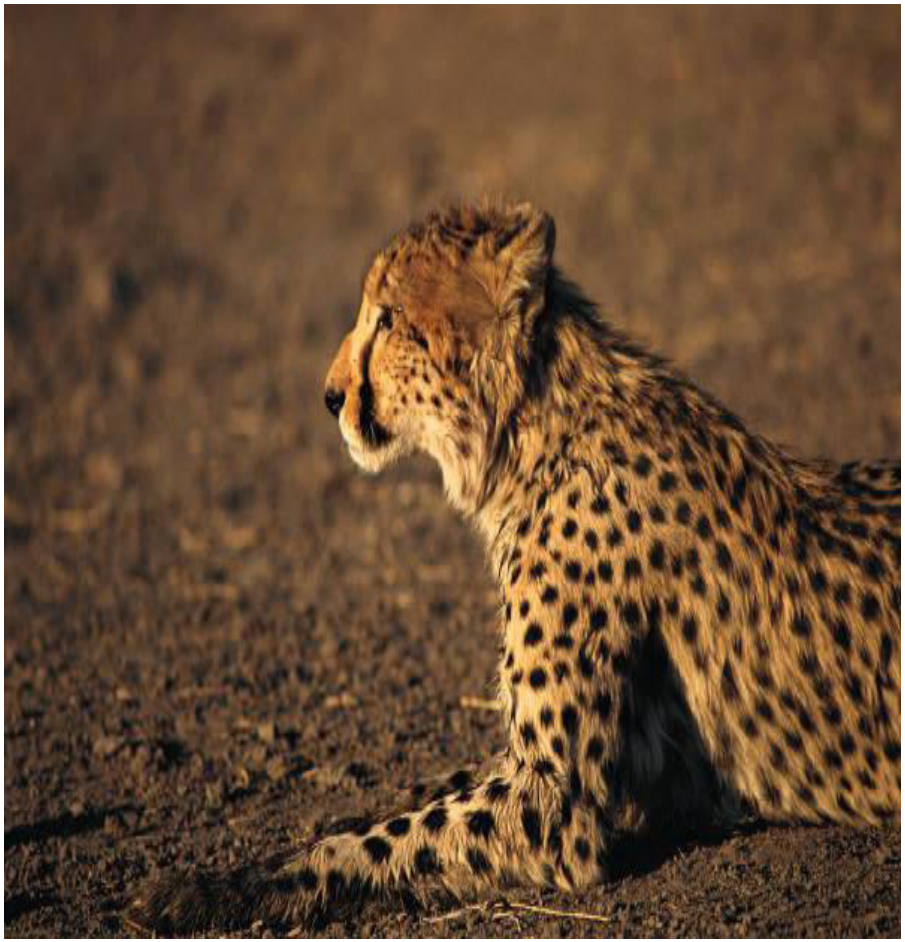
Prions

Etiology

- Caused by prions
- Neurodegenerative disease
- Same agent responsible for Bovine Spongiform encephalopathy

Prions

Host species



- Domesticated cats
- Cheetahs
- Puma
- Tiger
- Golden cat
- Lion
- Leopard

Prions

Geographical distribution



- Most cases in UK
- Australia
- North America

Prions

Transmission

- Eating BSE contaminated tissue
- No horizontal transmission
- Possible maternal transmission
- Iatrogenic transmission in cheetah

Prions



Prions

Symptoms

- Aggression
- Hiding
- Gait abnormalities
- Hyperesthesia
- Excessive salivation
- Tremors
- Death

END

Virology

TSE BY PRIONS

Chronic Wasting Disease

Chronic Wasting Disease

Etiology

- Caused by prions
- Contagious neurological disease
- Causes a characteristic spongy degeneration of the brains of infected animals resulting in emaciation, abnormal behavior, loss of bodily functions and death.

Chronic Wasting Disease



Host

- Mule deer
- white-tailed deer
- elk
- moose
- reindeer

Chronic Wasting Disease

Signs & symptoms

- Difficulties in movement
- Weight loss over time
- Listlessness
- Lowering of head
- Excessive salivation
- Nervousness
- tremors

Chronic Wasting Disease

Transmission

- Contact with infected animals
- Their body tissues
- Body fluids

END

Virology

TSE BY PRIONS

Bovine Spongioform Encephalopathy

Bovine Spongioform Encephalopathy

Etiology:

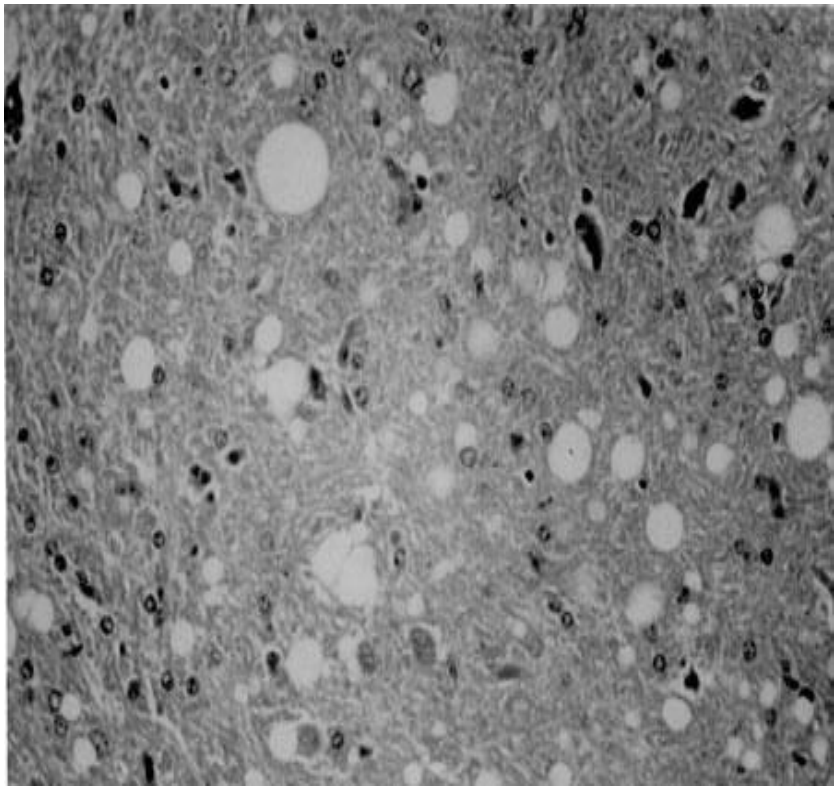
- Caused by prions
- Affects nervous system
- Also called **mad cow disease**

Bovine Spongioform Encephalopathy

HOST



Bovine Spongioform Encephalopathy



BSE shows spongy appearance of the brain tissue of infected cattle when sections are examined under a microscope.

Bovine Spongioform Encephalopathy

Signs:

- Apprehension
- Hyperaesthesia
- Frequent licking with progressive paresis, & ataxia
- Circling

Bovine Spongioform Encephalopathy

Transmission

- from the carcasses of scrapie-infected sheep.
- sheep by products infected with scrapie is used to feed cattle with the meat and bone mill (MBM)

END

Virology

TSE BY PRIONS

In Human

In Human

Classic CJD (Creutzfeldt-Jakob disease)

- Fatal Familial Insomnia
- Kuru

In Human

Classic CJD (Creutzfeldt-Jakob disease): disease):

- Most prevalent of spongiform diseases
- 1 out of million people
- 10% cases are inherited mutations

  in PRPN gene
- Strikes peopleage 50-75.

In Human

Symptoms:

dementia, muscle twitching, vision problems

Diagnosis:

- Post mortem definitive diagnosis
- Amyloid plaques surrounded by vacuoles
- Prion protein accumulation in cerebellum
- Spongiform appearance in gray matter

In Human

Fatal familial

- All case are inherited mutations in Prp gene
- Strikes people age 36-61
- Disruption of sleep/wake cycle leads to coma, then death.

In Human

Kuru:

Struck members of Fore tribe in 1950s and 1960s

In Human

Signs:

- Muscle weakness
- loss of coordination
- tremors
- inappropriate episodes of laughter or crying

Transmission:

Transmitted by ritual cannibalism as part of funeral ceremonies.

END

Virology

MOLECULAR BIOLOGY OF PRIONS

Molecular Biology of Prions

Prion protein:

- Abnormal isoform of normal protein
- Located in cell surface (brain, nervous system)
- Protects cell against oxidative stress (free radicals lacking electrons)

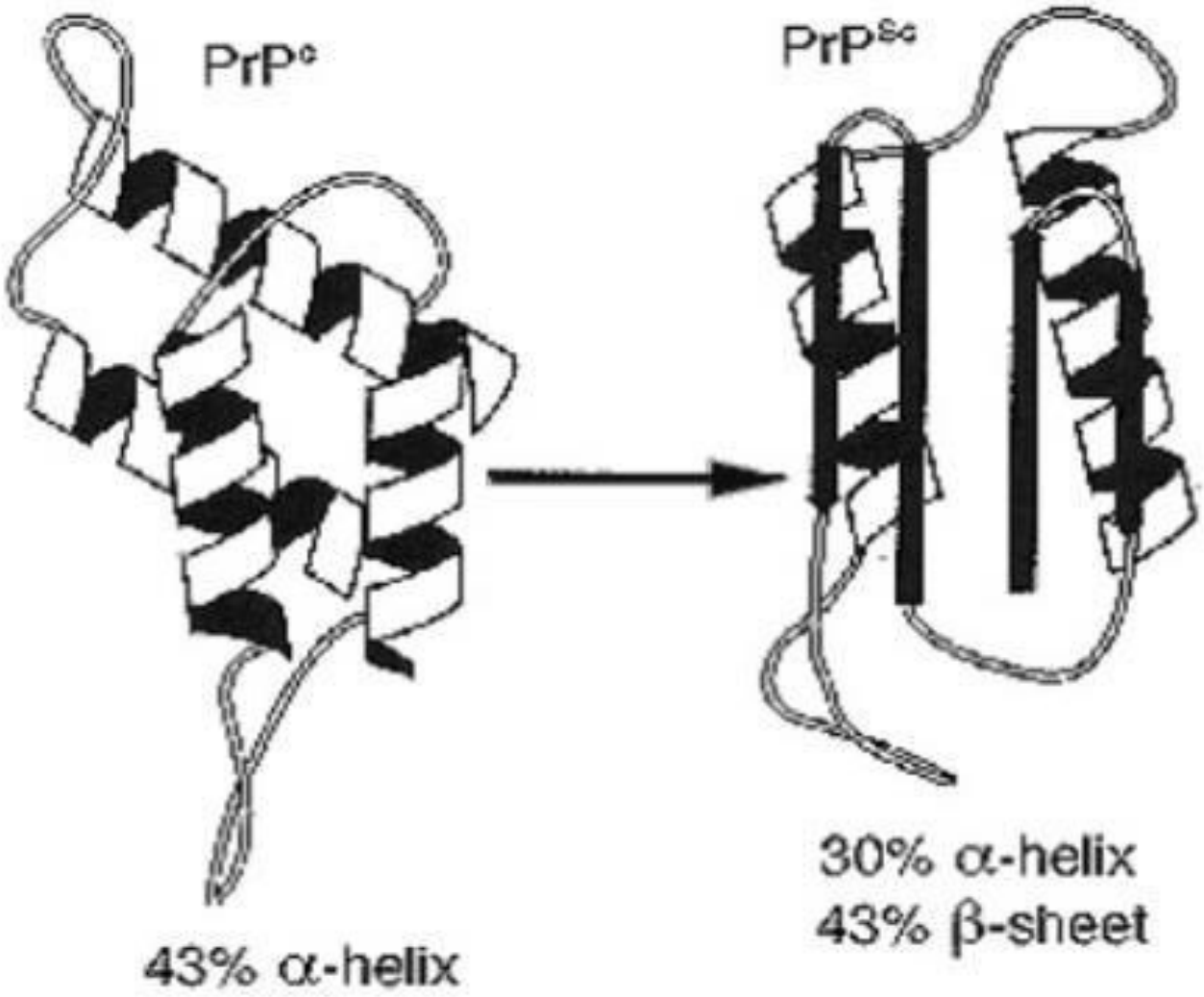
Molecular Biology of Prions

- PrP^{c} =prion protein cellular(normal form)
- PrP^{sc} =scrapie form (infectious form)

Molecular Biology of Prions

PrP^{c}	PrP^{Sc}
Rich in alpha helices	Rich in beta sheets
Soluble	Insoluble
PK (protease) sensitive • Definitive shelf life • Degrades when no longer needed	PK resistant
No aggregation • Cannot polymerize into aggregates	Aggregation • Allow polymerization of fibrils • Due to beta sheets

Molecular Biology of Prions



Virology

DEVELOPMENT OF CULTURING METHOD FOR VIRUSES

Developing Culture Method

Introduction

- First cell line developed in 1912 by Carrel and his coworkers
- From fragments of explanted chicken embryo heart
- Mantained by washing with Ringer solution

Developing Culture Method

Introduction

- Developed in 1940s and 1950s
- By John Franklin Enders, Thomas Huckle Weller and Frederick Chapman Robbins

Developing Culture Method

First culture

- Received Nobel Prize
- Culturing viruses in Monkey kidney cells

Developing Culture Method



Media

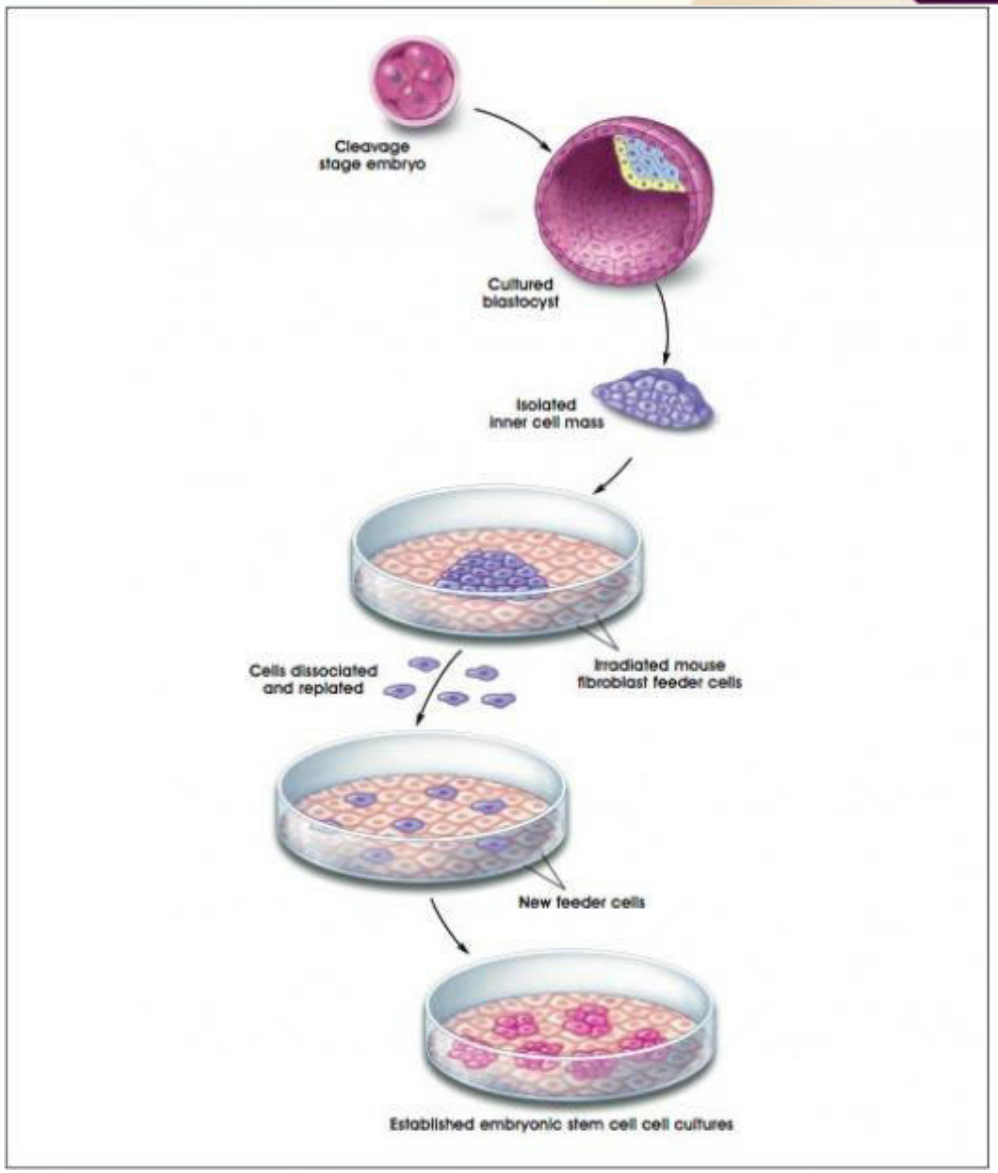
- Early use was naturally derived products from serum
- Later use of chemical-based synthetic media

Developing Culture Method

Observation

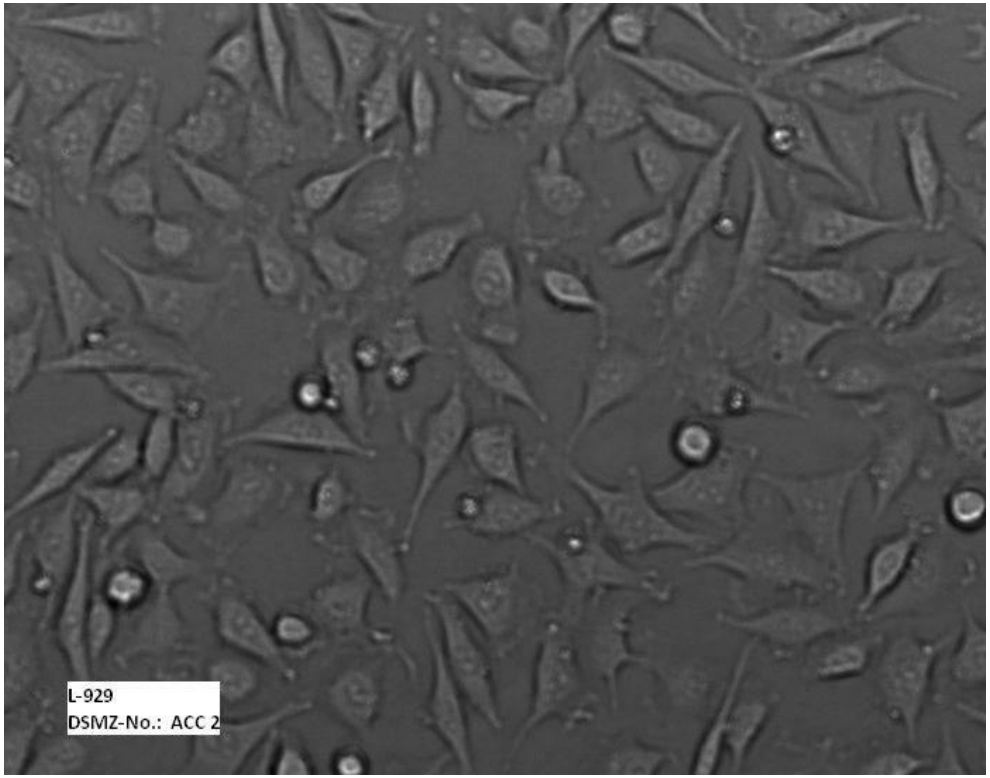
- Using modern microscopy techniques
- Possibility to observe inclusions in cells as a result of virus

Developing Culture Method



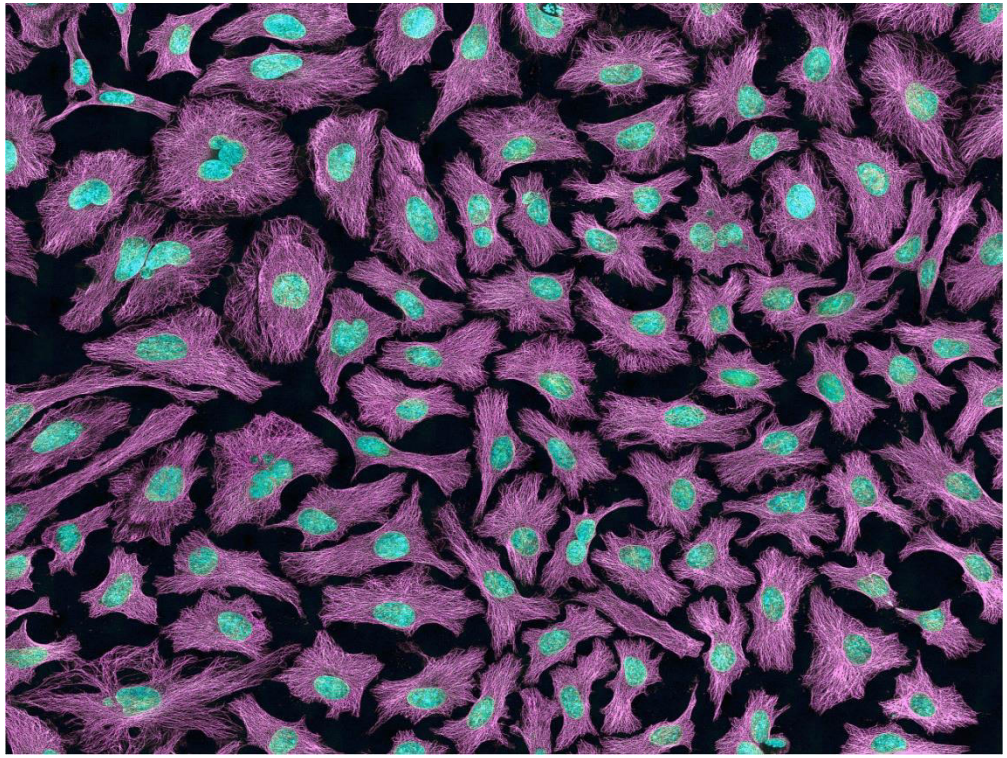
Developing Culture Method

L- cell line



- Developed by Earle in 1948
- From subcutaneous mouse tissue

Developing Culture Method



HeLa cell line

- In 50s and 60s
- Diploid cell lines
- From human tissue

Developing Culture Method

Vero cell line

- In 50s and 60s
- Diploid cell lines
- From Simian tissue

Developing Culture Method

Earliest derived cell lines

Name	Species and tissues	Morphology	Year of origin
L929	Mouse connective tissue	Fibroblast	1948
HeLa	Human cervix	Epithelial-like	1951
CHO	Chinese Hamster Ovary	Epithelial	1957
MDCK	Canine kidney	Epithelial	1958
WI-38	Human lung	Fibroblast	1961
BHK-21	Syrian Hamster Kidney	Fibroblast	1961
Vero	African Green monkey kidney	Epithelial	1962

Virology

INTRODUCTON OF CELL CULTURE

Cell Culture

introduction

- Cell culture is the process by which prokaryotic, eukaryotic or plant cells are grown under controlled conditions. But in practice it refers to the culturing of cells derived from animal cells.

Cell Culture

- **Major development's in cell culture technology**
- First development was the use of antibiotics which inhibits the growth of contaminants.
- Second was the use of trypsin to remove adherent cells to subculture further from the culture vessel
- Third was the use of chemically defined culture medium.

Cell Culture

importance

- Virology

  Cultivation of virus for vaccine production, also used to study there infectious cycle.

- Genetic Engineering

  Production of commercial proteins, large scale production of viruses for use in vaccine production e.g. polio, rabies, chicken pox, hepatitis B & measles

Cell Culture

importance

- Gene therapy

  Cells having a functional gene can be replaced to cells which are having non-functional gene

END

Virology

SLIDE CULTURE

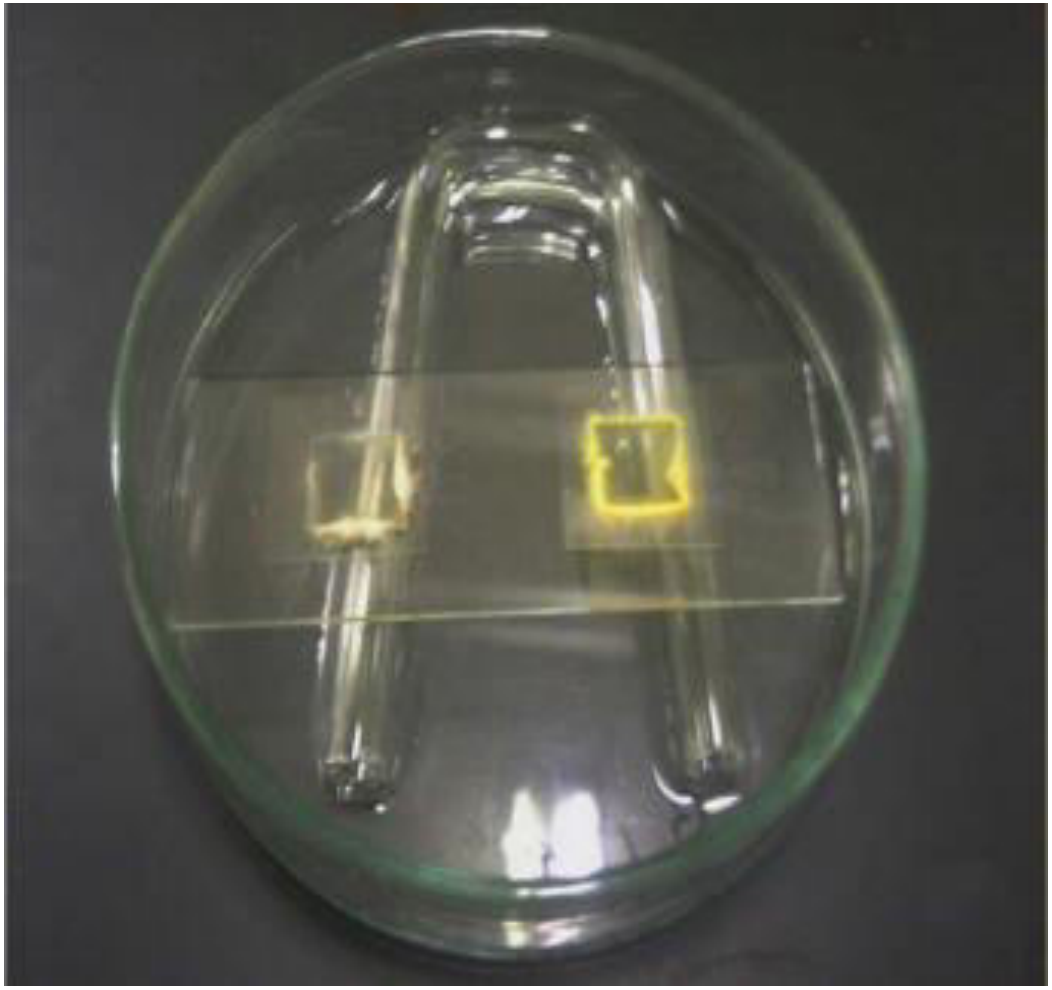
Slide Culture

- Rapid **method** of preparing fungal colonies for examination and identification.
- Permits the fungi to be studied virtually insitu with as little disturbance as possible.

Slide Culture

Procedure:

- A) Slide Culture Preparation



Slide Culture

B) Application of Stain

- Place a drop of lactophenol cotton blue stain on a clean microscope slide.
- Remove the cover glass from the slide culture and discard the block of agar.
- Add a drop of 95% ethanol to the hyphae on the cover glass.

Slide Culture

B) Application of Stain

- when alcohol has evaporated place the co ver glass, mold sidedo wn, on the drop of l actophenol cotton blue stain on the slide.
- Examine the slide under microscope

END

Virology

Flask Culture

Flask Culture



Laboratory flask

Laboratory flasks are vessels or containers that fall into the category of laboratory equipment known as glassware.

Flask Culture



Erlenmeyer flask

- An Erlenmeyer flask, also known as a conical flask or titration flask, is a type of laboratory flask which features a flat bottom, a conical body

Flask Culture



Round-bottom flask

Round-bottom flasks (also called round-bottomed flasks or RB flasks) are types of flasks having spherical bottoms

Flask Culture



Fernbach flask

- A Fernbach flask is a type of flask suited for large volume cell culture where the culture requires a large surface area to volume ratio.

Virology

ROLLER TUBE CULTURES

Roller Tube Culture

INTRODUCTION

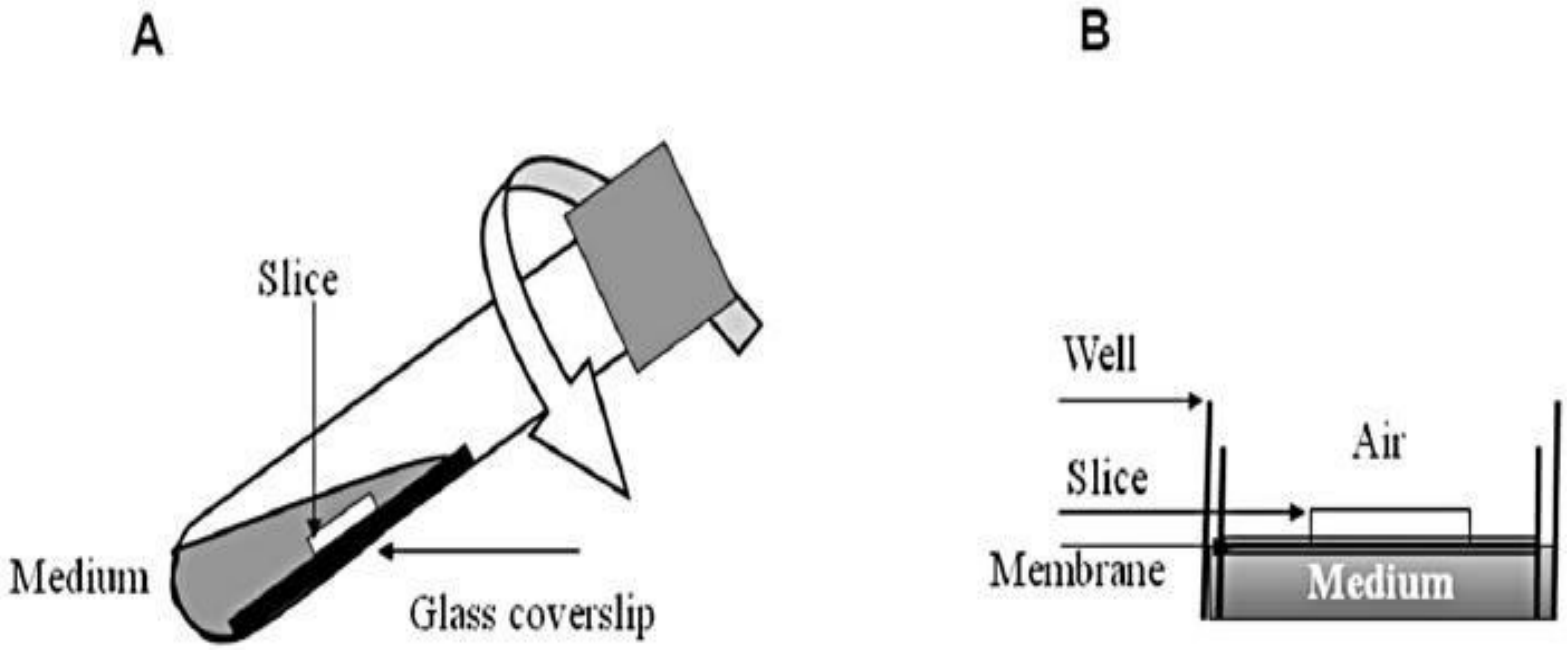
- Grown in test tubes
- Rotated during incubation

Roller Tube Culture

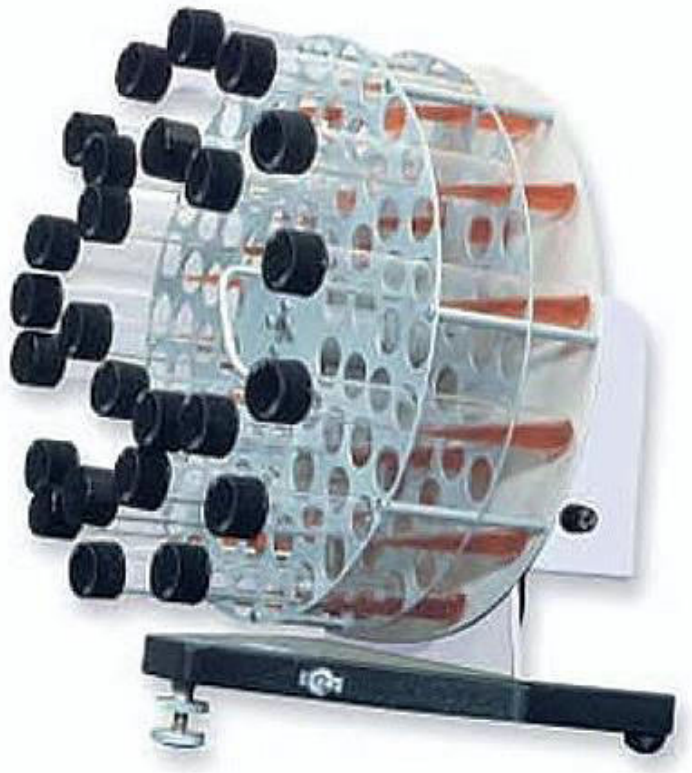
PROCEDURE

- Tissue explants placed in test tubes
- Small amount of plasma and agar distributed over the inside surface of tube, so fragments stay in place

Roller Tube Culture



Roller Tube Culture



PROCEDURE

- As the tube is rotated 8 to 12 rpm
- Tissue explants submerged in nutrient media, only a portion of time

Roller Tube Culture

PROCEDURE

- Some cells don't grow satisfactorily if entirely submerged
- Other cells, especially those cultivated through many transfers will stick to glass surface and grown when completely submerged

END

Virology

MONOLAYER TISSUE CULTURE

Monolayer tissue culture

https://www.youtube.com/watch?v=CMRKKl9XSDU

Virology

CYTOPATHIC EFFECTS ON TISSUE CULTURE

Cytopathic Effects

- **Definition:**

Cytopathic effect or **cytopathogenic effect CPE**) refers to structural changes in host cells that are caused by viral invasion.

Cytopathic Effects

- **Total destruction of the cell monolayer**
- most severe form of CPE. All cells in the monolayer rapidly shrink, become dense (pyknosis), and detach from the glass within 72 hours.
- typical of most enteroviruses.

Cytopathic Effects

- **Subtotal destruction**
- detachment (death) of some but not all of the cells in the monolayer.
- togaviruses (alphaviruses), picornaviruses,
- can be observed using the 20X objective

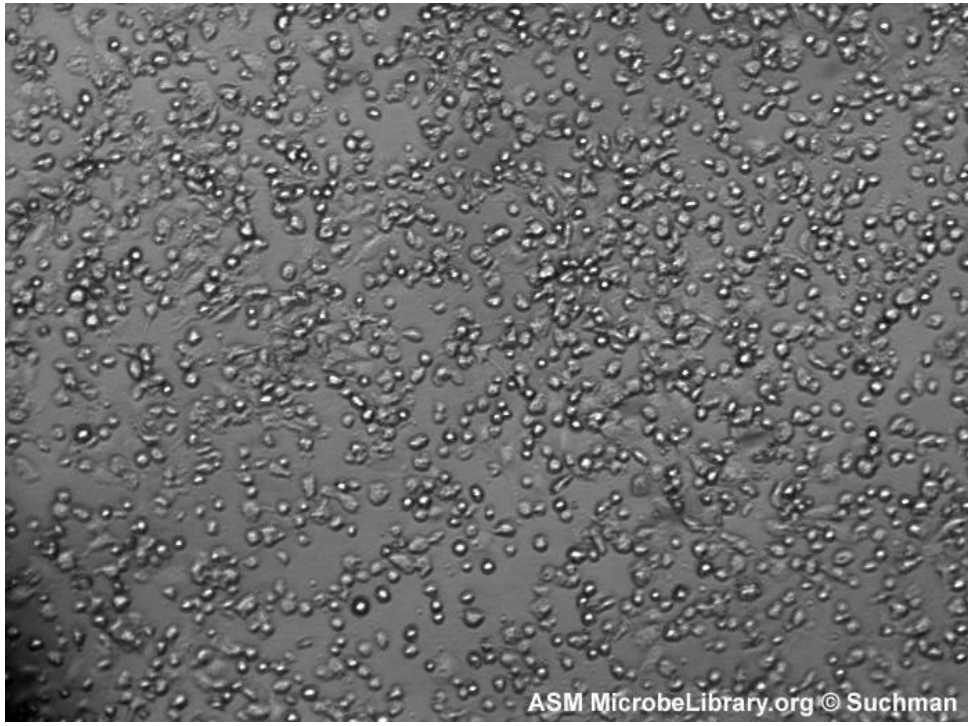
Cytopathic Effects

- **Focal degeneration**
- Herpesviruses and poxviruses.
- viruses produce localized areas (foci) of infection.
- The focal nature of these lesions is due to direct cell-to-cell transfer of virus rather than diffusion through the extracellular medium.

Cytopathic Effects

- **Focal degeneration**
- Cells initially become enlarged, rounded, and refractile (more easily seen),then eventually detach from the growth surface leaving cleared areas surrounded by rounded up cells as the infection spreads concentrically

Cytopathic Effects



- **Swelling and clumping**
- typical of adenoviruses Infected cells greatly enlarge and clump together in "grape-like" clusters.

Cytopathic Effects



- **Foamy degeneration (vacuolization)**
- production of large and/or numerous cytoplasmic vacuoles.
- retroviruses, paramyxoviruses, and flaviviruses (pestiviruses) may cause vacuolization.
- Vacuolization is difficult to visualize without staining.

Cytopathic Effects



- **Cell fusion (syncytium or polykaryon formation)**
- fusion of the plasma membranes of four or more cells to produce an enlarged cell with four or more nuclei.
- paramyxoviruses, herpesviruses may produce syncytia as well as other forms of CPE

Cytopathic Effects



- Inclusion bodies:

cytoplasmic stranding (arrow) and nuclear inclusion bodies (dashed arrow)

Virology

VIRUSES AND CANCER

Viruses and Cancer

Etiology of cancer:

- Viruses (papilloma,Epstein-Barr,Hepatitis B, retrovirus)
- Radiation
- Environmental/industrial carcinogens
- Immunodeficiency syndromes
- Genetic susceptibility syndromes

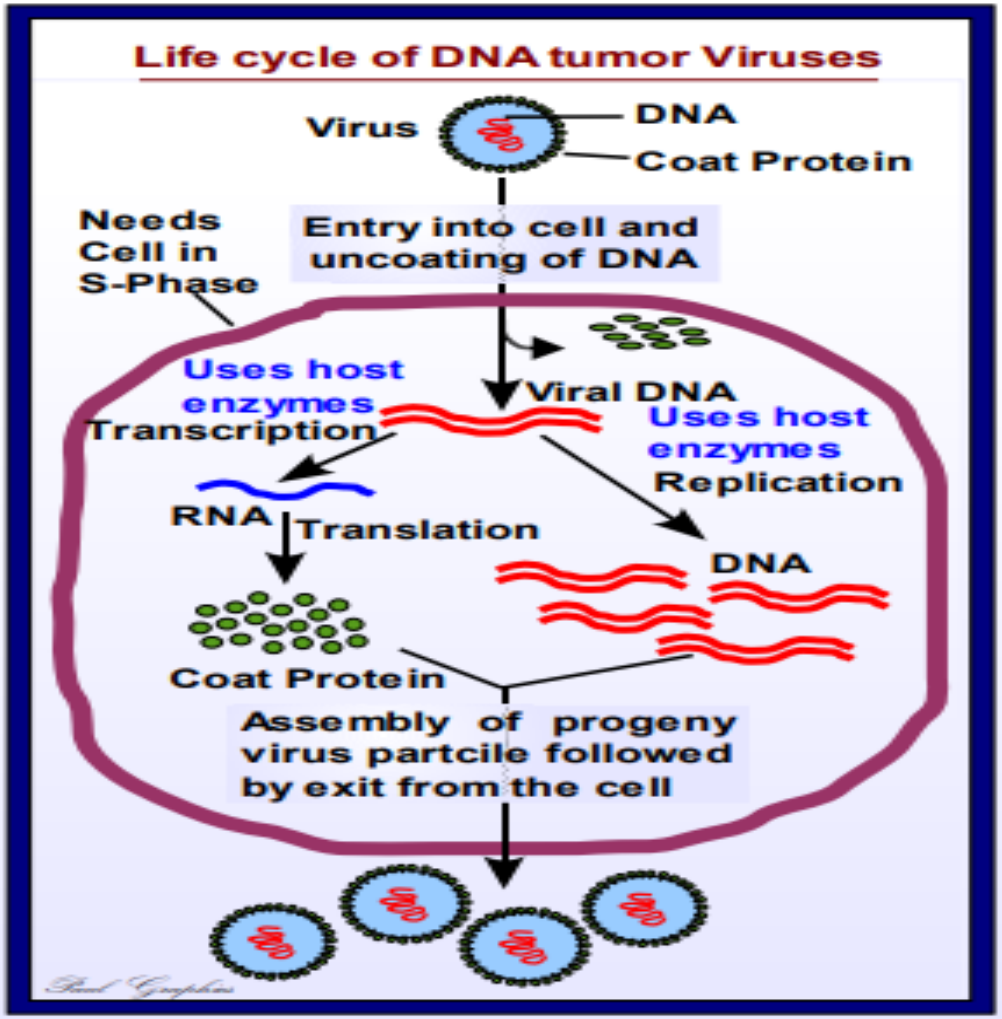
Viruses and Cancer

History:

- 1876: russian researcher reported transmission of a tumor from one dog to another
- 1908, **Wilhelm** and **Olaf Bang** in Copenhagen reported the extraact from chicken leukemia can cause tumor.
- In 1909 **Rous** found that extract from chicken sarcoma can cause tumor.

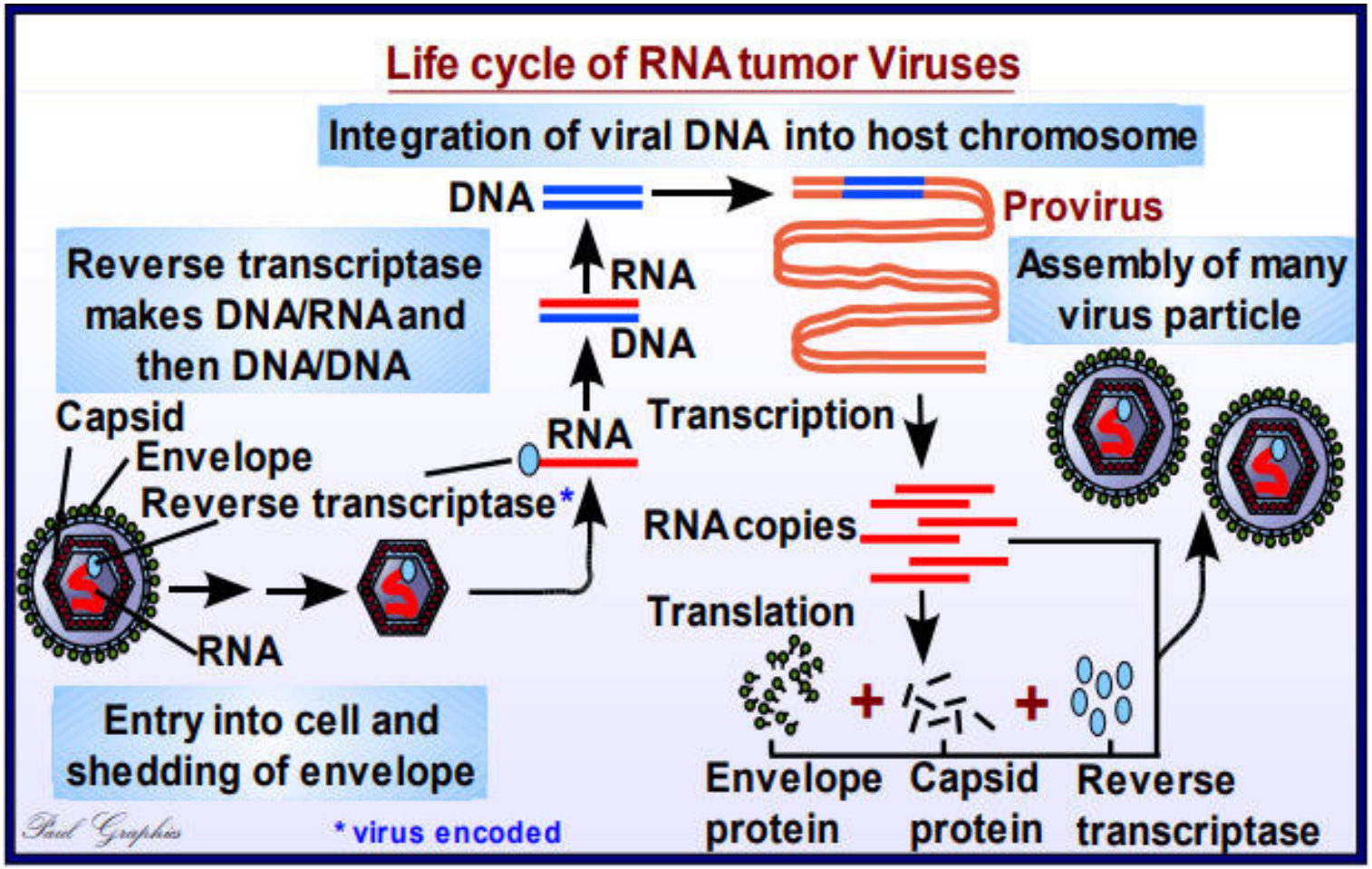
Viruses and Cancer

DNA tumor Viruses



Viruses and Cancer

RNA tumor Viruses



Virology

TRANSFORMATION OF NORMAL CELL TO CANCEROUS CELL

Normal to Cancerous Cell

- Oncogene – a gene that when mutated or expressed at abnormally high levels contributes to converting a normal cell into a cancer cell
- Proto-oncogene – the “normal” cellular progenitors of oncogenes that function to promote the normal growth and division of cells

Normal to Cancerous Cell

- **Proto-oncogene to oncogene:**
- An alteration occurs in a normal cellular gene (proto-oncogene) that makes the protein hyper-functional (oncogene)

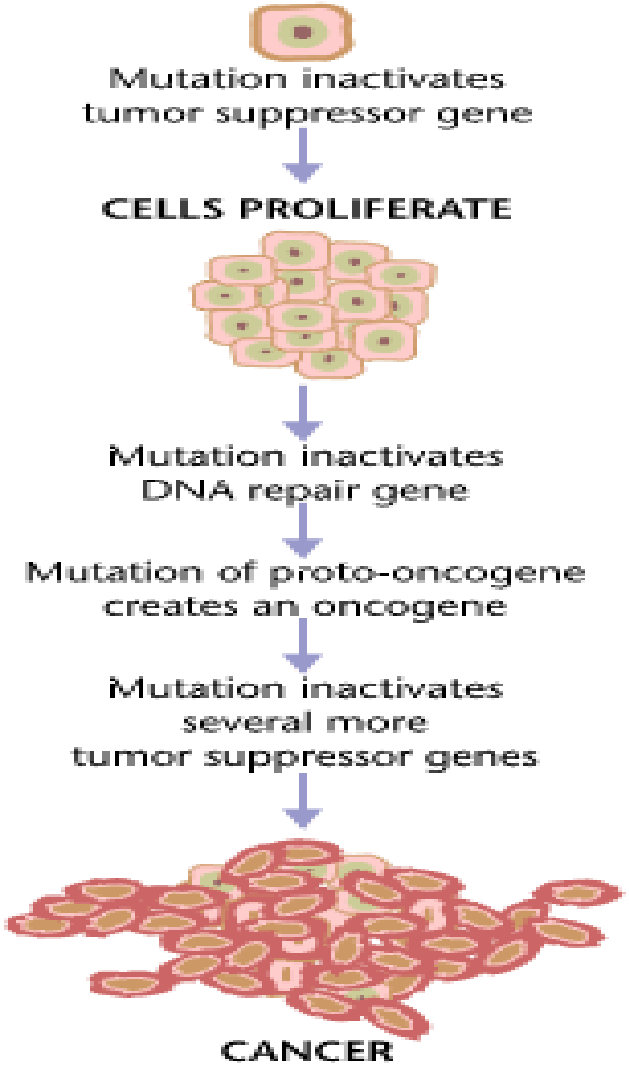
- Proteins involved in the cell signaling pathways are products of proto-oncogenes
  - Proliferative
  - Anti-apoptotic (survival)
  - Angiogenic

Normal to Cancerous Cell

- Tumor suppressors are recessive –
- require mutation of both alleles

- Oncogenes are dominant –
- mutation of 1 allele is sufficient

Normal to Cancerous Cell



Virology

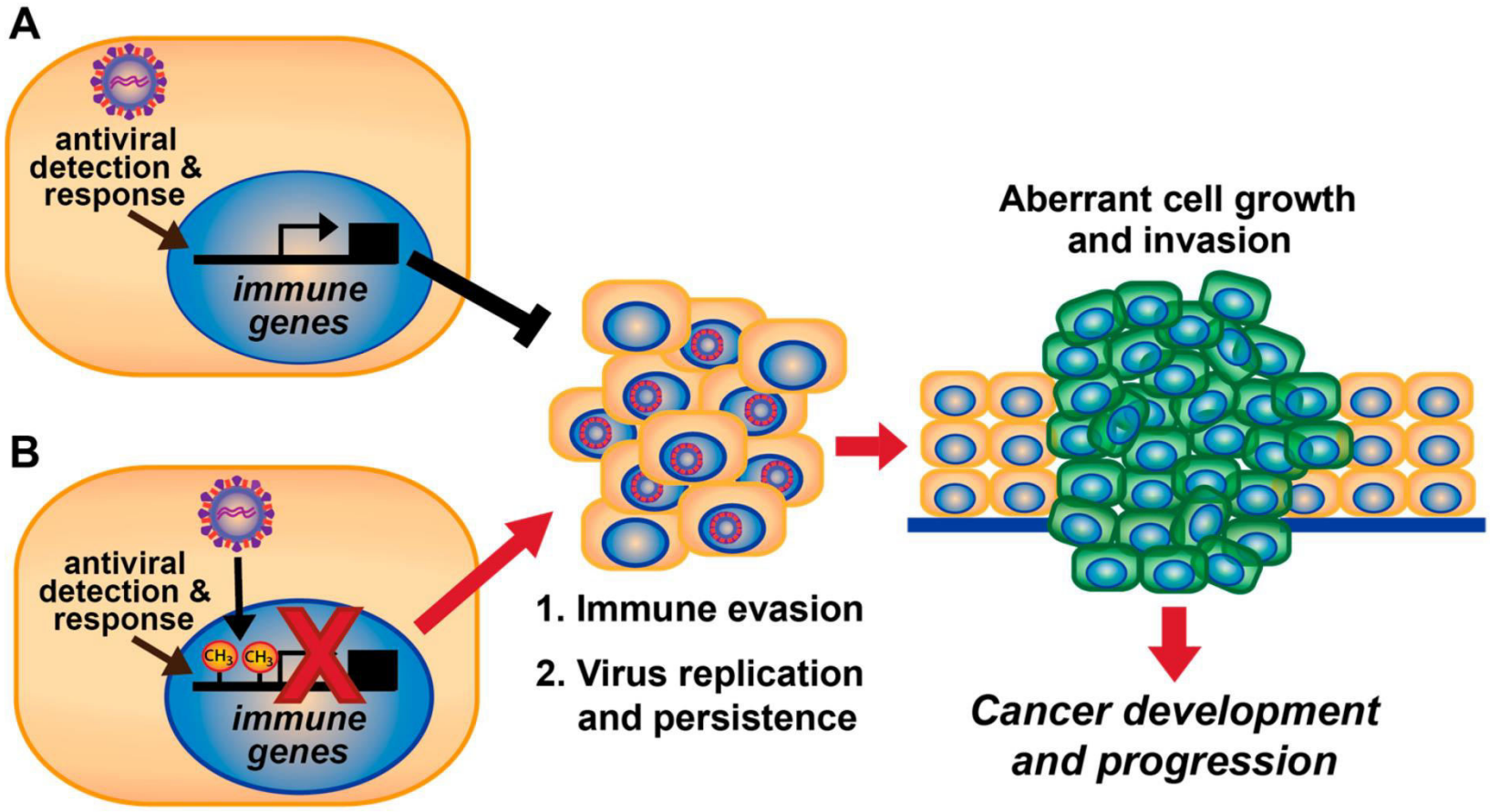
DNA TUMOR VIRUSES

DNA Tumor Viruses

DNA Oncogenic viruses

- Papilloma virus
- Polyoma virus
- Adenovirus
- Herpes virus
- Hepatitis B virus

DNA Tumor Viruses



DNA Tumor Viruses

PAPILLOMA VIRUS

- DNA complexed with histone proteins
- Human penile, uterine, cervical and anal carcinomas
- Warts

DNA Tumor Viruses

PAPILLOMA VIRUS



DNA Tumor Viruses

POLYOMA VIRUS

- Mouse Polyoma virus
- Simian Polyoma virus
- Human Polyoma Virus

DNA Tumor Viruses

POLYOMA VIRUS



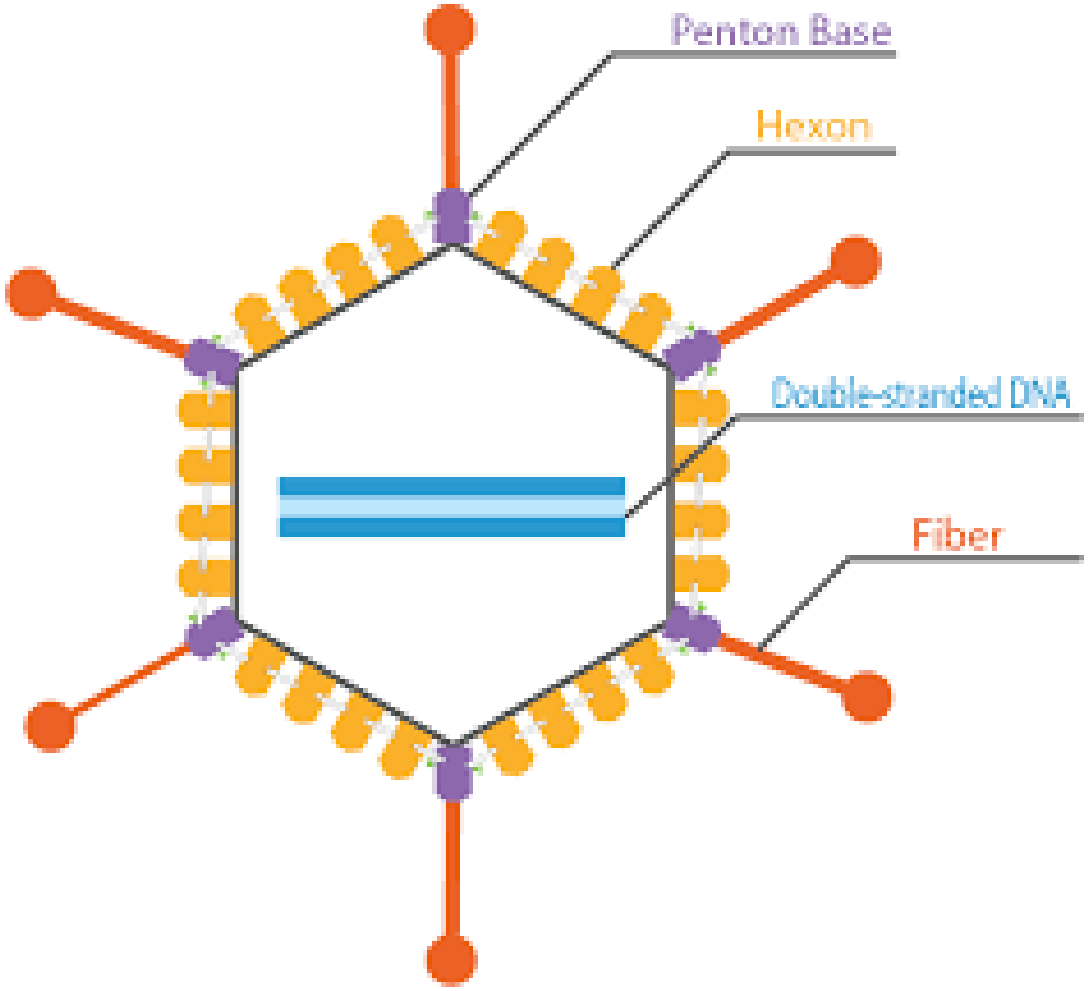
DNA Tumor Viruses

ADENO VIRUS

- Highly oncogenic
- Host makes antibodies against own T-cells

DNA Tumor Viruses

ADENO VIRUS



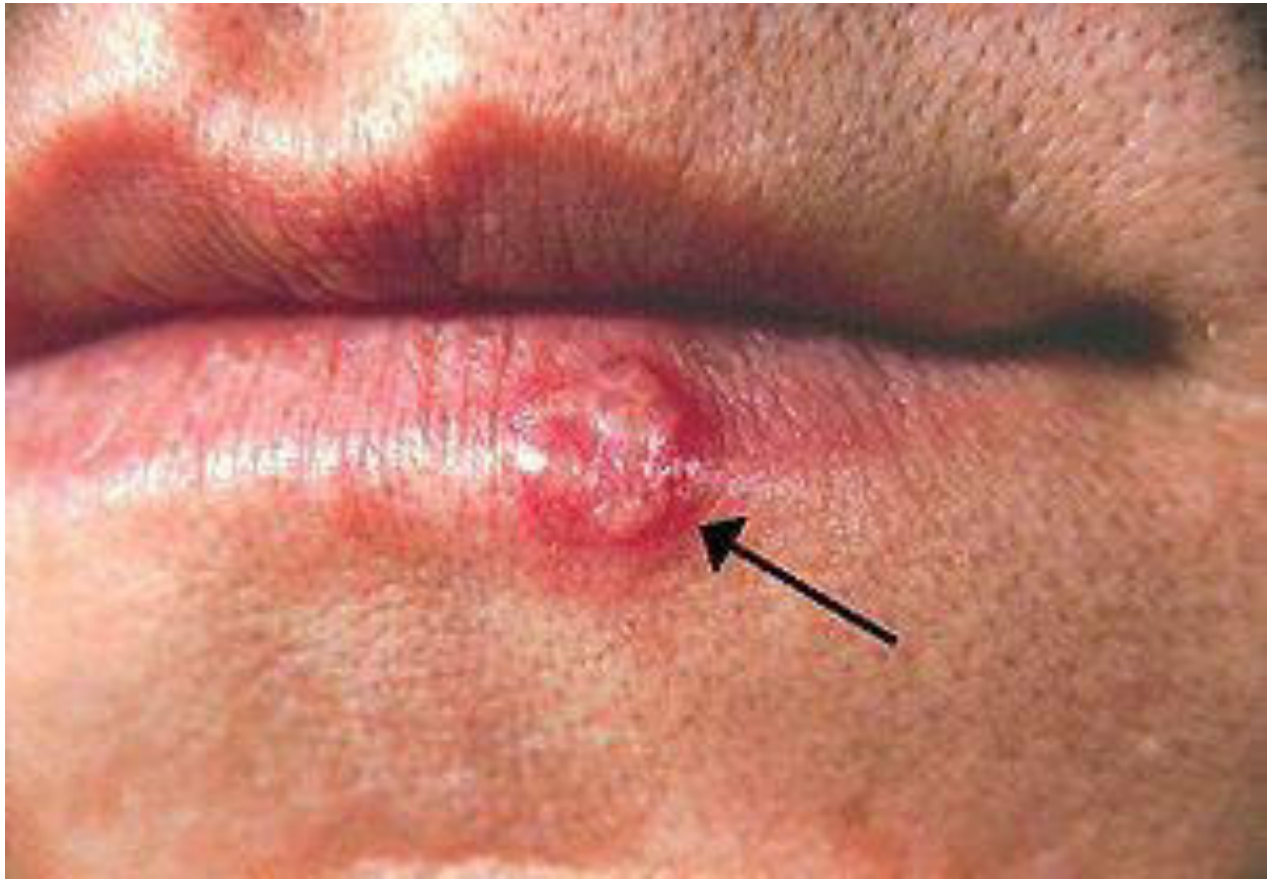
DNA Tumor Viruses

HERPES VIRUS

- Large size
- Epstien barr virus
- Kaposi Sarcoma virus
- Human cytomegalovirus

DNA Tumor Viruses

HERPES VIRUS



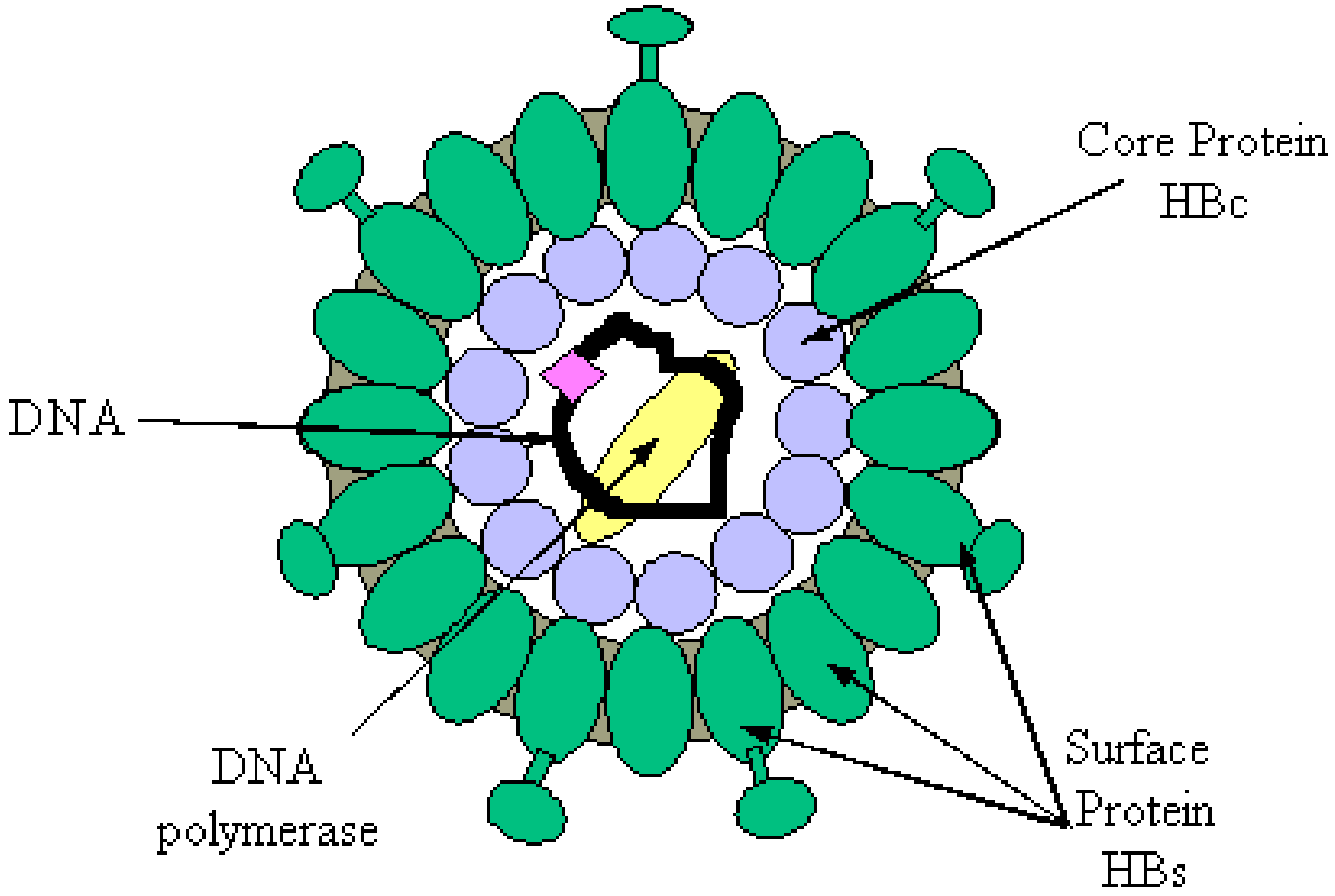
DNA Tumor Viruses

HEPADNA VIRUS

- Causes hepatitis B
- Hepatocellular carcinoma
- Reverse transcription like retroviruses

DNA Tumor Viruses

HEPADNA VIRUS



Virology

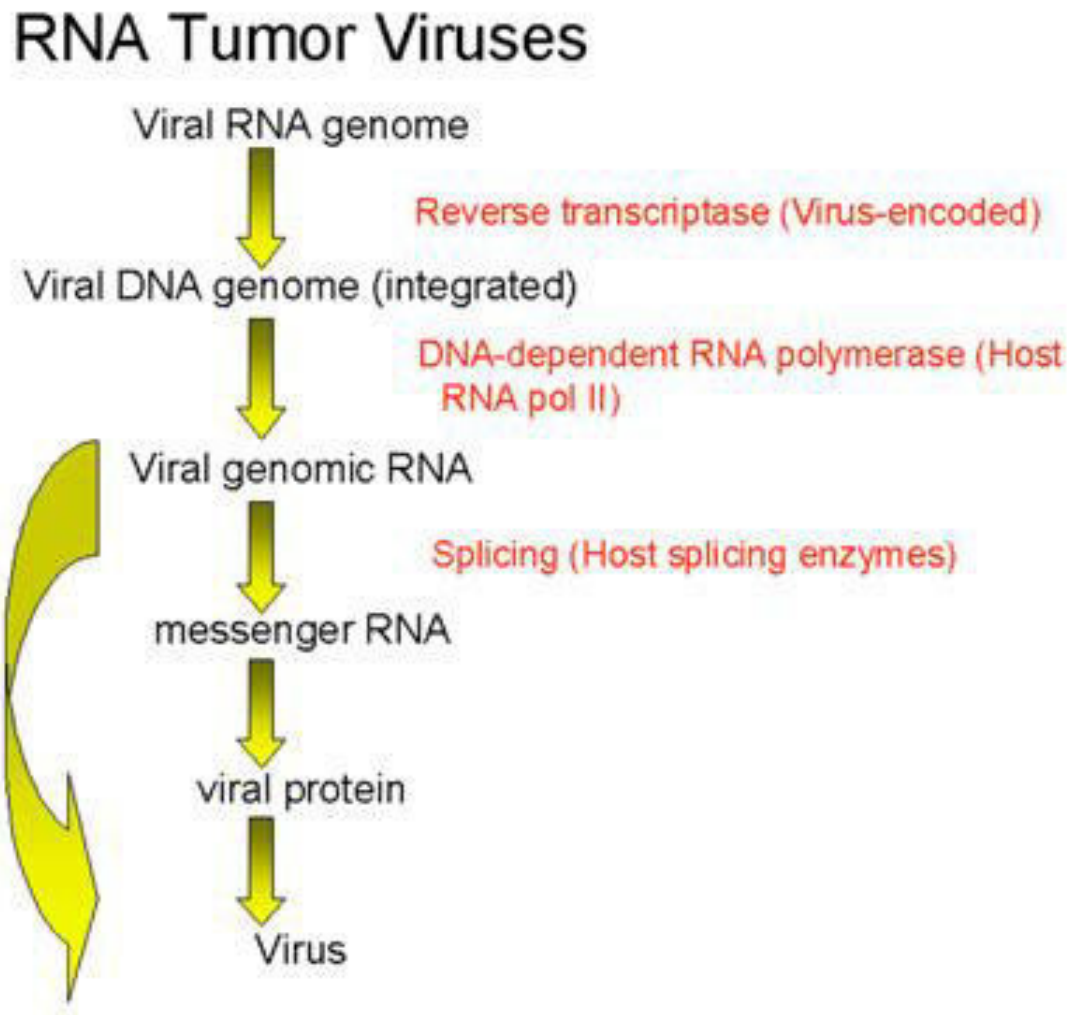
RNA TUMOR VIRUSES

RNA Tumor Viruses

RNA Oncogenic viruses

- Oncovirinae virus
- Lentivirinae virus
- Spumavirus

RNA Tumor Viruses

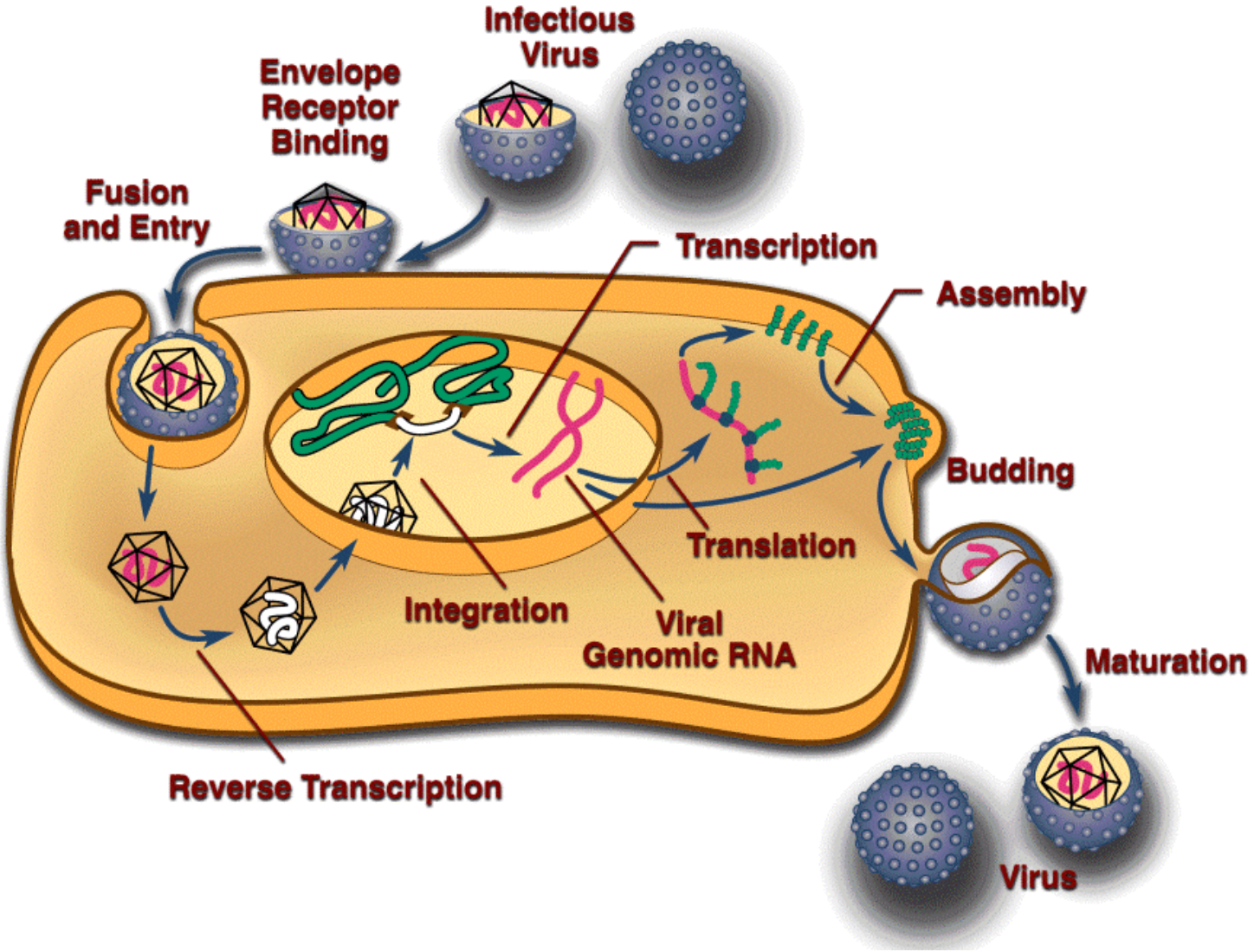


RNA Tumor Viruses

ONCOVIRINAE

- First discovered Rous- Sarcoma virus, showing slow neoplasms in chicken
- Human T-cell lymphotropic virus

RNA Tumor Viruses

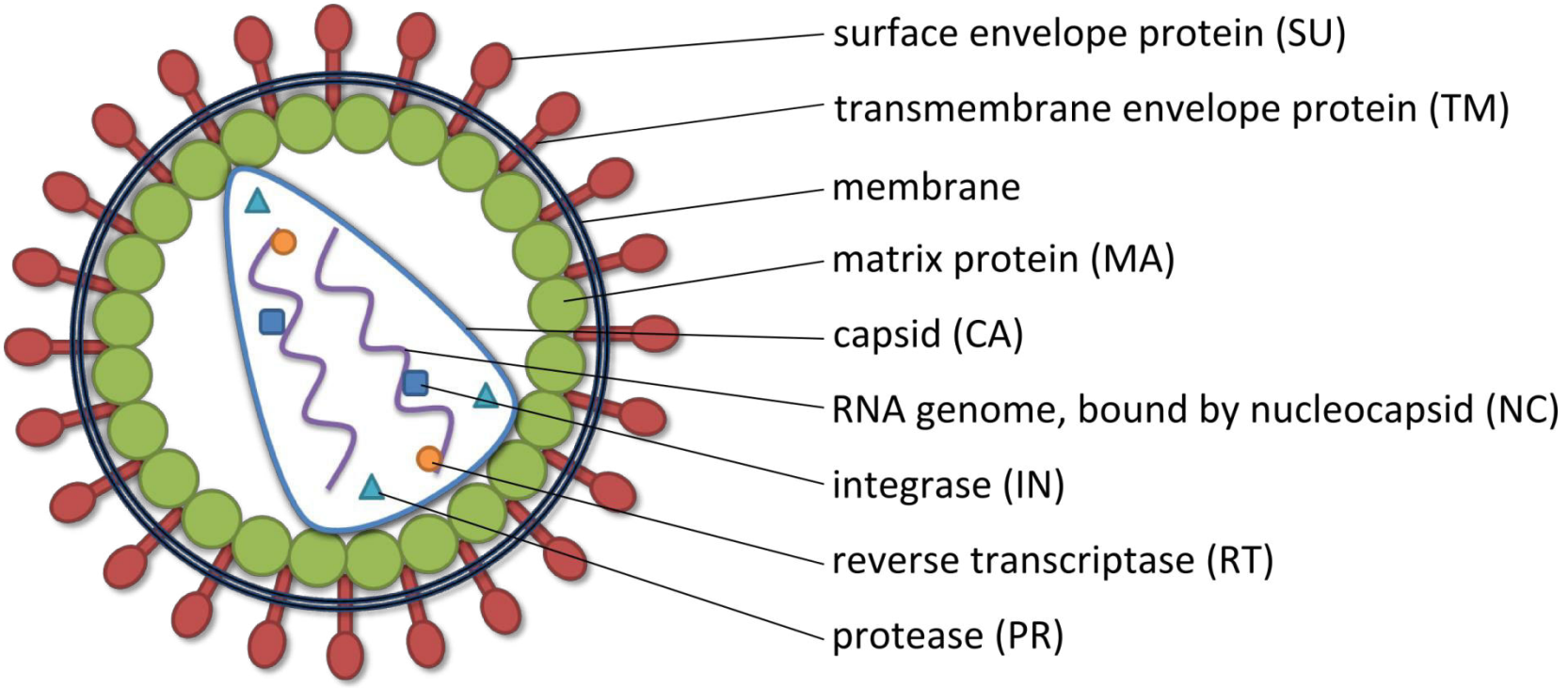


RNA Tumor Viruses

LENTIVIRINAE

- Long latent period of infections
- Disease of ungulates
- HIV that causes AIDS

RNA Tumor Viruses

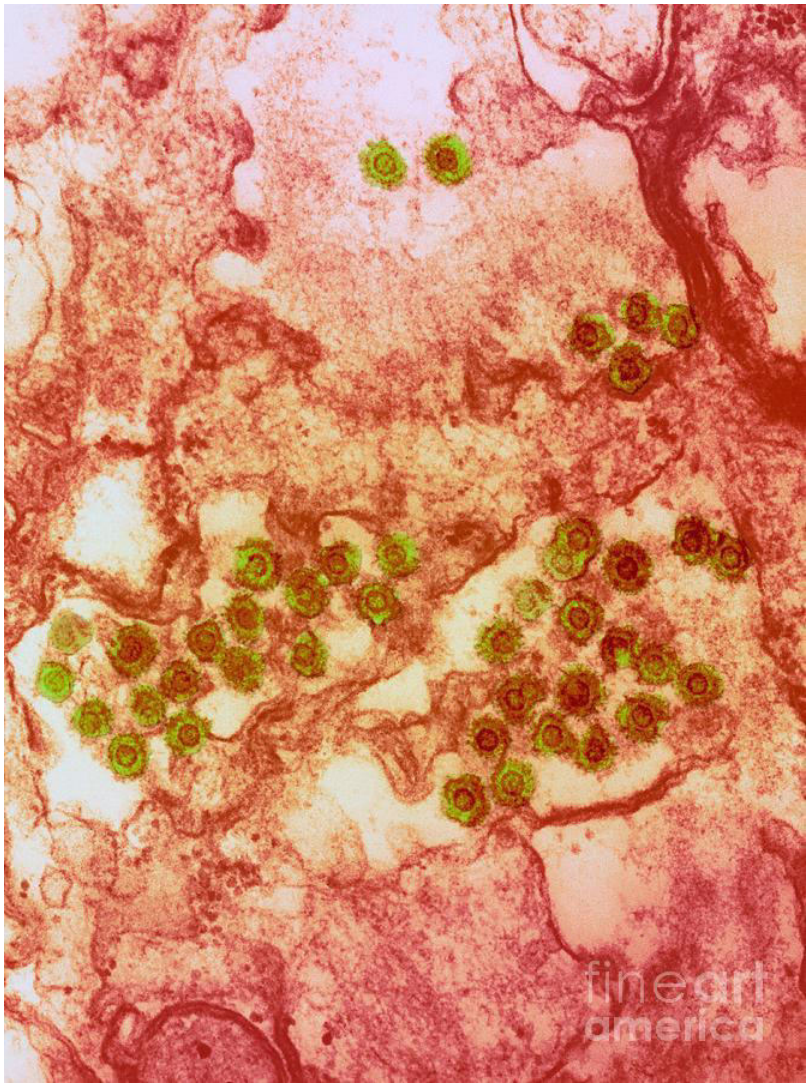


RNA Tumor Viruses

SPUMAVIRINAE

- Foamy appearance of cells
- Syncytia formation of giant multinucleate cells
- Human foamy virus
- Simian foamy virus

RNA Tumor Viruses



Virology

PERSISTENT VIRAL INFECTION

Persistent Viral Infection

- Characterized as those in which the virus is not cleared but remains in specific cells of infected individuals.
- Involve stages of both silent and productive infection without rapidly killing or even producing excessive damage of the host cells.

Persistent Viral Infection

- There are three types of overlapping persistent virus-host interaction
- latent,
- chronic
- slow infection

Persistent Viral Infection

- PATHOGENESIS:
- modulation of virus
- cellular gene expression
- modification of the host immune response.

➢ Host immunosuppression ➡ associated with reactivation of a number of persistent virus infections.

Persistent Viral Infection

Bodily sites of persistence:

- Nervous system
- Glands and body surfaces
- Germ cells
- Blood

Persistent Viral Infection

Importance:

- Reactivation
- Tumor causing
- Can cause progressive disease
- Persistence enables the parasite to be successful

Virology

Plant Viruses and Viroids

Plant Viruses and Viroids

- Plant viruses are viruses that affect plants.
- Like all other viruses, plant viruses are obligate intracellular parasites
- That do not have the molecular machinery to replicate without a host.
- Plant viruses can be pathogenic to higher plants.

Plant Viruses and Viroids





Pepper mild mottle virus and leaf curl virus

Plant Viruses and Viroids

General characteristics

- Most plant viruses are rod-shaped, with protein discs forming a tube surrounding the viral genome
- Isometric particles are another common structure
- They rarely have an envelope
- The great majority have an RNA genome, which is usually small and single stranded (ss), but some viruses have double-stranded (ds) RNA, ssDNA or dsDNA genomes

Plant Viruses and Viroids

- Plant viruses are not as well understood as their animal counterparts
- One plant virus has become iconic: tobacco mosaic virus (TMV), the first virus to be discovered
- This and other viruses cause an estimated U.S$60 billion loss in crop yields worldwide each year
- Plant viruses are grouped into 73 genera and 49 families
- However, these figures relate only to cultivated plants, which represent only a tiny fraction of the total number of plant species.

Plant Viruses and Viroids

Transmission

- To transmit from one plant to another and from one plant cell to another, plant viruses must use strategies that are usually different from animal viruses.
- Plants do not move, and so plant-to-plant transmission usually involves vectors (such as insects).

Plant Viruses and Viroids

Transmission

- Plant cells are surrounded by solid cell walls, therefore transport through plasmodesmata
- Plants have specialized mechanisms for transporting mRNAs through plasmodesmata

Plant Viruses and Viroids

Defense system

- Plant defenses against viral infection include, among other measures, the use of siRNA in response to dsRNA.
- Most plant viruses encode a protein to suppress this response.
- Plants also reduce transport through plasmodesmata in response to injury.

END

Virology

Infection in Plants-I

Infection in Plants-I/Structure

Structure of plant Viruses Viruses

- These are extremely small and can only be observed under an electron microscope.
- The structure of a virus is given by its coat of proteins, which surround the viral genome.
- Assembly of viral particles takes place spontaneously.

Infection in Plants-I/Structure

Structure of plant Viruses

- Over 50% of known plant viruses are rod-shaped (flexuous or rigid).
- The length of the particle is normally dependent on the genome but it is usually between 300–500 nm with a diameter of 15–20 nm.
- Protein subunits can be placed around the circumference of a circle to form a disc.
- In the presence of the viral genome, the discs are stacked, then a tube is created with room for the nucleic acid genome in the middle

Infection in Plants-I/Structure

Structure of plant Viruses

- The second most common structure amongst plant viruses are isometric particles.
- They are 25–50 nm in diameter.
- In cases when there is only a single coat protein, the basic structure consists of 60 T subunits, where T is an integer.
- Some viruses may have 2 coat proteins that associate to form an icosahedral shaped particle.

Infection in Plants-I/Structure

Structure of plant Viruses

- There are three genera of Geminiviridae that consist of particles that are like two isometric particles stuck together.
- A very small number of plant viruses have, in addition to their coat proteins, a lipid envelope.
- This is derived from the plant cell membrane as the virus particle buds off from the cell

Virology

Infection in Plants-II

Infection in Plants-II/Transmission

Through sap

- Viruses can be spread by direct transfer of sap by contact of a wounded plant with a healthy one.
- During agricultural practices, as by damage caused by tools or hands, or naturally, as by an animal feeding.
- Examples, TMV, potato viruses and cucumber mosaic viruses

Infection in Plants-II/Transmission

Insects

- Plant viruses need to be transmitted by a vector, most often insects such as leafhoppers.
- One class of viruses, the Rhabdoviridae, has been proposed to actually be insect viruses that have evolved to replicate in plants

Infection in Plants-II/Transmission

Nematodes

- Soil-borne nematodes also have been shown to transmit viruses.
- They acquire and transmit them by feeding on infected roots.
- Viruses can be transmitted both non-persistently and persistently, but there is no evidence of viruses being able to replicate in nematodes.
- Examples; tobacco ringspot virus and tobacco rattle virus.

Infection in Plants-II/Transmission

Plasmodiophorids

- A number of virus genera are transmitted, both persistently and non-persistently, by soil borne zoosporic protozoa.
- These protozoa are not phytopathogenic themselves, but parasitic.
- Transmission of the virus takes place when they become associated with the plant roots.
- Examples; Polymyxa graminis and Polymyxa betae.

Infection in Plants-II/Transmission

- Plant virus transmission from generation to generation occurs in about 20% of plant viruses.
- When viruses are transmitted by seeds, the seed is infected in the generative cells and the virus is maintained in the germ cells and sometimes, but less often, in the seed coat.
- There does not seem to be a correlation between the location of the seed on the plant and its chances of being infected.
- Examples; families Leguminosae, Solanaceae, Compositae, Rosaceae, Cucurbitaceae, Gramineae.
- Bean common mosaic virus is transmitted through seeds.

Infection in Plants-II/Transmission

Direct plant-to-human transmission

- The Pepper Mild Mottle Virus (PMMoV) may have moved on to infect humans.
- This is a very rare and highly unlikely event
- One possibility is that the virus does not infect human cells directly.
- Instead, the naked viral RNA may alter the function of the cells through a mechanism similar to RNA interference, in which the presence of certain RNA sequences can turn genes on and off

Virology

Macroscopic Signs of Plant Viral Diseases

Macroscopic Signs of Plant Diseases

- Important for virus identification
- Used to name the disease
- Solely can not be used to characterize the virus as affected by many factors

Macroscopic Signs of Plant Diseases

local lesions

Necrotic

Chlorotic

Macroscopic Signs of Plant Diseases

Systemic lesions

- Appear in different patterns depending upon virus-host combinations
- Appear in sequential patterns and
- May comprise of different symptoms

Macroscopic Signs of Plant Diseases

Symptoms

- 1. Mosaic
- 2. Mottle
- 3. Yellowing
- 4. Chlorosis
- 5. Vein clearing

Macroscopic Signs of Plant Diseases

Symptoms

6. Vein banding
   - Green vein banding
   - Yellow vein banding
7. Leaf roll
8. Leaf curl
9. Streak
10. Blistering

END

Virology

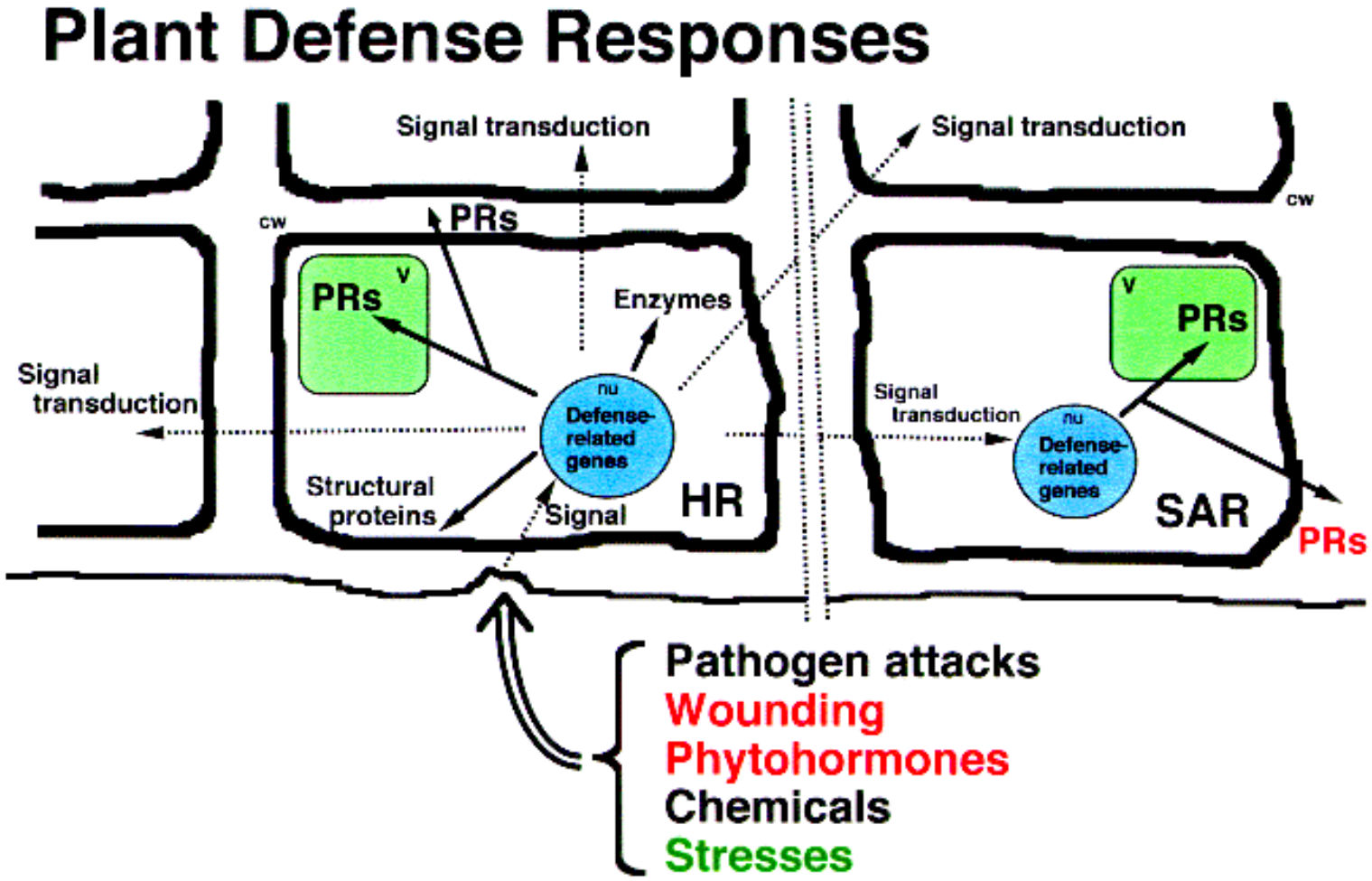
Synthesis of Pathogen Related Proteins

Synthesis of Pathogen Related Proteins

Pathogen Related (PR) Proteins

- Produced as a result of pathogen attack
- Infection activate genes that produce PR proteins

Synthesis of Pathogen Related Proteins

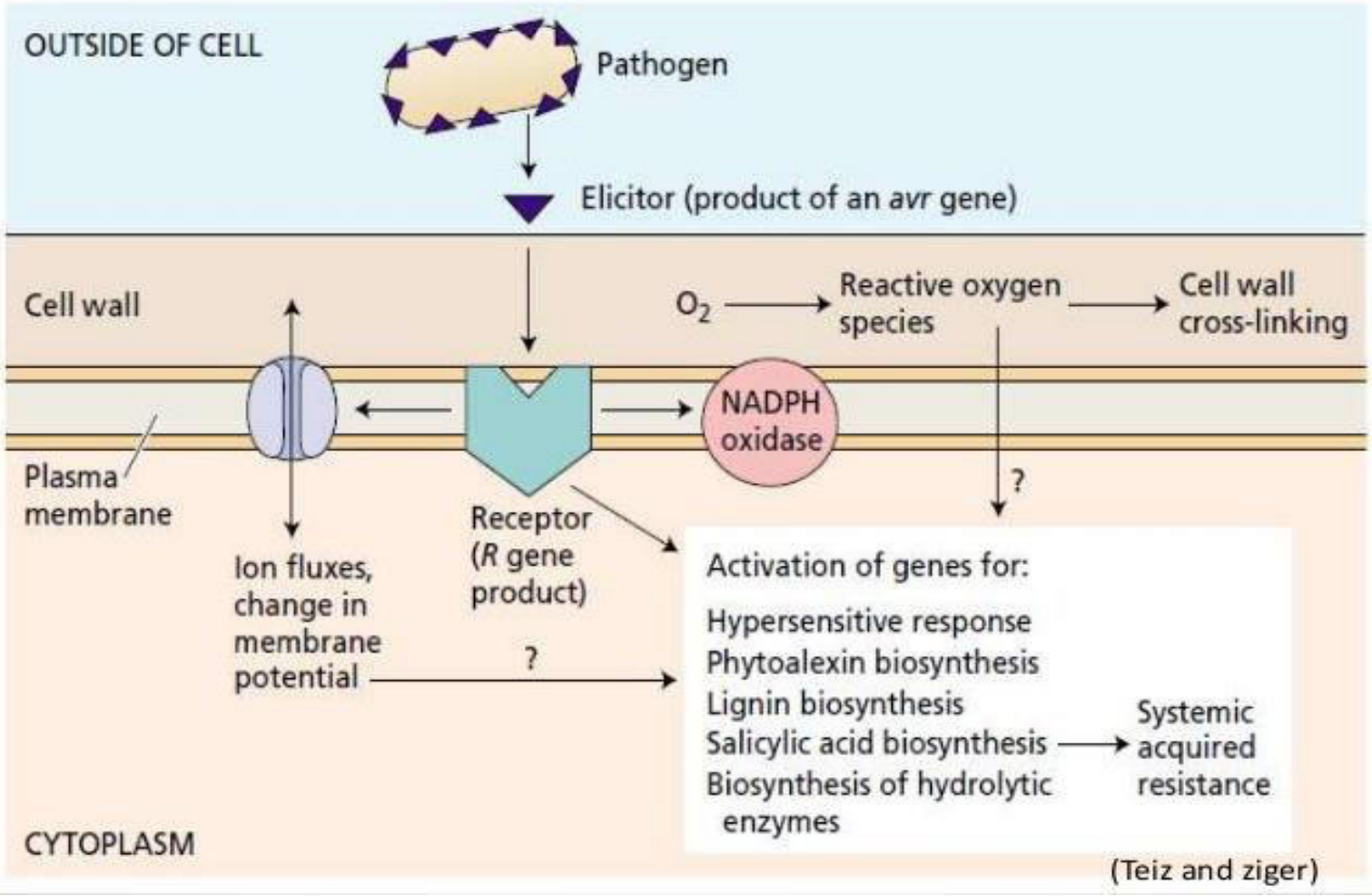


Synthesis of Pathogen Related Proteins

Nature

- Some are antimicrobial
- Some function as signal to spread news
- Salicyclic acid induces production of proteins

Synthesis of Pathogen Related Proteins



Synthesis of Pathogen Related Proteins

Functions

- Antiviral action
- As well as antifungal and antibacterial
- Signals the message to nearby cells and tissues
- Stimulate cross-linkage of molecules

Synthesis of Pathogen Related Proteins

Functions

- Peroxidase
- Ribonuclease
- Lysozyme activity
- Hydrolytic and proteinase-inhibitory
- Inactivate viral proteins

END

Virology

Virus Resistant Plants

Virus Resistant Plants

Plant viruses



{Tulip breaking virus (TBV)}
(https://en.wikipedia.org/wiki/Tulip_breaking_virus)



{Tobacco mosaic virus (TMV) symptoms on tobacco leaf}

Virus Resistant Plants

Modes of Transmission of plant viruses

- To transmit from one plant to another plant and from one plant cell to another, plant viruses must use strategies that are usually different from animal viruses.
- Plant do not move so plant plants transmission usually involves vectors (such as insect).

Virus Resistant Plants

Modes of Transmission of plant viruses

- Insect – this forms the largest and most significant vector group and particularly include; Aphids, Whiteflies, Hoppers, Beetles etc.
- Through sap
- Eg TMV, Potato virus
- Nematodes
- Eg Tobacco ring spot virus
- Through seed

Virus Resistant Plants

Mechanism of Plant resistance to viruses

- There are approximately 450 species of plant-pathogenic viruses, which cause a range of diseases.
- The plant show two types of antiviral defense resistanse mechanism are
- 1) R gene mediated
- 2) RNA Silencing

Virology

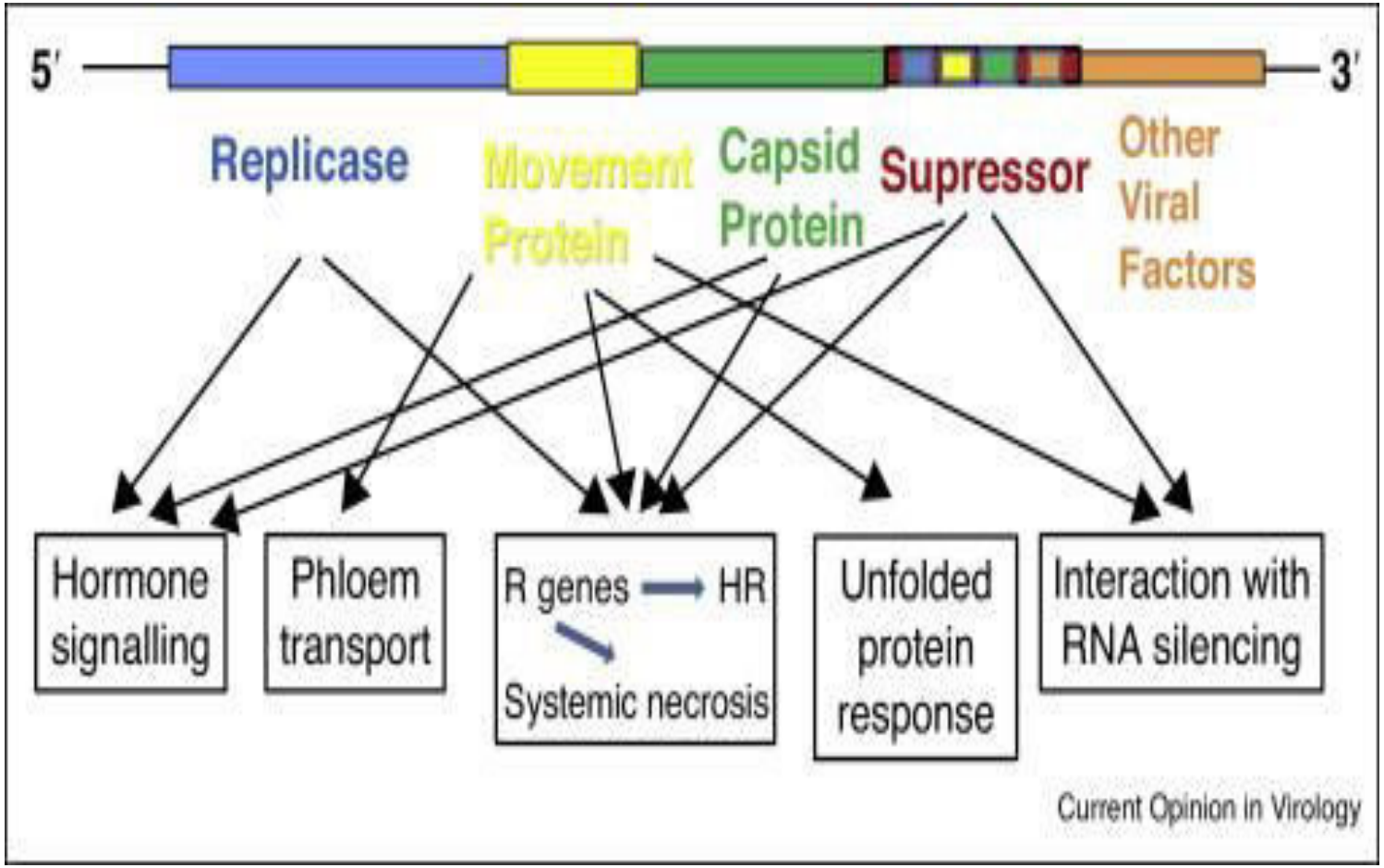
Pathogenesis of Plant Viral Diseases

Pathogenesis of Plant Viruses

Routes

- Seeds
- Grafting/ Vegetative propagation
- Vectors
- Mechanical

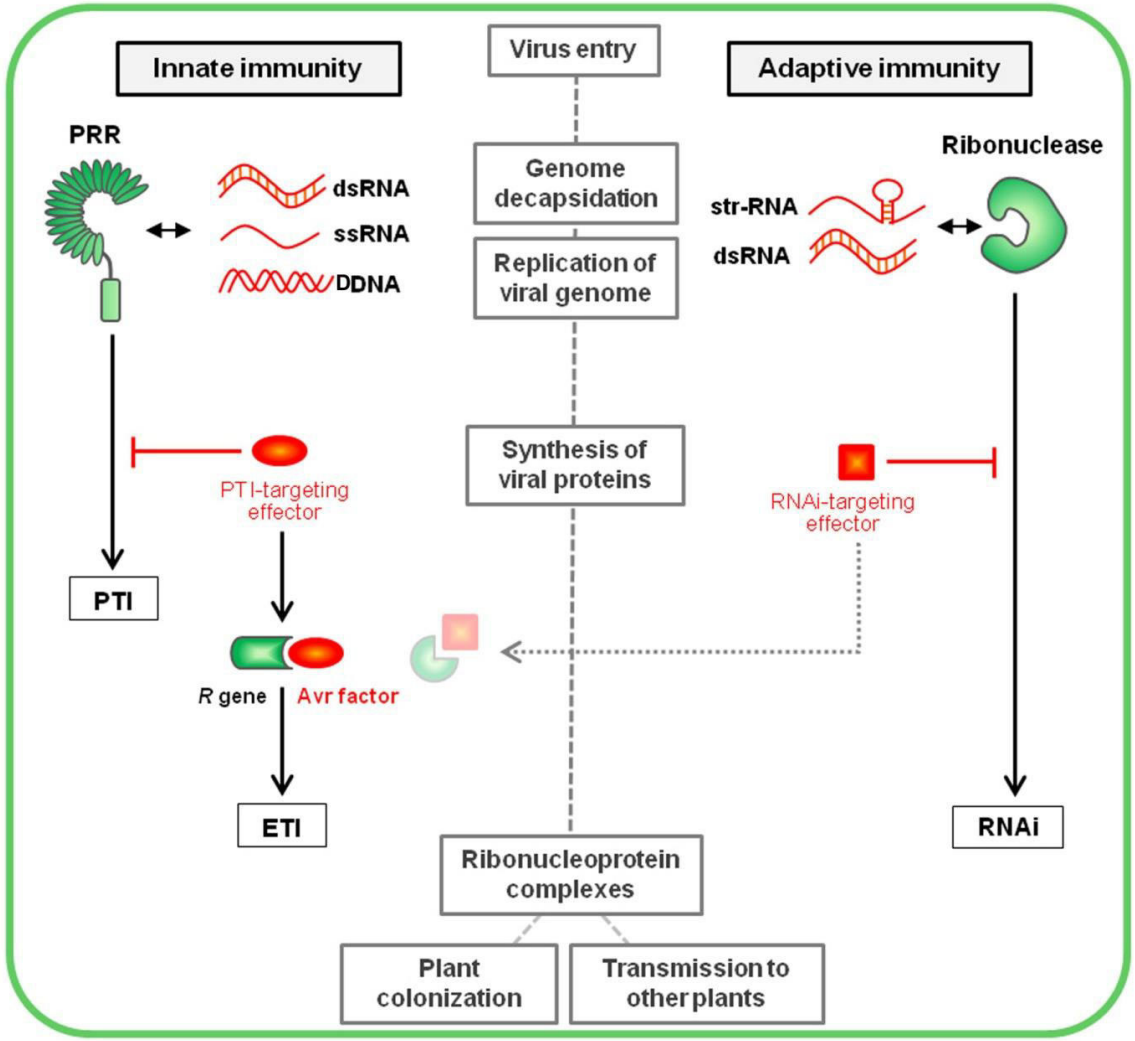
Pathogenesis of Plant Viruses



Pathogenesis of Plant Viruses

- No specific receptors on cells
- Rely on mechanical breach for virus to enter through cell wall
- Usually with the help of insects

Pathogenesis of Plant Viruses

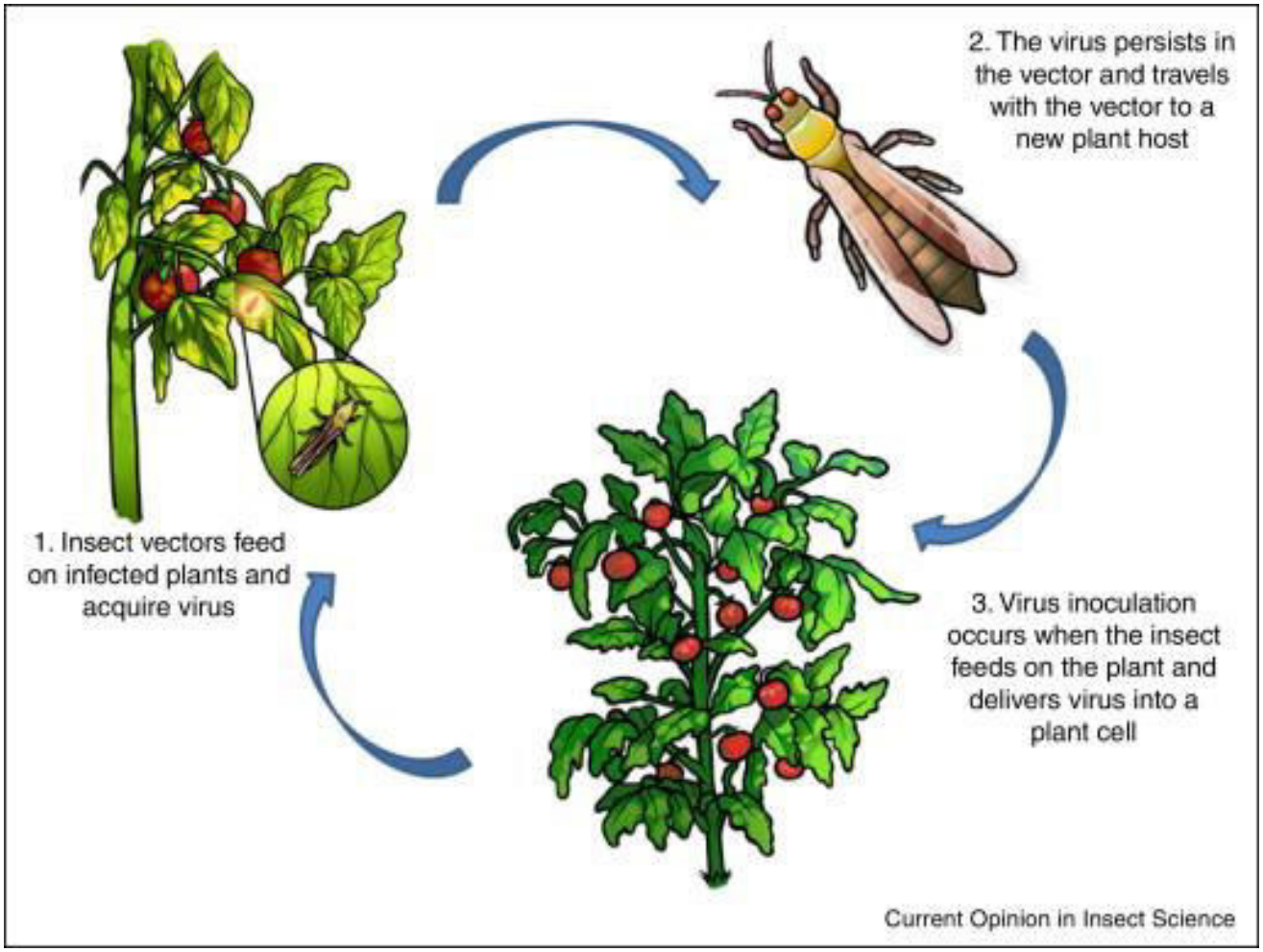


Pathogenesis of Plant Viruses

Propagative Transmission

- Multiply in tissues of insects as well as host
- Most multiplication at site of infection
- Localized symptoms

Pathogenesis of Plant Viruses



Pathogenesis of Plant Viruses

Spread

- Direct cell to cell
- Vascular system

Virology

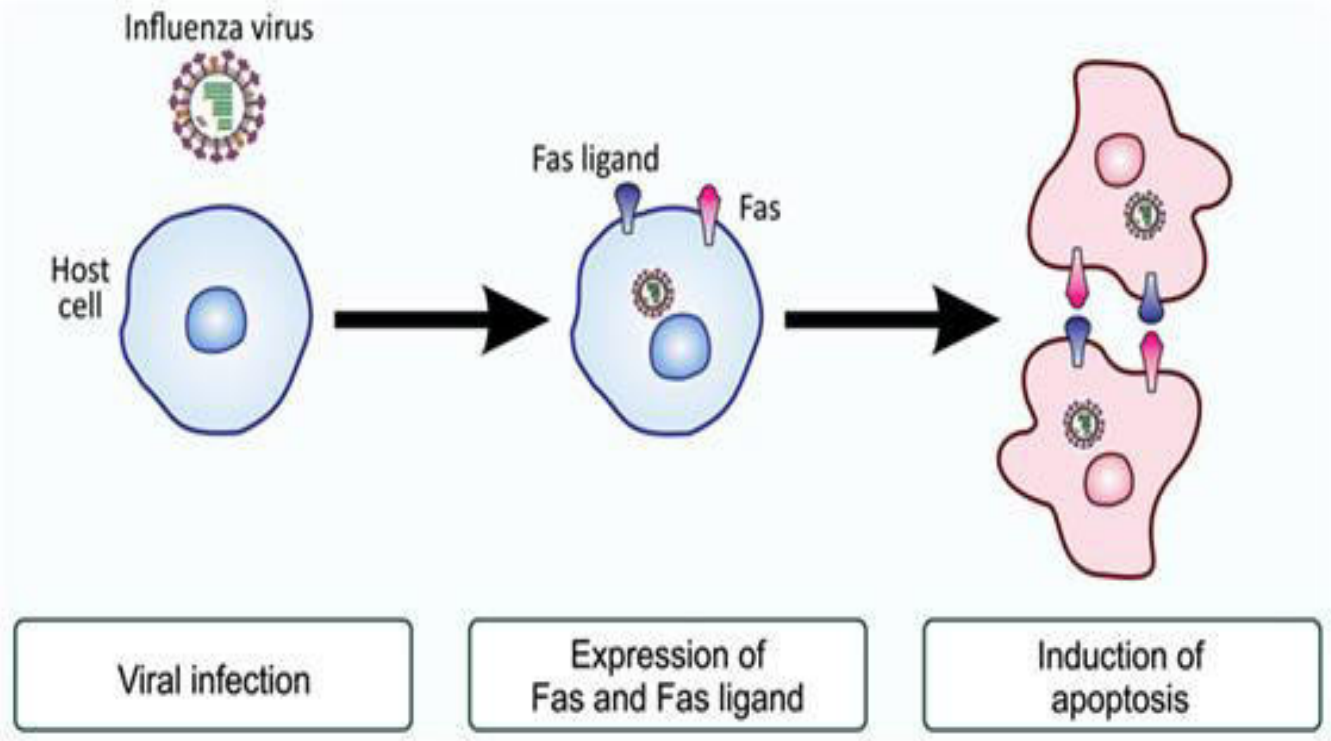
Apoptosis Caused by Viruses

Apoptosis by Viruses

Apoptosis

- Programmed cell death
- During cell development and cell killing by immune system

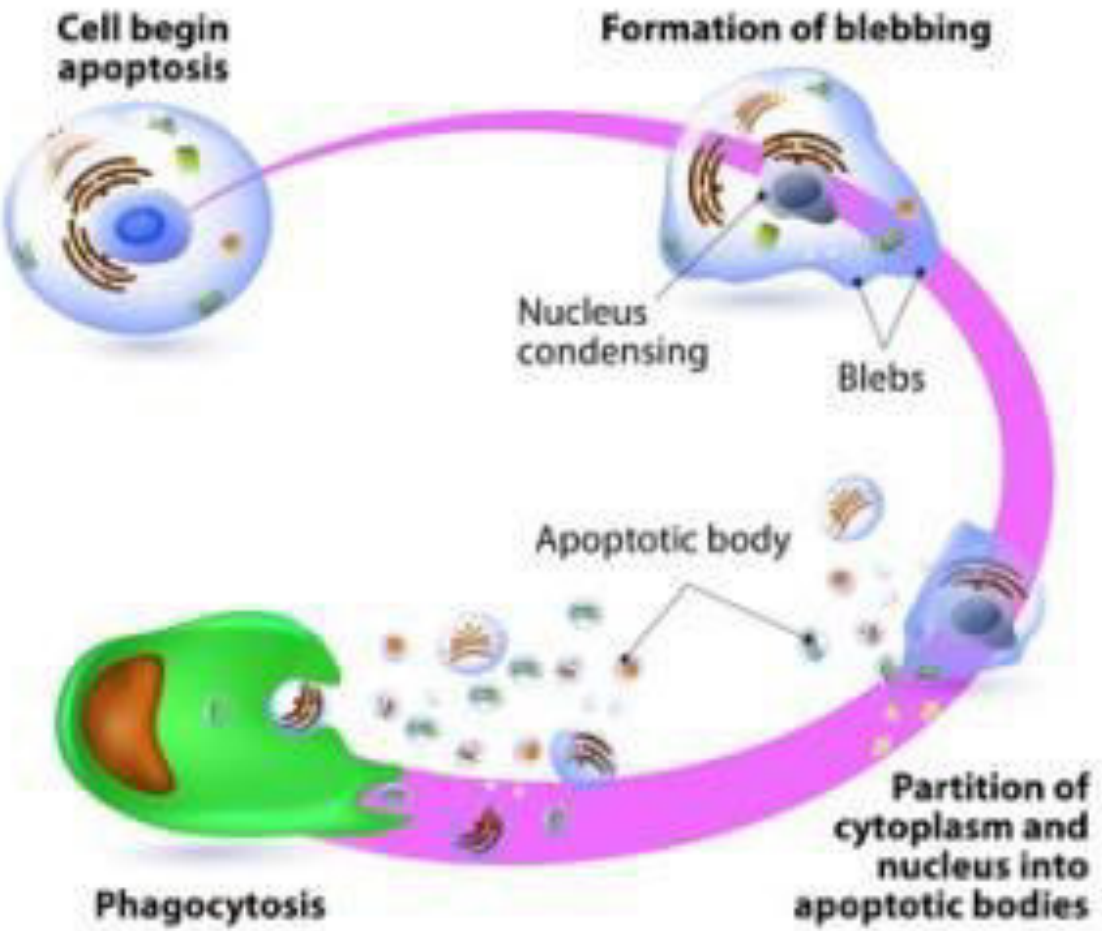
Apoptosis by Viruses



Apoptosis by Viruses

- Tightly regulated process
- Relies on complex molecule cascades for its control
- Shrinkage, condensation of chromatin, DNA fragmentation and blebbing

Apoptosis by Viruses

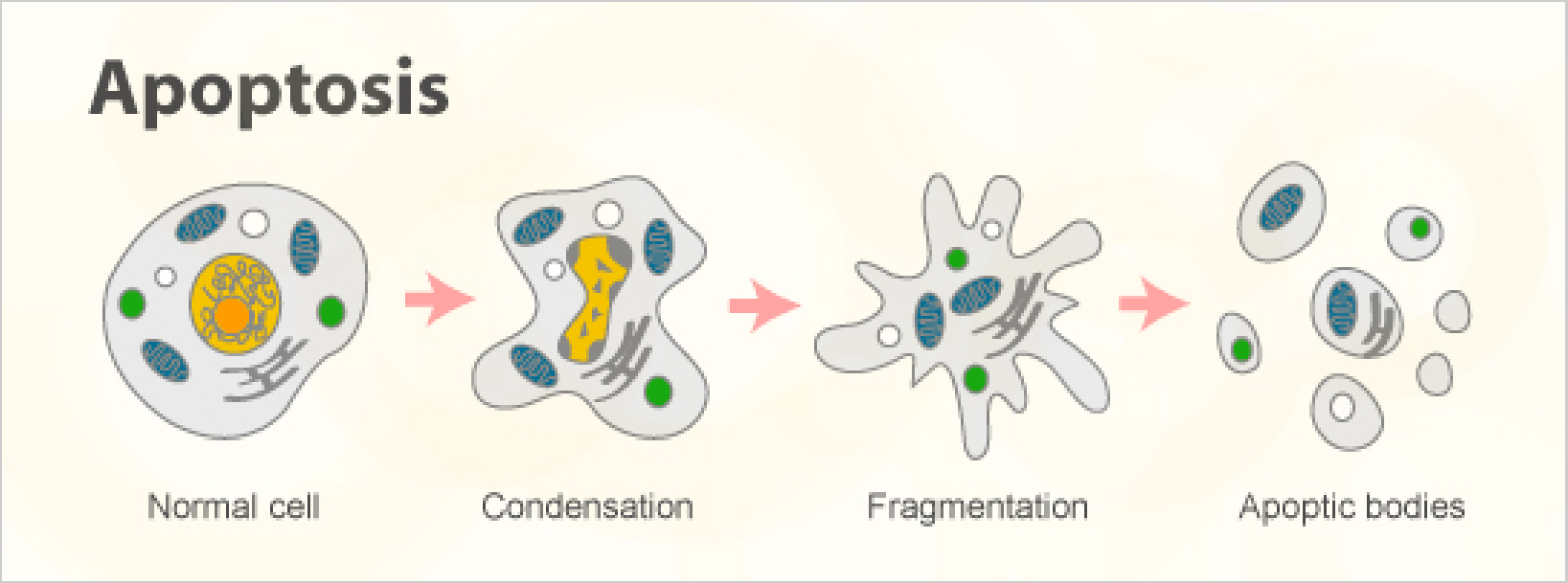


Apoptosis by Viruses

Activation

- Immune effector cells like CTL's and NK cells release lytic granules
- Act on target cells by two mechanisms

Apoptosis by Viruses



Apoptosis by Viruses

MECHANISM 1

- Release of cytokinins such as, perforin: peptide related to C9, forms a polyperforin and affects permeability of cell membrane
- Granzyme: serene proteases, affect cell pores

MECHANISM 2

- CTLs express a ligand, that binds on surface on target cell
- Triggers apoptosis
- Triggers cascade to initiate apoptosis

Virology

Lytic Cycle in Viruses

Lytic Cycle in Viruses



HOW DO PHAGES REPLICATE?

1. Lysogenic Cycle
2. Lytic Cycle

Virology

Lysogenic Cycle in Viruses

Lysogenic Cycle in Viruses

LYSOGEN OR LYSOGENIC

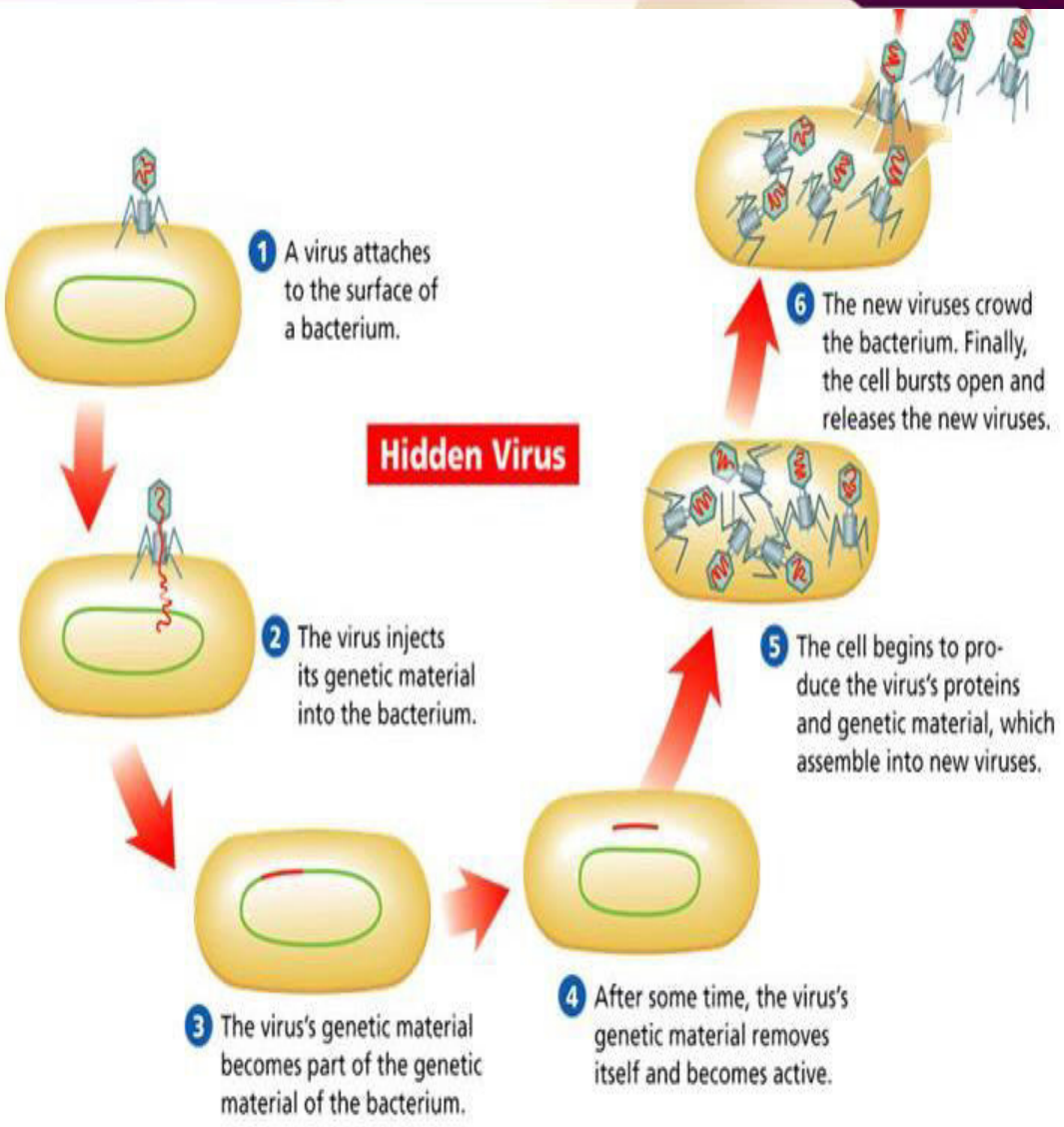
A **lysogen** or **lysogenic** bacterium is a bacterial cell in which a phage exists as DNA in its dormant state (prophage)

Lysogenic Cycle in Viruses

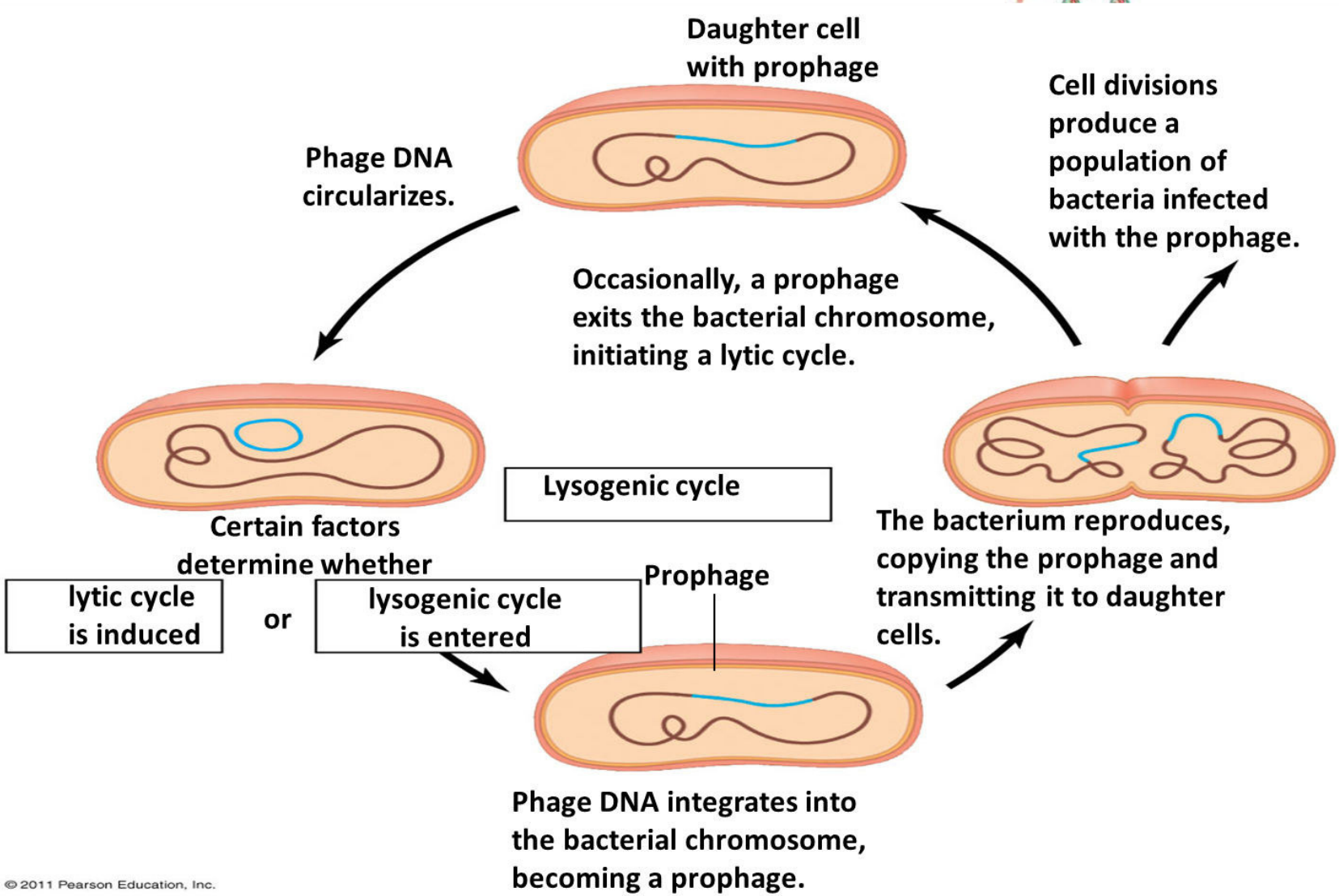
STEPS

1. Viral genome enter cells
2. Viral Genome integrates into host cell genome
3. Host DNA polymerase copies viral chromosomes
4. Host cell divides and viral chromosomes divide in to daughter cells
5. When virus is triggered, viral genome detaches and enters lytic cell

Lysogenic Cycle in Viruses



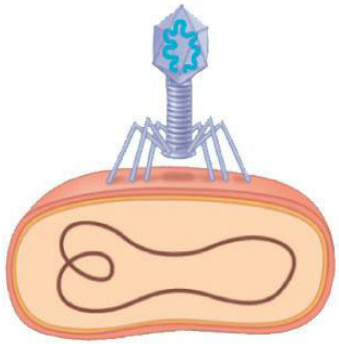
Lysogenic Cycle in Viruses



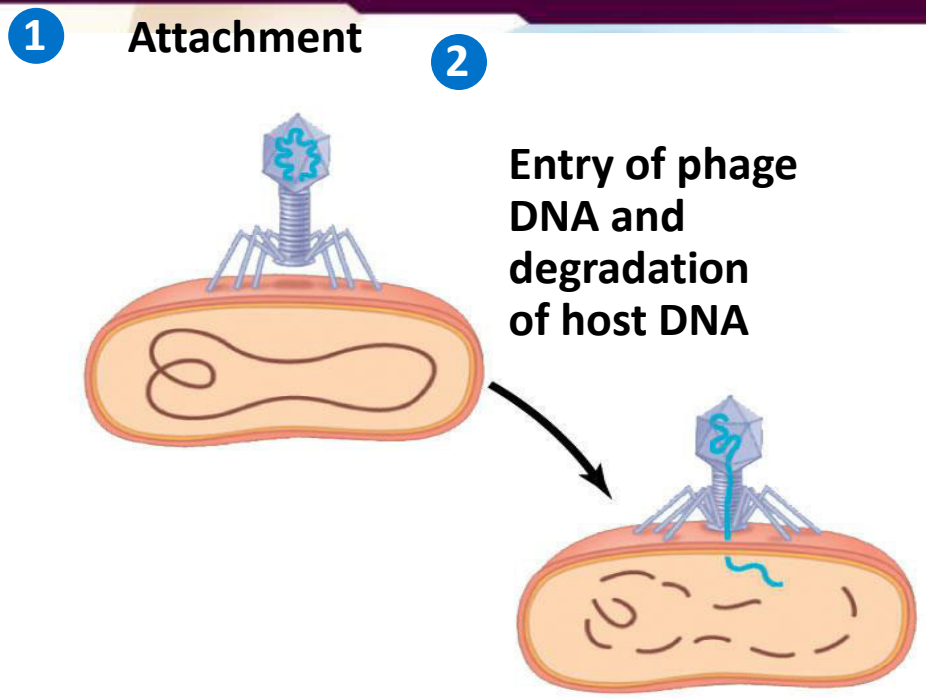
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Lytic Cycle

1 **Attachment**

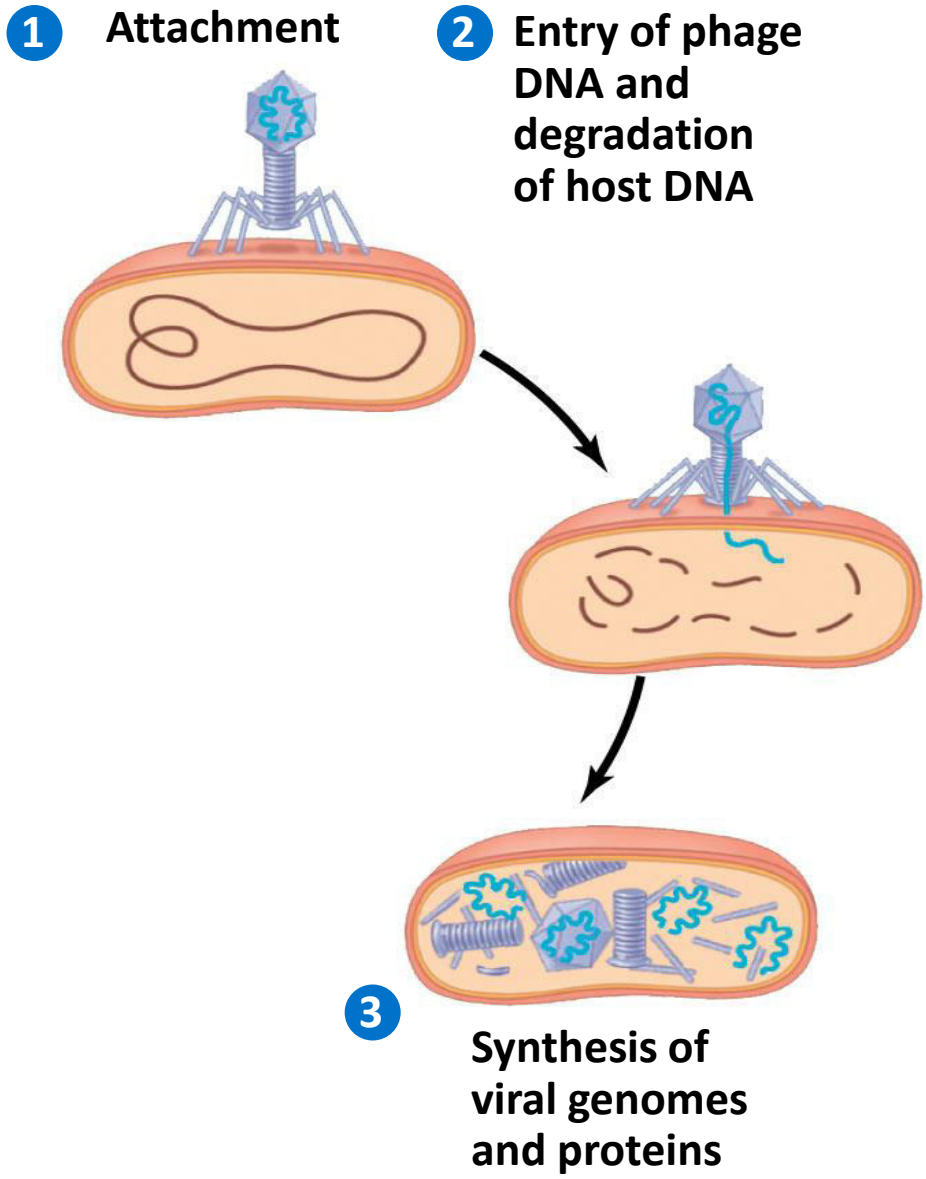


Lytic Cycle



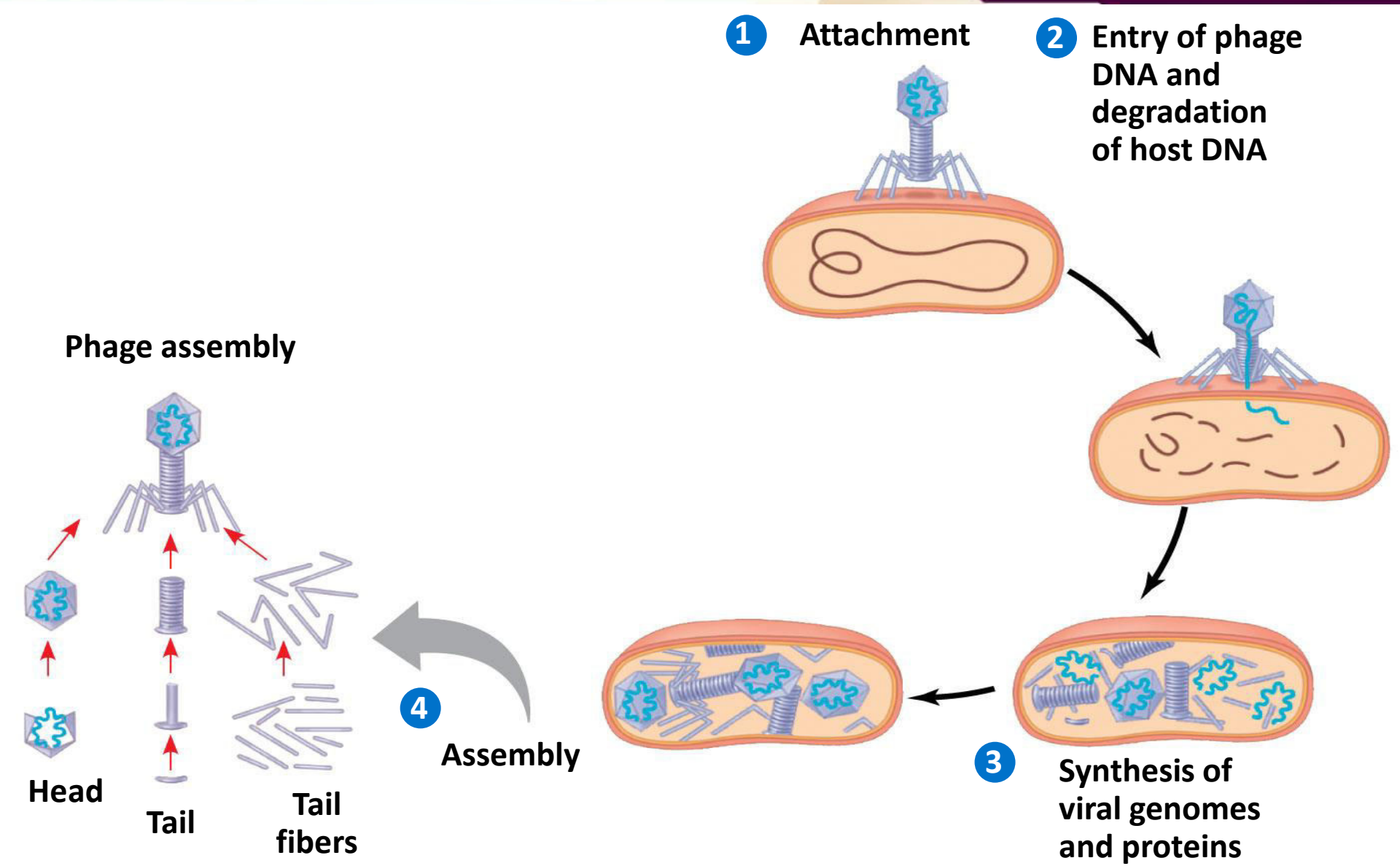
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Lytic Cycle



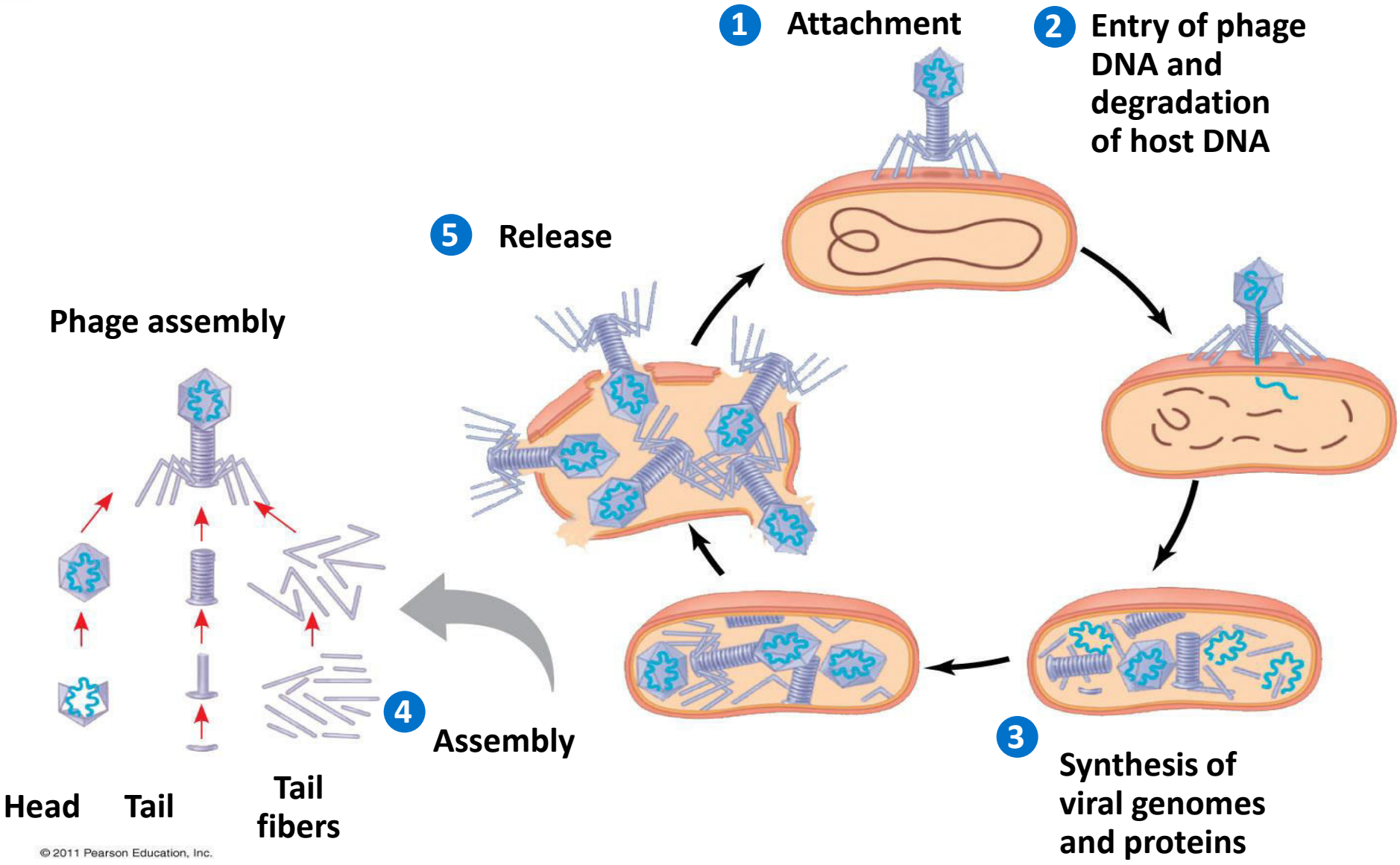
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Lytic Cycle



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Lytic Cycle



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Virology

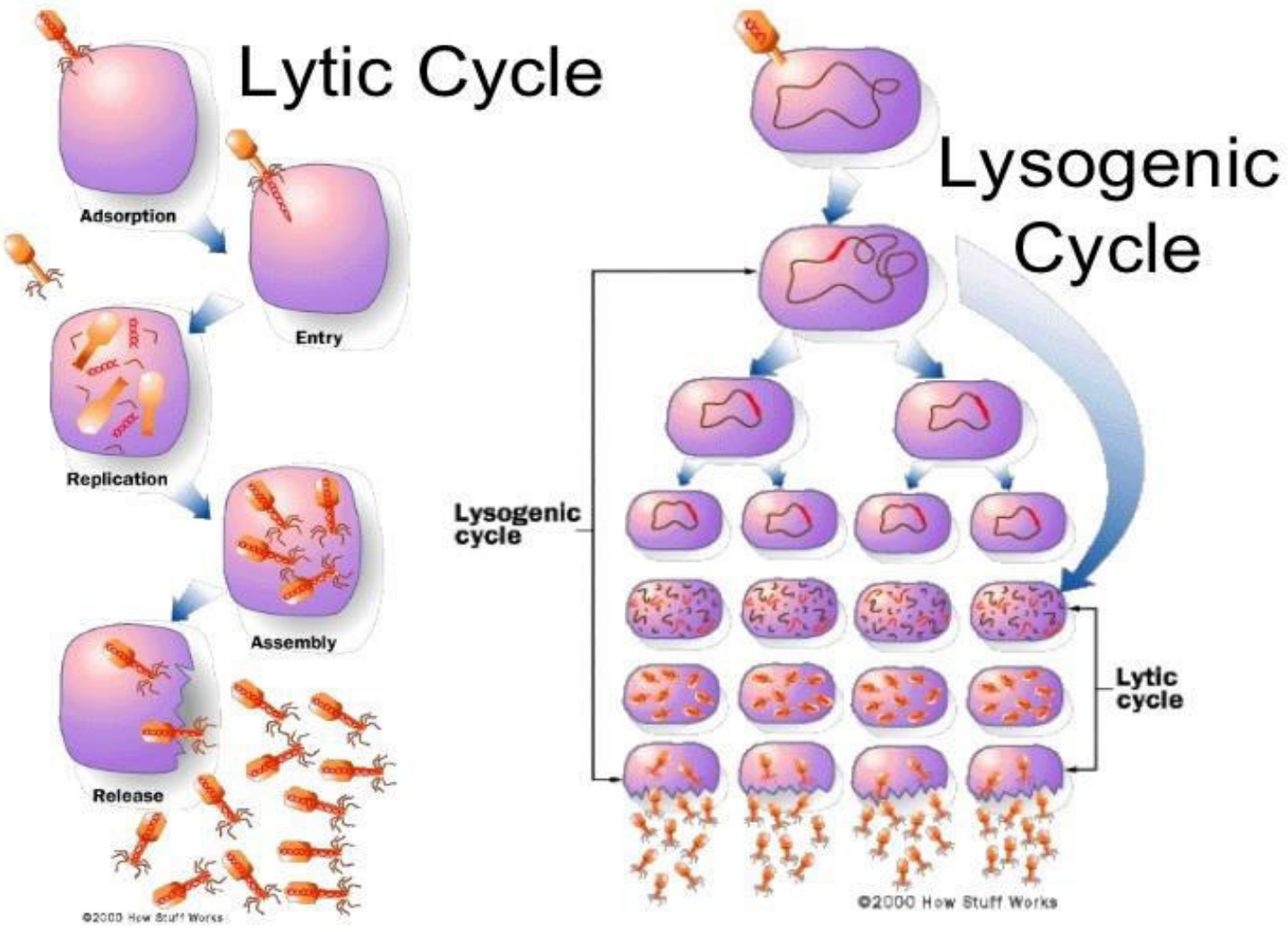
Comparison of Lytic and Lysogenic

Lytic Vs Lysogenic Cycle

- Lytic cycle (virulent phage)
- Release of virus bursts and kills host cells (lysis)

- Lysogenic cycle (temperate phage)
- Viral DNA integrates into host genome (provirus)
- Can be transmitted to daughter cells.
- Can initiate lytic cycle in response to environmental signals (stress)

Lytic Vs Lysogenic Cycle



Lytic Vs Lysogenic Cycle

Advantages and Limits

Lytic cycle

- Replication of new viruses is fast
- However, the host is also immediately killed preventing the viral genome from passing onto the next generation of host cells

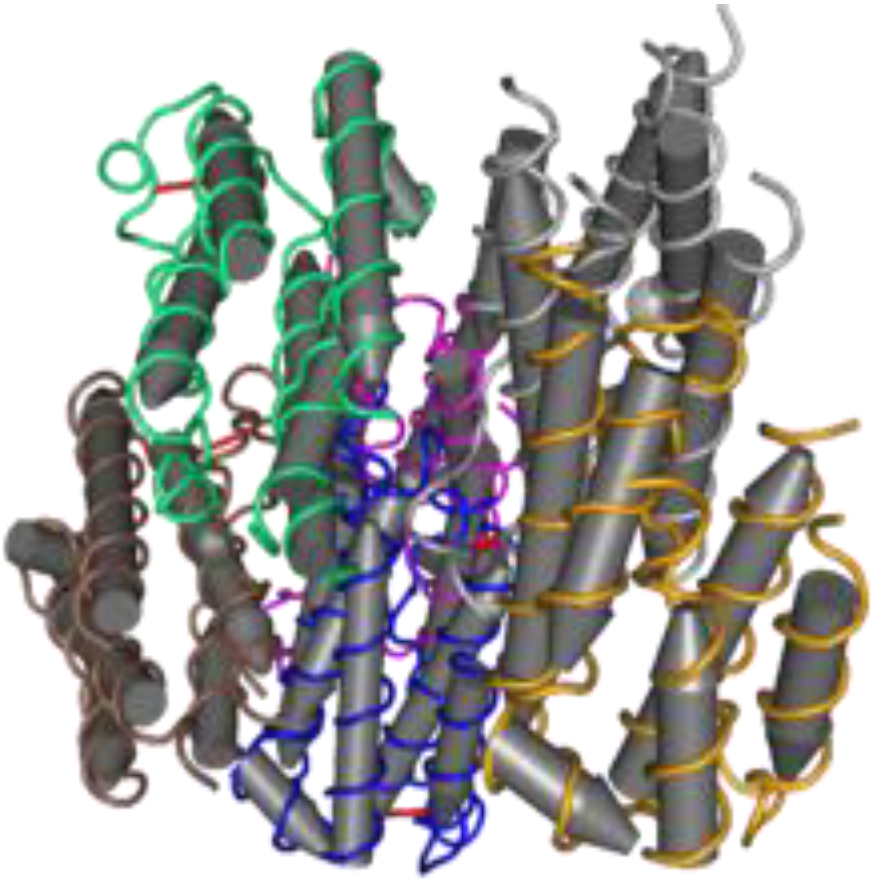
Lysogenic cycle

- Many more viruses can be made because the viral genome is passed onto future generation of host cells
- However, replication is takes longer because it is dependent on the host cell's replication

Virology

Interferons

Interferons



- Natural antiviral compounds
- substances that have antiviral properties in adjacent, noninfected cells adjacent

Interferons

Types of Interferons

Type I:

- (1) Interferon alpha = maximal antiviral activity
- (2) Interferon Beta = intermediate antiviral activity

Type II:

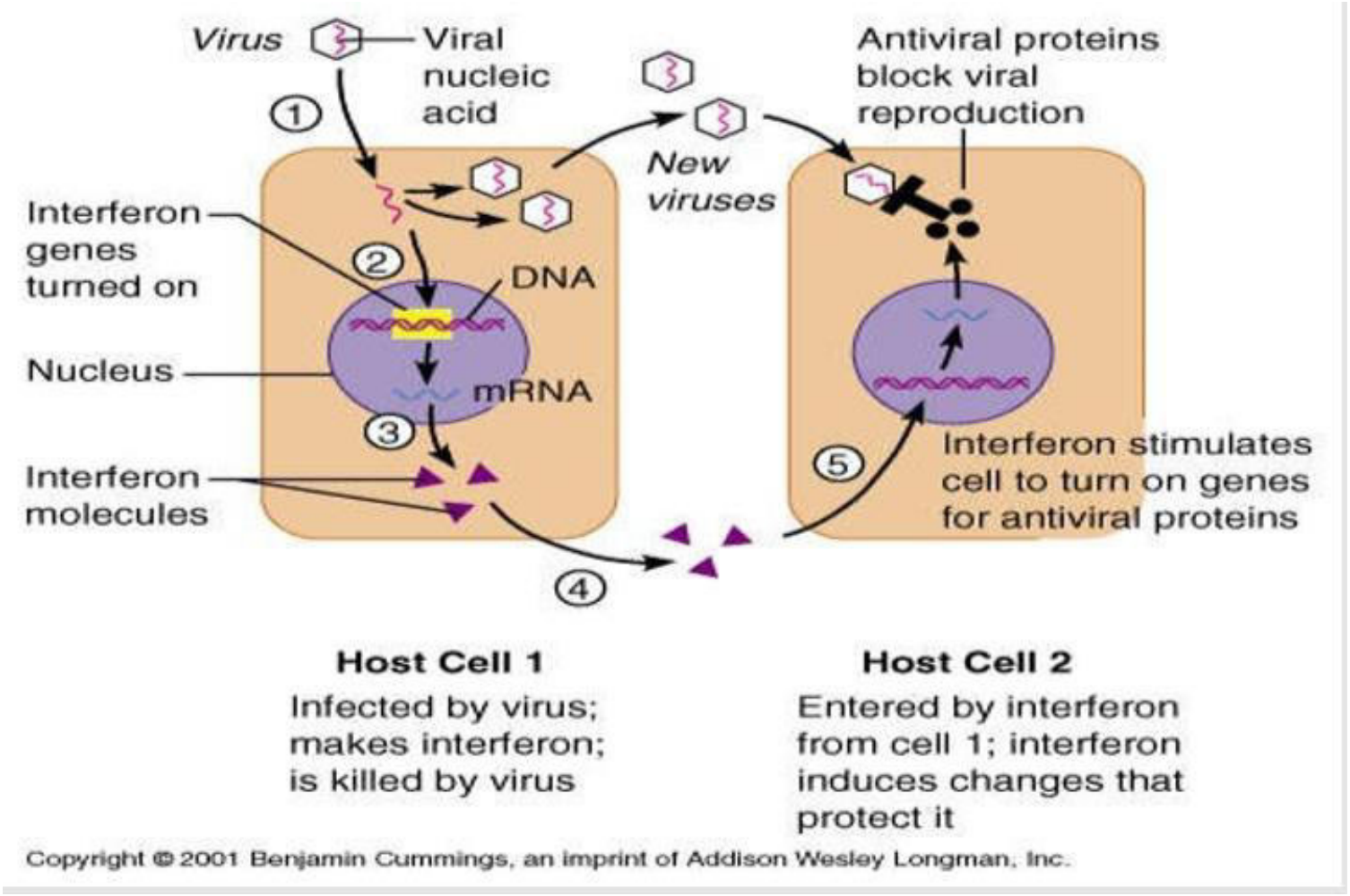
- Interferon Gamma = more lymphokine than antiviral

Interferons

Regulation of Interferon Expression

- not expressed in a normal resting cell
- labile repressors bind to promoter elements, block transcription
- production of labile suppressors drop in viral infection and allows in viral infection and allows interferon synthesis to occur

Interferons



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Interferons

Clinical Uses:

IFN-A :

- treatment of viral infections: condylomata acuminata and chronic hepa B and C
- prophylactic or therapeutic agent in immunocomp. hosts (VZV, HSV 1 and 2)
- prophylaxis vs CMV in renal transplant
- treatment of AIDS-associated Kaposi's sarcoma and hairy cell leukemia

IFN-G: IFN-G:

- immunostimulant in oncologic and immunodeficiency disorders

Virology

Evasion of Immune Response by Viruses

Evasion of Immune Response by Viruses

- When a virus infects a host, a complex immune response develops to eliminate the invading pathogen.
- Viruses, in turn, have evolved a profusion of strategies to escape from the immune system

Evasion of Immune Response by Viruses

Mechanisms of viral evasion occur at

- cellular level
- systemic level.

Evasion of Immune Response by Viruses

Viruses can avoid detection by both innate and adaptive immune responses

Evasion of Immune Response by Viruses

Targets of viral evasion proteins

- Pattern recognition receptors in infected cells
- Interferons
- dendritic cells
- T cell receptors
- Antibodies

Evasion of Immune Response by Viruses

- Viruses can express proteins that directly interfere with host processes or mimic host proteins and compete to bind specific receptors.
- Pathogens are able to directly deplete immune cells, and prevent their recruitment to the site of infection.
- In a state of latency, viruses remain dormant and undetectable in host cells

Virology

Prevetion of Viral Infection

Prevention of Viral Infection

- Viruses cause a variety of diseases in animals, including humans, ranging from the common cold to potentially fatal illnesses like meningitis

Prevention of Viral Infection

- Vaccines may be prepared using live viruses, killed viruses, or molecular subunits of the virus.
- The killed viral vaccines and subunit viruses are both incapable of causing disease.
- Live viral vaccines are designed in the laboratory to cause few symptoms in recipients while giving them protective immunity against future infections.
- Polio vaccines.

Prevention of Viral Infection



- Antiviral drugs.
- have limited success in curing viral disease, but in many cases, they have been used to control and reduce symptoms for a wide variety of viral diseases.
- For most viruses, these drugs can inhibit the virus by blocking the actions of one or more of its proteins.

Prevention of Viral Infection

- Acyclovir-for genital herpes
- Tamiflu (oseltamivir) – for influenza

Virology

Viral Vaccines

Introduction

Viral Vaccines

A vaccine is a biological preparation that provides active acquired immunity to a particular disease.

Viral Vaccines

- A vaccination is the injection of a killed or weakened organism that produces immunity in the body against organism.
- An immunization is the process by which a person or animal becomes protected from a disease

Viral Vaccines

History:

- 1796: Edward Jenner prevented smallpox by vaccination with cowpox

- 1885 Louis Pasteur successfully prevented rabies in Joseph Meister by post-exposure vaccination.

Viral Vaccines

Why Vaccines?

- Vaccines are safe.
- Vaccines are effective.
- Vaccines work with immune system.
- Vaccines protect individuals and the community.

Virology

Viral Vaccines

Types

Viral Vaccines

When scientists create vaccines, they consider:

- How your immune system responds to the germ
- Who needs to be vaccinated against the germ
- The best technology or approach to create the vaccine

Viral Vaccines

There are 4 main types of vaccines:

- Live-attenuated vaccines
- Inactivated vaccines
- Subunit, recombinant, polysaccharide, and conjugate vaccines
- Toxoid vaccines

Viral Vaccines

Live-attenuated vaccines

- Live vaccines use a weakened (or attenuated) form of the germ that causes a disease.
- Because these vaccines are so similar to the natural infection that they help prevent, they create a strong and long-lasting immune response.

Viral Vaccines

Live vaccines are used to protect against:

- Measles, Mumps, Rubella (MMR combined vaccine)
- Rotavirus
- Smallpox
- Chickenpox
- Yellow fever

Viral Vaccines

Inactivated vaccines

- Inactivated vaccines use the killed version of the germ that causes a disease.
- Inactivated vaccines usually don't provide immunity (protection) that's as strong as live vaccines.
- Inactivated vaccines are used to protect against:
  - Hepatitis A
  - Flu (shot only)
  - Polio (shot only)
  - Rabies

Viral Vaccines

Subunit, recombinant, polysaccharide, and conjugate vaccines

- Use specific pieces of the virus — like its protein, sugar, or capsid
- They give a very strong immune response that's targeted to key parts of the virus.
- One limitation- need booster shots
  - Hepatitis B
  - HPV (Human Papillomaviurs)
  - Whooping cough (part of the DTaP combined vaccine)
  - Pneumococcal disease
  - Meniggococcal disease

Viral Vaccines

Toxoid vaccines

- Use a toxin (harmful product) made by the virus that causes a disease.
- They create immunity to the parts of the virusthat cause a disease instead of the virus itself. That means the immune response is targeted to the toxin instead of the whole virus.
- You may need booster shots
- Toxoid vaccines are used to protect against:
- Diphtheria
- Tetanus

Virology

Viral Vaccines

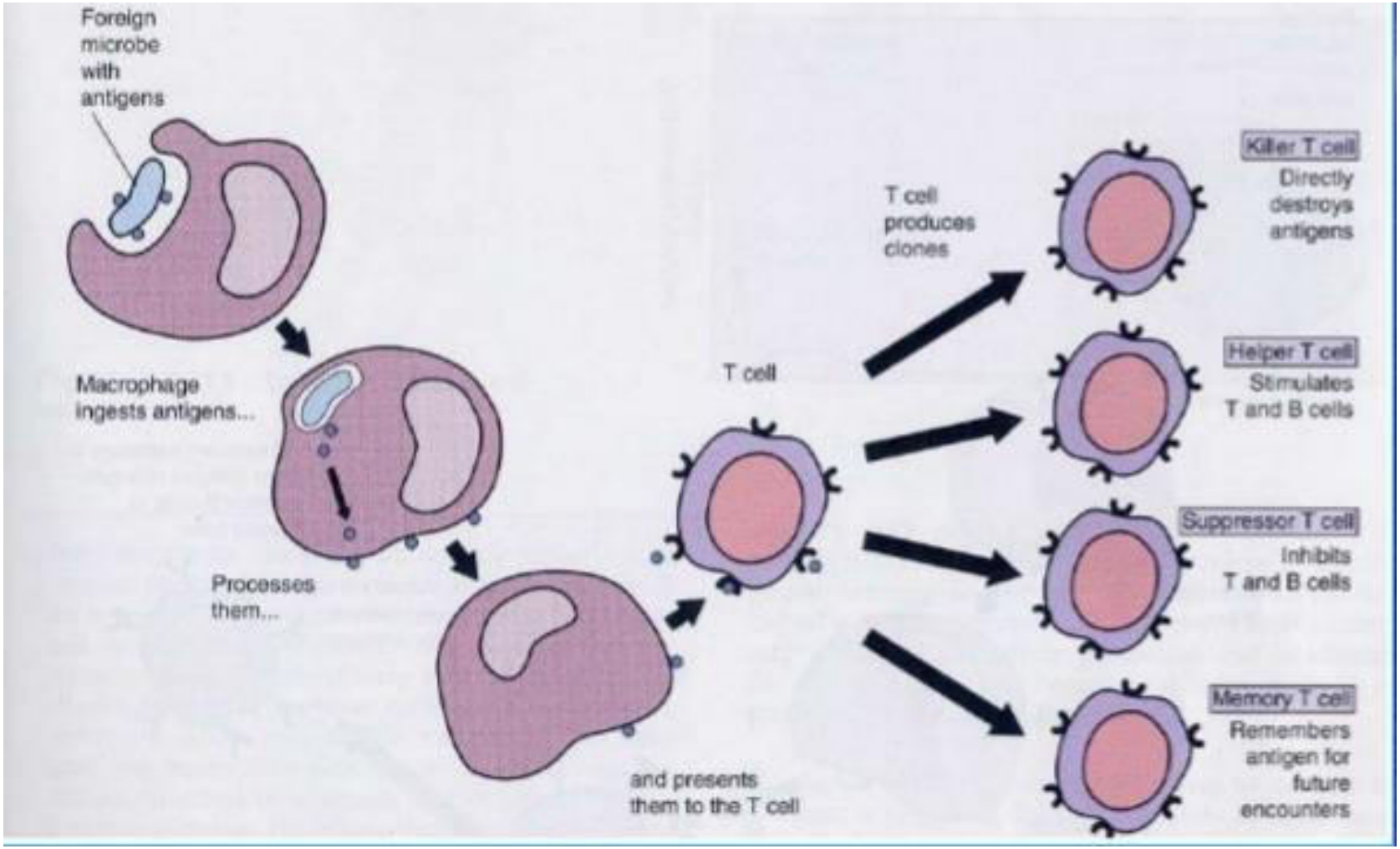
Mechanism

Viral Vaccines

Mechanism of vaccines

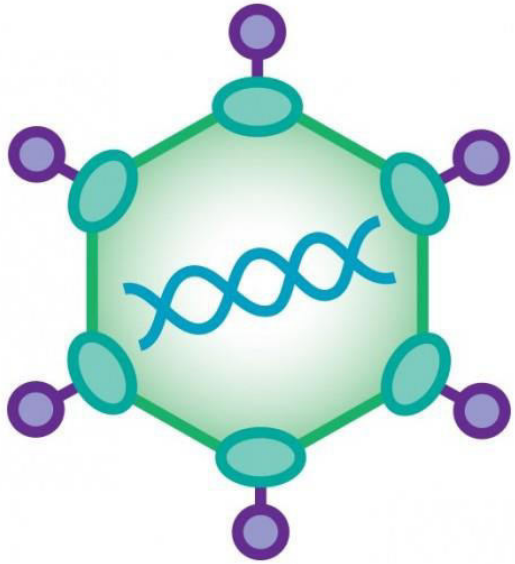
- Vaccines are made from same organism or parts of them that causes disease.
- Introduced in body.
- Immune system reacts to the vaccine-make antibodies.
- Antibodies destroy vaccine germs
- Stay in body, giving specific immunity.

Viral Vaccines



Virology

Virus Vectors



Virus Vectors

SV 40 Based Vectors

- First animal virus based vector
- Small virus with capsid and circular dsDNA
- Bidirection replication

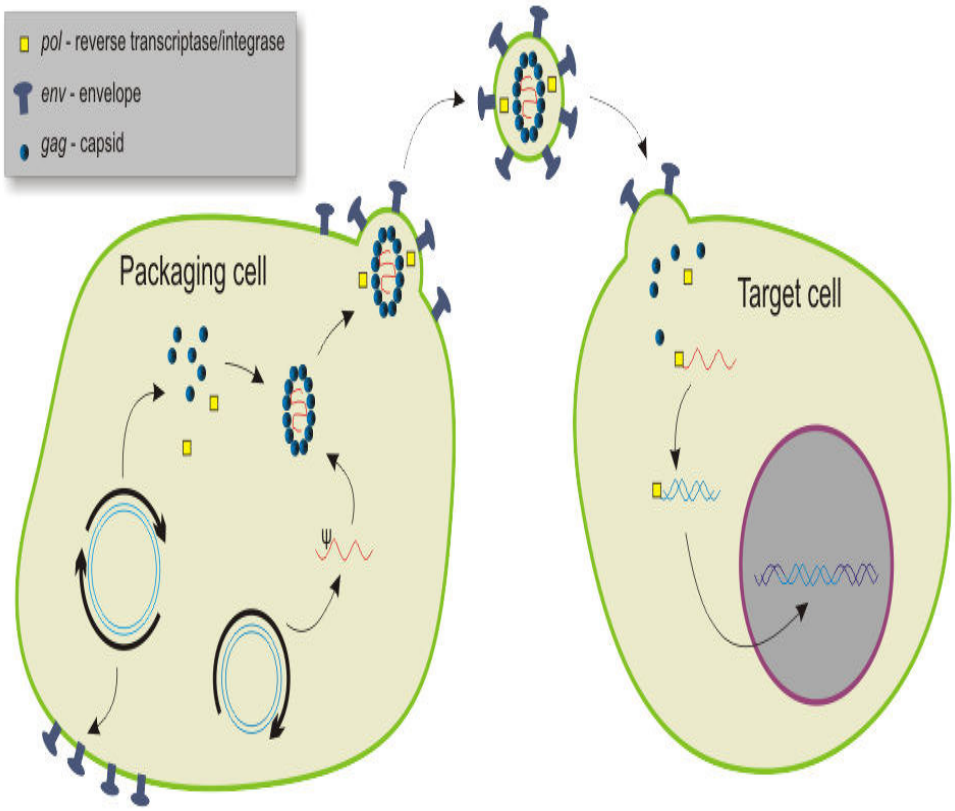
Virus Vectors

Lytic Infection of Sv40

Three steps:

- Early mRNA synthesis
- Viral DNA replication
- Late mRNA synthesis

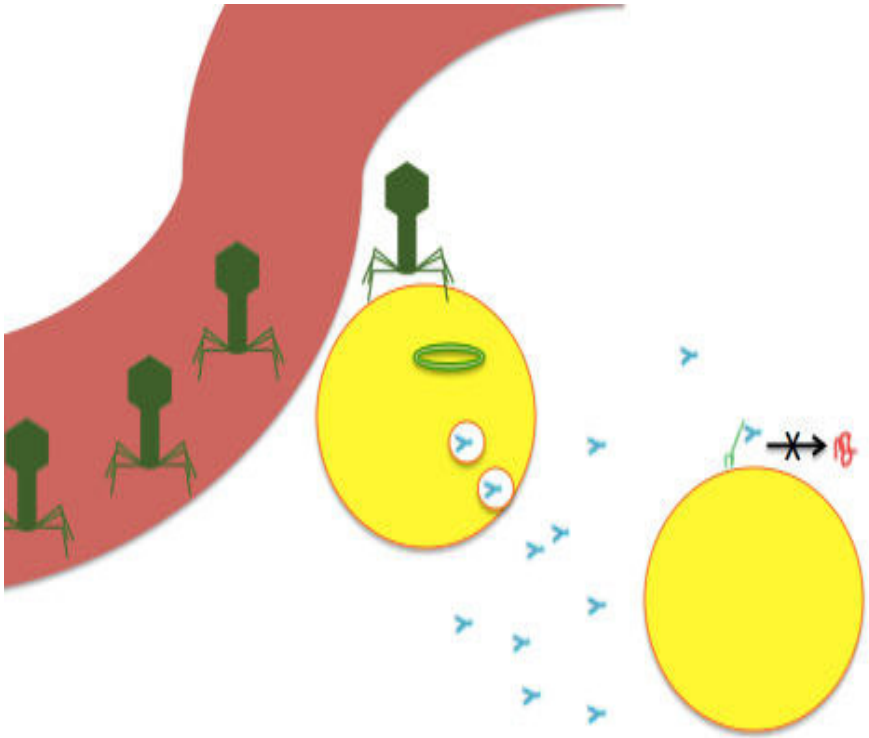
Virus Vectors



Vaccinia Virus Vector

- Large ds DNA
- Contains enzymes for replication
- Generated in vivo

Virus Vectors



Eppstein Barr Virus

- Herpes virus with large genome
- Infects primates and canines
- Used for construction of episomal cDNA

Virology

Gene Therapy

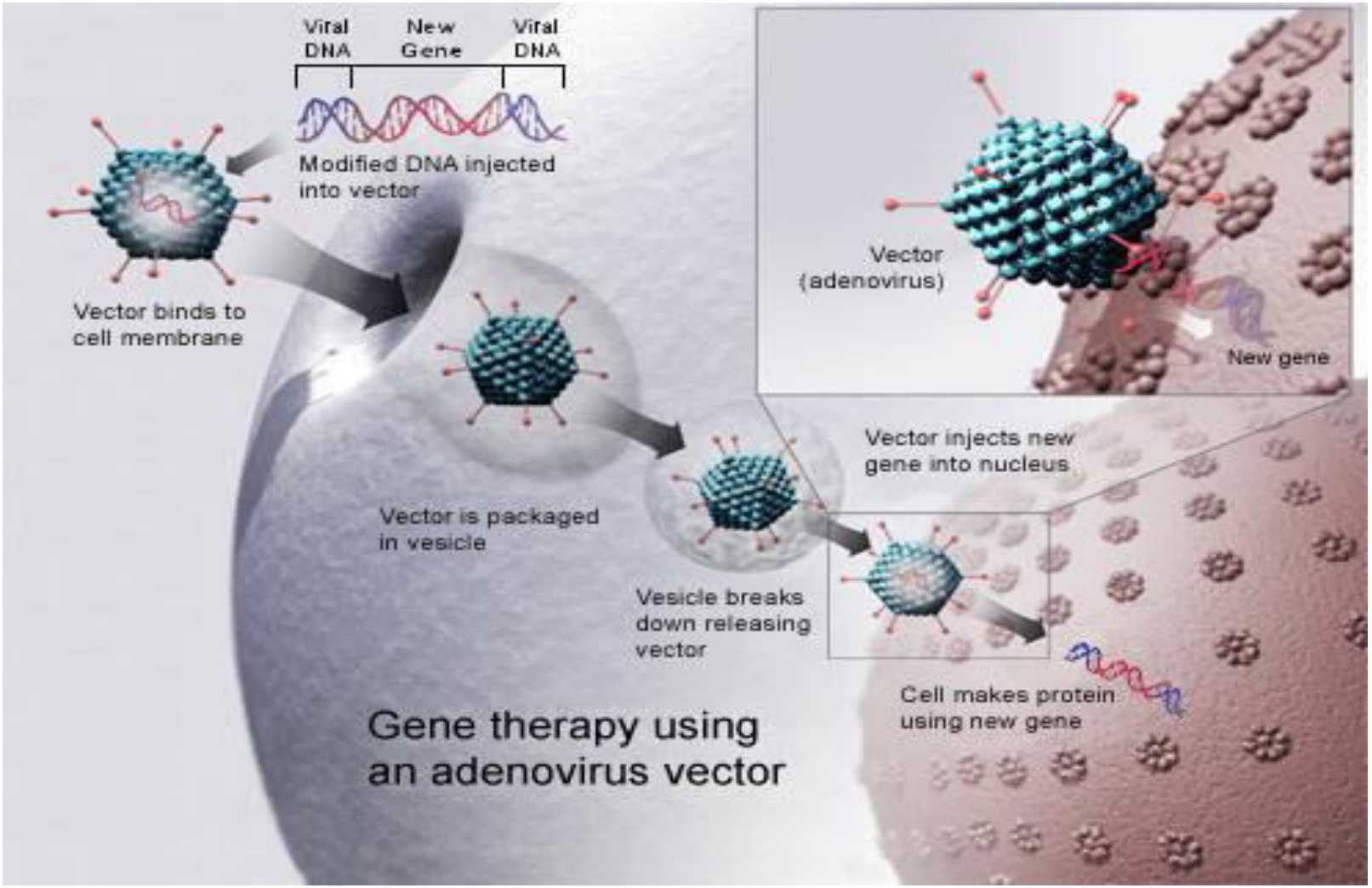
Gene Therapy

- Experimental technique that uses genes to treat or prevent disease.
- Therapeutic delivery of nucleic acid into a patient's cells as a drug to treat disease

Gene Therapy

- First attempt at modifying human DNA was performed in 1980 by Martin Cline.

Gene Therapy



Gene Therapy

Vectors

- Viral
- Non viral
  - electroporation,
  - the gene gun,
  - sonoporation,
  - magnetofection,
  - oligonucleotides, lipoplexes, dendrimers, and inorganic nanoparticles

Gene Therapy

Hurdles:

- Short-lived nature
- Immune response
- Problems with viral vectors
- Multigene disorders
- Insertional mutagenesis
- Cost effective

END

Chemotherapy Against Viral Diseases

Chemotherapy

IDEAL DRUG

- Water soluble
- Chemically and metabolically stable
- Easily absorbed (apolar)

NOT

- Toxic
- Carcinogenic
- Allergenic
- Mutagenic
- Teratogenic

Chemotherapy

A successful drug must interfere with:

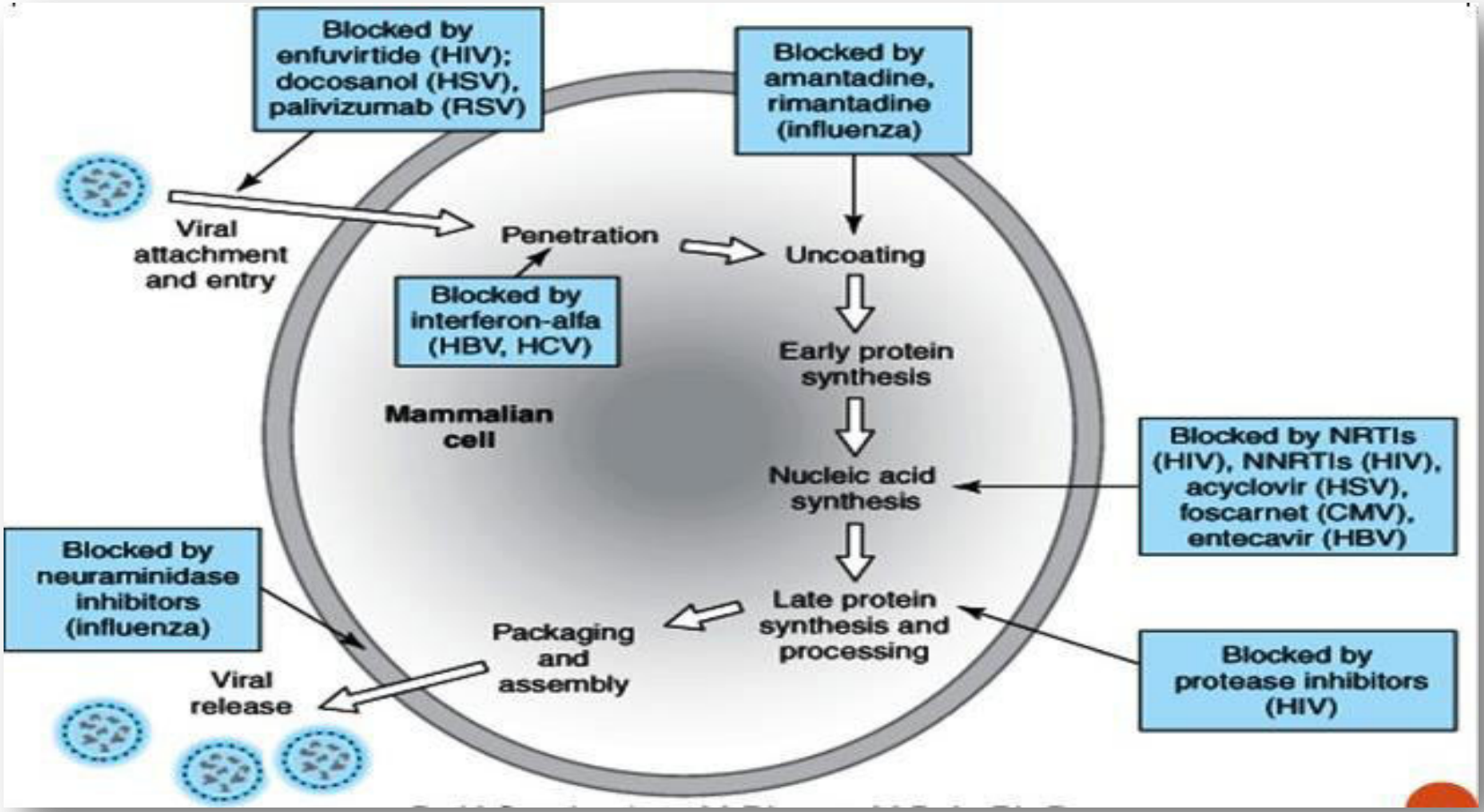
- A specific viral function e.g. enzyme necessary for viral life cycle Anti-Viral Chemotherapy
- A cellular function that the virus needs in order to replicate

If interfere with cellular function either:

- It must be crucial to virus but not the cell
- Only the virus-infected cell must be killed.

Chemotherapy

MECHANISM OF ANTIVIRAL AGENTS



Chemotherapy

Approved Antiviral Drugs

- Amantadine and rimantadine (influenza A virus)
- Ribavirin (respiratory syncytial virus)
- Idoxuridine and trifluridine (topical treatment of herpetic keratitis)
- Vidarabine and acyclovir (systemic treatment of herpes simplex virus and varicella-zoster virus infections)

Chemotherapy

- Famciclovir and valaciclovir (oral treatment of varicella-zoster virus infections)
- Ganciclovir and foscarnet (cytomegalovirus)
- Zidovudine, didanosine, zalcitabine and stavudine (human immunodeficiency virus)

Chemotherapy

Limitations of Antiviral Drugs

- Narrow antiviral spectrum
- Ineffectiveness against the latent virus
- Development of drug-resistant mutants
- Toxic side effects

Virology

Virus Titration

Introduction

Virus Titration

Introduction

- To understand the concentration of virus
- Infectivity titer
- Reed and Muench method or Karber formula

Virus Titration

Standard

- LD50/mL
- Based on mortality of lab mice
- Infectivity titer
- Computation

Virus Titration

Titer

- Frequency of production of specific response of host
- Necessary for viral reproduction
- Smallest amount of virus capable of producing manifestation in host

Virus Titration

Reed and Muench method

Table 1 Calculation of virus titre in mice using the Reed and Muench method

Log_{10} virus dilution	Mice		Cumulative total			Percent mortality
	Died	Survived	Died	Survived	Total	
-1	10	0	57	0	57	57/57 × 100 = 100
-2	10	0	47	0	47	47/47 × 100 = 100
-3	10	0	37	0	37	37/37 × 100 = 100
-4	10	0	27	0	27	27/27 × 100 = 100
-5	10	0	17	0	17	17/17 × 100 = 100
-6	6	4	7	4	11	7/11 × 100 = 63
-7	1	9	1	13	14	1/14 × 100 = 7

Difference of logarithms = (63-50)/(63-7) = 0.23; $\log_{10}$ 50% end point dilution = -6 - (0.23 × 1) = -6.23; 50% end point dilution = $10^{-6.23}$; the titre of the virus = $10^{6.23}$ LD_{50}/mL.

Virus Titration

Karber method

Midlength (mm)	Log Midlength (x_i)	Number of fish (n_i)	Mature (r_i)	Proportion of mature fish (p_i)	$X = X_{i+1} - X_i$	$q_i = 1 - p_i$	$(p_i x q_i)/(n_i - 1)$
85	1.9294	4	0	0	0.0483	1	0
95	1.9777	142	0	0	0.0435	1	0
105	2.0212	155	0	0	0.0395	1	0
115	2.0607	64	4	0.0625	0.0362	0.9375	0.0009
125	2.0969	51	25	0.4902	0.0334	0.5098	0.0050
135	2.1303	59	24	0.4068	0.0310	0.5932	0.0042
145	2.1614	29	11	0.3793	0.0290	0.6207	0.0084
155	2.1903	40	23	0.5750	0.0272	0.4250	0.0063
165	2.2175	27	15	0.5556	0.0256	0.4444	0.0095
175	2.2430	13	11	0.8462	0.0241	0.1538	0.0108
185	2.2672	4	2	0.5000	0.0229	0.5000	0.0833
195	2.2900	7	6	0.8571	0.0217	0.1429	0.0204
205	**2.3118**	4	4	**1**		0	0
Total		**599**	**125**	**5.6726**			**0.1488**

$m = 2.3118 + (0.0362/2) - (0.0362 \times 5.6726) = 2.1244$

Virology

Factors Affecting Titration Procedures

Factors Affecting Titration Procedures

Incubation Time

- Precise period
- Most viruses take precise time to reproduce
- Time between innoculum and appearance of symptom

Factors Affecting Titration Procedures

Type of virus

- Plant virus or animal virus
- Conditions provided
- Optimum media

Factors Affecting Titration Procedures

Procedure contamination

- Bacterial contamination
- Fungal contamination
- Careless lab work
- Improper cleanliness and sanitation
- Disruption of viral genome

Virology

Bacteriophage Titration

Bacteriophage Titration

Definition

- Determination of the number of phage particles in a stock

Bacteriophage Titration

Definition

- Simplest type of titration
- Several dilution preparation
- Plating one or more measured samples of each dilution

Bacteriophage Titration

Procedure

- Phage and bacteria introduced on agar surface
- Incubation
- Interruption of continuous bacterial layer by lysis of phage

Bacteriophage Titration

Charging

- Sometimes adhere to plastic or glass
- If particles adhere to pipettor tips during transfers, it may lower the amount of phage particles present
- One way to minimize this problem is to “charge” the pipettor by pipetting a sample of the solution up and down in the tip, and discharging the sample back into the original tube

Virology

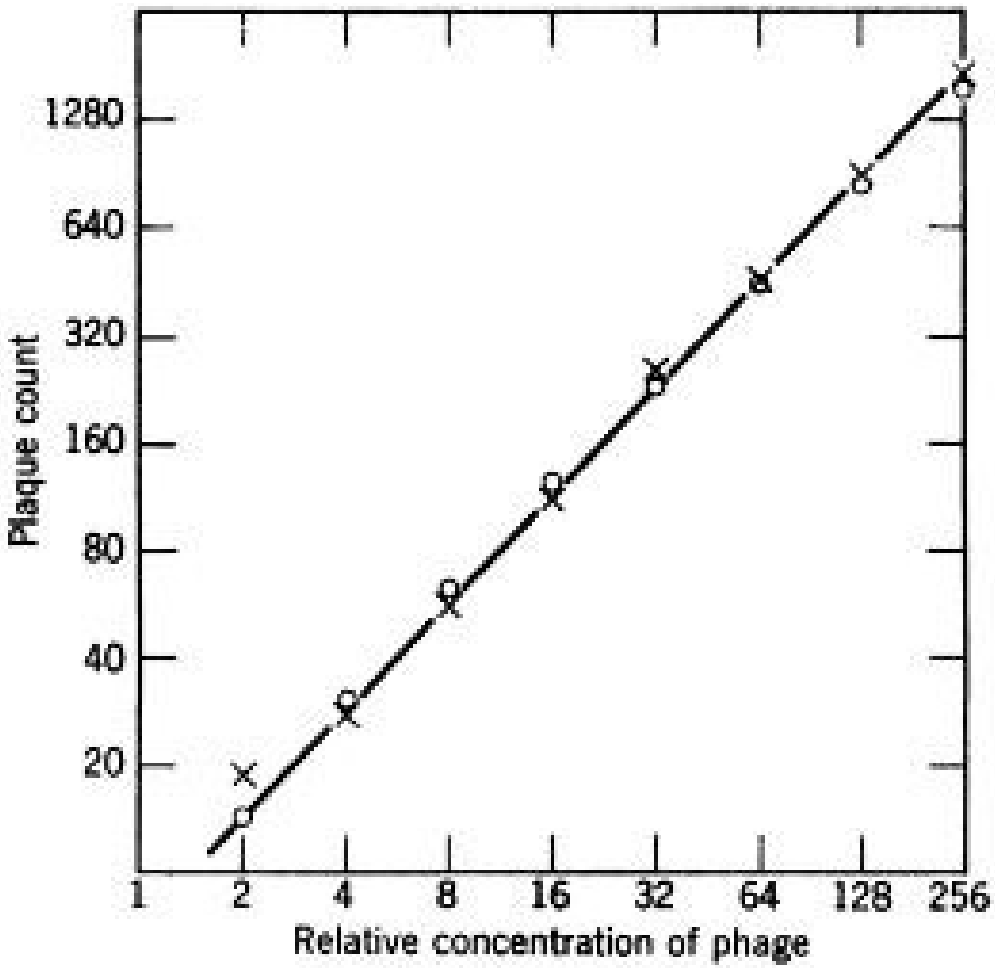
Plaque Count in Bacteriophage Titration

Plaque Count

Plaque values

- Plaque counts analogous to colony counts for bacteria
- Values directly proportional to amount of phage plated

Plaque Count



Plaque Count

- Plate cultures prepared by combining phage with host cells in **top agar**
- Top agar preparations contain lower conc. of agar (7 g/L) than normal solutions used to prepare agar plates (15 g/L)

Plaque Count

- Low agar conc. allows progeny phage from lysed cells to diffuse through the media and infect neighbouring cells
- After lysis, **plaque** (zone of lysed cells) is produced on the plate
- The size of the plaques depend on how soon the bacteria in the agar reach stationary phase and stop reproducing
- Plaques will stop spreading at this point

Final Term Syllabus
Lecture 119 to 213

Efforts By

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Contact us to join our esteemed group

“All About Biotech”

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

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